

Aspirin (Lexi-Drugs Multinational)

Monograph

Images

Adult Patient Education

Pediatric Patient Education

Brand Names: International

Find brand name(s) by country (for United States and Canada see separate Brand Names sections)

Q Enter a Country or Country Code

For country code abbreviations ( [show table](#))

Brazilian Nonproprietary Names (DCB)

Ácido Acetilsalicílico

Japanese Accepted Name (JAN)

アスピリン

Anatomic Therapeutic Chemical (ATC) Classification

- B01AC06
- N02BA01
- N02BA51

Pharmacologic Category

Analgesic, Nonopioid; Antiplatelet Agent; Nonsteroidal Anti-inflammatory Drug (NSAID), Oral; Salicylate

Dosing: Adult

Dosage guidance:

Safety: Avoid regular or frequent use of nonsteroidal anti-inflammatory drugs (NSAIDs) in patients taking aspirin for cardiovascular protection as these may reduce the cardioprotective effects of aspirin (Capone 2005; Catella-Lawson 2001; MacDonald 2003).

Dosing: Dosing information below is based on the IR formulations; do not use the ER formulation in situations when a rapid onset of action is necessary.

Clinical considerations: In patients who require concomitant therapeutic anticoagulation, antiplatelet selection and/or duration of therapy may differ in order to balance risks for thrombosis and bleeding (ACC [Kumbhani 2021]).

Anti-inflammatory for arthritis associated with rheumatic disease

Anti-inflammatory for arthritis associated with rheumatic disease: Immediate release: Oral: 4 to 8 g/day in 4 to 5 divided doses as needed; titrate dose based on response and tolerability. Continue treatment until symptoms resolve (typically 1 to 2 weeks, but potentially up to 8 weeks). Use of aspirin at these high doses (4 to 8 g/day) may be limited by adverse effects (tinnitus, diminished auditory acuity, GI intolerance), making other available NSAIDs preferred (Abramson 2021; Carapetis 2012; Steer 2019).

Atherosclerotic cardiovascular disease

Atherosclerotic cardiovascular disease:

Acute coronary syndrome:

Note: For rapid onset, non-enteric-coated IR tablet(s) should be chewed and swallowed upon identification of clinical and ECG findings suggesting an acute coronary syndrome. Enteric-coated aspirin is not preferred, since onset of action may be delayed. If it is the only product available, enteric-coated IR tablet(s) may be chewed and

swallowed (ACCP [Eikelboom 2012]; Sai 2011). For maintenance therapy, any oral formulation is acceptable for use.

Non-ST-elevation acute coronary syndromes or ST-elevation myocardial infarction:

Note: For initial therapy, administer aspirin in combination with an IV anticoagulant and a P2Y₁₂ inhibitor (ACC/AHA [Lawton 2022]).

Initial:

Immediate release (non-enteric-coated): Oral: 162 to 325 mg administered once (chew and swallow) at the time of diagnosis (ACC/AHA [Lawton 2022]).

Rectal (alternative route): 600 mg administered once at the time of diagnosis if an IR oral formulation is unavailable or oral route is not feasible (Maalouf 2009).

Maintenance (secondary prevention):

Immediate release: Oral: 75 to 100 mg once daily (ACC/AHA [Lawton 2022]; Mehta 2001).

Duration of therapy:

Preferred approach: Aspirin plus a P2Y₁₂ inhibitor (dual antiplatelet therapy [DAPT]) should be continued for ≥12 months, unless major bleeding is a concern. For patients at high risk of bleeding or who experience overt bleeding, DAPT for 6 months may be reasonable. If there have been no major bleeding complications after 12 months, continuation of DAPT may be considered. Reevaluate the need for DAPT at regular intervals based on bleeding and thrombotic risks. When DAPT is complete, discontinue the P2Y₁₂ inhibitor and continue aspirin indefinitely for secondary prevention (ACC/AHA [Lawton 2022]; Bonaca 2015; Mauri 2014; Mehta 2001; Wallentin 2009; Wiviott 2007; Yusuf 2001).

Alternative approach in select patients to minimize bleeding events: Aspirin in combination with a P2Y₁₂ inhibitor (DAPT) should be continued for 1 to 3 months after PCI, then discontinue aspirin and continue P2Y₁₂ inhibitor monotherapy. When the P2Y₁₂ inhibitor is discontinued, restart aspirin for secondary prevention (ACC/AHA [Lawton 2022]; Hahn 2019; Kim 2020; Mehran 2019; O'Donoghue 2020; Vranckx 2018; Watanabe 2019).

Percutaneous coronary intervention for stable ischemic heart disease (off-label use):

Initial:

Note: For initial therapy, non-enteric-coated IR tablet(s) should be administered. Enteric-coated aspirin is not preferred since onset of action is delayed. For patients who receive a coronary stent during percutaneous coronary intervention, administer aspirin in combination with a parenteral anticoagulant and a P2Y₁₂ inhibitor (clopidogrel or ticagrelor; prasugrel is **not** recommended in this patient population) (ACC/AHA [Lawton 2022]).

Immediate release (non-enteric-coated): Oral: 162 to 325 mg given ≥2 hours (preferably 24 hours) before the procedure (ACC/AHA [Lawton 2022]).

Maintenance:

Immediate release: Oral: 75 to 100 mg once daily in combination with clopidogrel **or** ticagrelor (DAPT) (ACC/AHA [Lawton 2022]). Refer to clopidogrel or ticagrelor monograph for information on duration of DAPT and subsequent management.

Atherosclerotic cardiovascular disease, primary prevention (off-label use):

Note: Use should be a shared decision between health care professionals and patients ≥40 years of age who are at increased risk for cardiovascular disease, after weighing risks versus benefits. Some experts recommend against this use for patients ≥60 years of age (USPSTF 2022), while others recommend against this use for patients >70 years of age (ACC/AHA [Arnett 2019]).

Immediate release: Oral: 75 to 100 mg once daily (ACC/AHA [Arnett 2019]).

Atherosclerotic cardiovascular disease, secondary prevention:

Carotid artery atherosclerosis, asymptomatic or symptomatic (off-label use): Immediate release: Oral: 75 to 325 mg once daily (ACCP [Alonso-Coello 2012]; Walker 1995).

Coronary artery bypass graft surgery: Immediate release: Oral: 75 to 81 mg once daily beginning preoperatively; continue indefinitely following surgery (AHA [Kulik 2015]; Aranki 2021). Following surgery, some experts restart aspirin (with or without a one-time, 325 mg loading dose) approximately 6 hours after completion of the procedure or after extubation, whichever comes first (Aranki 2021).

Off-pump coronary artery bypass graft surgery: Following surgery, consider adding clopidogrel in combination with aspirin for 12 months then discontinue clopidogrel and continue aspirin indefinitely (AHA [Kulik 2015]).

Patients with acute coronary syndrome followed by coronary artery bypass graft surgery: Administer aspirin in combination with a P2Y₁₂ inhibitor for 12 months then continue aspirin indefinitely (AHA [Kulik 2015]). Some experts do not use P2Y₁₂ inhibitors postoperatively in these patients (Aranki 2021).

Ischemic stroke/transient ischemic attack:

Note: In patients who receive IV thrombolytic therapy, antiplatelet therapy is generally delayed for at least 24 hours, but administered as soon as possible thereafter (AHA/ASA [Powers 2019]; Oliveira-Filho 2022).

Due to intracranial large artery atherosclerosis (50% to 99%), secondary prevention:

Note: Avoid dual antiplatelet therapy (DAPT) in patients with hemorrhagic transformation (Oliveira-Filho 2022).

Immediate release: Oral:

Monotherapy: 325 mg once daily. May consider a short course of DAPT as outlined below. (AHA/ASA [Kleindorfer 2021]).

Short-term (90 days) DAPT with concomitant clopidogrel in patients with recent (≤ 30 days) stroke or TIA due to 70% to 99% stenosis of an intracranial large artery: 325 mg once daily followed by long-term aspirin monotherapy (AHA/ASA [Kleindorfer 2021]; Turan 2024).

Short-term (21 days) DAPT with concomitant clopidogrel in patients with recent (≤ 30 days) minor stroke or high-risk TIA due to 50% to 69% stenosis of an intracranial large artery: 160 to 325 mg loading dose followed by 50 to 100 mg once daily (Turan 2024).

Short-term (21 to 30 days) DAPT with concomitant ticagrelor in patients with recent (≤ 24 hours) minor stroke or high-risk TIA and $>30\%$ stenosis of ipsilateral major intracranial artery: 300 to 325 mg loading dose followed by 75 to 100 mg once daily (AHA/ASA [Kleindorfer 2021]; Oliveira-Filho 2022; ticagrelor manufacturer's labeling).

Due to other noncardioembolic causes (eg, small vessel disease), secondary prevention:

Note: For patients with recent (eg, ≤ 24 hours) minor stroke or high-risk TIA, may consider short-term DAPT with concomitant clopidogrel (21 to 90 days) or ticagrelor (21 to 30 days) followed by long-term single-agent antiplatelet therapy with aspirin, clopidogrel, or aspirin/ER dipyridamole. Avoid DAPT in patients with hemorrhagic transformation (AHA/ASA [Kleindorfer 2021]; Oliveira-Filho 2022).

Initial:

Immediate release: Oral: 160 to 325 mg administered once at the time of diagnosis (AHA/ASA [Powers 2019]; Oliveira-Filho 2022).

Rectal (alternative route): 300 mg administered once at the time of diagnosis if oral route is not feasible (AHA/ASA [Powers 2019]; IST 1997; Sandercock 2014).

Maintenance:

Immediate release: Oral: 50 to 325 mg once daily; some experts recommend 50 to 100 mg once daily when used with clopidogrel for DAPT and as long-term single-agent therapy (ACCP [Lansberg 2012]; AHA/ASA [Kleindorfer 2021]; Cucchiara 2022). When used with ticagrelor for DAPT, 75 to 100 mg once daily is recommended (ticagrelor manufacturer's labeling).

Peripheral atherosclerotic disease (upper or lower extremity; with or without a revascularization procedure) (off-label use): Immediate release: Oral: 75 to 100 mg once daily (ACCP [Alonso-Coello 2012]; AHA/ACC [Gerhard-Herman 2017]).

Stable ischemic heart disease: Immediate release: Oral: 75 to 100 mg once daily (AHA/ACC [Virani 2023]; Kannam 2019).

Carotid artery stenting

Carotid artery stenting (off-label use):

Percutaneous approach:

Initial:

Initiation ≥ 48 hours before procedure: Immediate release: Oral: 325 mg twice daily in combination with clopidogrel (or ticagrelor if clopidogrel cannot be used) (Brott 2010, Jim 2024a, Marcaccio 2022).

Initiation < 48 hours before procedure: Immediate release: Oral: 650 mg once at least 4 hours before procedure in combination with clopidogrel (or ticagrelor if clopidogrel cannot be used) (Brott 2010, Jim 2024a, Marcaccio 2022).

Maintenance:

Immediate release: Oral: 75 to 325 mg once daily in combination with clopidogrel (or ticagrelor if clopidogrel cannot be used) for at least 4 weeks, then discontinue clopidogrel and continue aspirin 75 to 325 mg once daily indefinitely thereafter. In patients with history of neck irradiation, some experts recommend continuing aspirin plus clopidogrel indefinitely (ASA/ACCF/AHA [Brott 2011]; Brott 2010; Jim 2024a, Marcaccio 2022).

Transcarotid approach:

Initial:

Initiation ≥ 72 hours before procedure: Immediate release: Oral: 75 to 325 mg once daily in combination with clopidogrel (or ticagrelor if clopidogrel cannot be used) (Jim 2024b, Kwolek 2015, Marcaccio 2022).

Initiation < 72 hours before procedure: Immediate release: Oral: 650 mg once at least 4 hours before procedure in combination with clopidogrel (or ticagrelor if clopidogrel cannot be used) (Jim 2024b, Kwolek 2015, Marcaccio 2022).

Maintenance:

Immediate release: Oral: 75 to 325 mg once daily in combination with clopidogrel (or ticagrelor if clopidogrel cannot be used) for at least 4 weeks, then discontinue clopidogrel and continue aspirin 75 to 325 mg once daily indefinitely thereafter. In patients with history of neck irradiation, some experts recommend continuing aspirin plus clopidogrel indefinitely (Brott 2010; Jim 2024b; Kwolek 2015, Marcaccio 2022).

Carotid endarterectomy

Carotid endarterectomy: Immediate release: Oral: 75 to 325 mg once daily starting prior to surgery and continued indefinitely (ACCP [Alonso-Coello 2012]; Fairman 2021).

Colorectal cancer risk reduction, primary prevention

Colorectal cancer risk reduction, primary prevention (off-label use):

Note: The optimal dose and duration of therapy for colorectal cancer risk reduction are unknown. Utilization should be a shared decision between health care professionals and patients that weighs the risk versus benefits of treatment (Chan 2020).

Immediate release: Oral: 75 to 325 mg once daily (Chan 2020; Rothwell 2010; Ye 2013).

Migraine, acute treatment

Migraine, acute treatment (off-label use):

Note: For mild to moderate attacks not associated with vomiting or severe nausea (Schwedt 2021).

Immediate release: Oral: 900 mg or 1 g once (Lipton 2005; MacGregor 2002).

Pain or fever

Pain or fever:

Immediate release:

Oral: 325 mg to 1 g every 4 to 6 hours as needed; usual maximum daily dose: 4 g/day (Abramson 2021).

Note: If patient cannot take orally, rectal suppositories (300 or 600 mg) are available.

IM/IV [International product]: Note: Dose is expressed as acetylsalicylic acid (900 mg lysine acetylsalicylate is equivalent to 500 mg acetylsalicylic acid [aspirin]); 500 mg to 1 g every 4 hours as needed; maximum: 3 g/day.

Note: Reserve IM/IV administration for severe pain in patients who are unable to tolerate oral formulations.

Pericarditis, acute or recurrent

Pericarditis, acute or recurrent (off-label use):

Note: Aspirin is preferred over other nonsteroidal anti-inflammatory drugs (NSAIDs) in patients with ischemic heart disease. If pericarditis occurs secondary to acute myocardial infarction (MI), avoid NSAIDs (other than once-daily, low-dose aspirin for treatment of MI) and use acetaminophen for analgesia. If this does not suffice or if additional pericarditis treatment is necessary after 7 to 10 days, aspirin is preferred and other NSAIDs should continue to be avoided (ACCF/AHA [O'Gara 2013]; LeWinter 2022).

Immediate release: Oral: Initial: 650 mg to 1 g every 8 hours until resolution of symptoms for at least 24 hours and normalization of inflammatory biomarkers (eg, C-reactive protein) if monitored; initial therapy typically lasts for ≥ 1 to 2 weeks. Gradually taper over several weeks by decreasing each dose by 250 to 500 mg every 1 to 2 weeks; during taper, ensure patient remains asymptomatic and inflammatory biomarkers remain normal (if monitored). Use in combination with colchicine. In patients at risk of NSAID-related GI toxicity, prophylaxis (generally with a proton pump inhibitor) is recommended (ESC [Adler 2015]; Imazio 2022).

Polycythemia vera, prevention of thrombosis

Polycythemia vera, prevention of thrombosis (off-label use):

Note: Use with caution in patients with concurrent acquired von Willebrand syndrome due to risk for bleeding (Tefferi 2024).

Immediate release: Oral: 75 to 100 mg once daily (Barbui 2006; BSH [McMullin 2019]; Landolfi 2004; Tefferi 2024).

Preeclampsia prevention

Preeclampsia prevention (off-label use):

Note: Consider for use in pregnant women with ≥ 2 moderate risk factors or ≥ 1 high risk factor for preeclampsia (ACOG 743 2018).

Immediate release: Oral: 81 to 162 mg once daily, ideally beginning between 12 to 16 weeks' gestation but may be started up to 28 weeks' gestation; continue therapy until delivery (ACOG 743 2018; Rolnik 2017).

Valvular heart disease

Valvular heart disease:

Surgical prosthetic heart valve replacement, thromboprophylaxis:

Bioprosthetic aortic or mitral heart valve replacement (off-label use): Note: Timing of aspirin initiation is based on bleeding risk. For patients at low risk of bleeding, treat with warfarin monotherapy for 3 to 6 months, then transition to aspirin monotherapy. For patients at elevated risk of bleeding, initiate aspirin within 24 hours of surgery (ie, no initial treatment with warfarin) (ACC/AHA [Otto 2021]; Nkomo 2022).

Immediate release: Oral: 75 to 100 mg once daily (ACC/AHA [Otto 2021]).

Mechanical aortic or mitral heart valve replacement (off-label use): Note: Aspirin is required, in combination with warfarin, for patients who receive an On-X mechanical aortic valve. For other mechanical valve types, consider risks of thromboembolism and bleeding before adding aspirin; warfarin monotherapy is typically sufficient. Addition of aspirin is not routinely required in the absence of another indication for antiplatelet therapy (ACC/AHA [Otto 2021]; Puskas 2014).

Immediate release: Oral: 75 to 100 mg once daily in combination with warfarin (ACC/AHA [Otto 2021]; Puskas 2014).

Transcatheter *aortic* valve replacement, thromboprophylaxis (off-label use):

Note: Refer to institutional policies and procedures on use of antiplatelet therapy for patients who require therapeutic anticoagulation for a different indication.

Immediate release: Oral: 75 to 100 mg once daily; may use in combination with clopidogrel for 3 to 6 months after transcatheter aortic valve replacement, depending on type of valve implanted (ACC/AHA [Otto 2021]). To minimize risk of bleeding complications, may give aspirin **or** clopidogrel alone and reserve dual antiplatelet therapy during the first 3 to 6 months for patients at high risk of a thrombotic event; for either strategy, continue aspirin indefinitely after the initial 3 to 6 months of therapy (Brouwer 2020; Kuno 2019).

Transcatheter *mitral* valve repair with MitraClip device, thromboprophylaxis (off-label use):

Note: Patients are generally treated with antithrombotic therapy (antiplatelet or anticoagulant if there is a concurrent indication) for at least 6 months following the procedure.

Immediate release: Oral:

Loading dose: 325 mg once immediately following MitraClip insertion or within 24 hours prior to the procedure; use in combination with clopidogrel (Stone 2018).

Maintenance: 81 mg once daily for at least 6 months; may use as monotherapy or in combination with clopidogrel (Stone 2018).

Venous thromboembolism prevention, indefinite therapy

Venous thromboembolism prevention, indefinite therapy (off-label use):

Note: For use in select patients to prevent recurrent venous thromboembolism (VTE) if unable to take an anticoagulant. In patients who have completed ≥ 6 months of anticoagulation and in whom indefinite therapeutic anticoagulation is indicated, aspirin is not recommended since it is less effective (ACCP [Stevens 2021]).

Immediate release: Oral: 100 mg once daily after completion of a conventional treatment course with therapeutic anticoagulation (Becattini 2012; Brighton 2012; Simes 2014).

Venous thromboembolism prophylaxis for total hip or total knee arthroplasty

Venous thromboembolism prophylaxis for total hip or total knee arthroplasty (off-label use):

Note: Some experts limit this strategy to lower-risk patients who undergo elective unilateral total hip arthroplasty (THA) or total knee arthroplasty (TKA), ambulate within 24 hours after surgery, and do not have additional risk factors for VTE, indications for long-term anticoagulation, lower limb or hip fracture in the previous 3 months, or expected major surgery in the upcoming 3 months (Douketis 2023).

Immediate release: Oral: Begin postoperative prophylaxis with rivaroxaban or a low-molecular-weight heparin, then discontinue therapy after 5 days for TKA or after 5 to 10 days for THA, and initiate aspirin 81 mg once daily. For TKA, continue aspirin through postoperative day 10 to 14 (if not fully ambulatory by day 14, may continue through postoperative day 35). For THA, continue aspirin through postoperative day 35 (Anderson 2013; Anderson 2018; Douketis 2023).

Venous thromboembolism prophylaxis in lower-risk patients with multiple myeloma receiving immunomodulatory therapy

Venous thromboembolism prophylaxis in lower-risk patients with multiple myeloma receiving immunomodulatory therapy (eg, lenalidomide, pomalidomide, thalidomide) (off-label use): Oral: 81 to 325 mg once daily during immunomodulatory therapy (Larocca 2012; Palumbo 2011; Rajkumar 2005).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

* See [Dosage and Administration in AHFS Essentials](#) for additional information.

Dosing: Older Adult

IM/IV [International product]: Analgesic and antipyretic: Note: Dose is expressed as acetylsalicylic acid (900 mg lysine acetylsalicylate is equivalent to 500 mg acetylsalicylic acid [aspirin]): 500 mg to 1 g every 4 hours as needed; maximum: 2 g/day.

Note: Reserve IM/IV administration for severe pain in patients who are unable to tolerate oral formulations.

Oral, rectal: Refer to adult dosing. Avoid for the primary prevention of cardiovascular disease. In older adults already taking for primary prevention, consider deprescribing. Unless alternative agents are ineffective and a gastroprotective agent can be administered, avoid short-term scheduled use of doses >325 mg in combination with corticosteroids, anticoagulants, or antiplatelet agents or chronic use with or without medications that increase risk for bleeding (Beers Criteria [AGS 2023]).

Dosing: Altered Kidney Function: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS

Antiplatelet uses:

Note: In general the benefit of low-dose aspirin outweighs any risk associated with nephropathy or other adverse effects even in the setting of severe kidney impairment; the recommended aspirin dose should not be reduced in any patient with suspected or documented cardiovascular disease, or other antithrombotic indication (^{Ref}).

Altered kidney function: No dosage adjustment necessary for any degree of kidney impairment (^{Ref}).

Hemodialysis, intermittent (thrice weekly): No dosage adjustment necessary (^{Ref}).

Peritoneal dialysis: No dosage adjustment necessary (^{Ref}).

CRRT: No dosage adjustment necessary (^{Ref}).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment necessary (^{Ref}).

Analgesia or anti-inflammatory uses:

Altered kidney function:

CrCl >10 mL/minute: No dosage adjustment necessary; however, high doses have been associated with acute kidney injury (AKI) ^(Ref). Use lowest effective dosage and limit duration of therapy, particularly for patients at high risk for developing AKI (eg, patients with chronic kidney disease, volume depletion, older age) ^(Ref).

CrCl <10 mL/minute: Avoid use (manufacturer's labeling; expert opinion). May exacerbate uremic GI and hematologic symptoms ^(Ref).

Hemodialysis, intermittent (thrice weekly): Avoid use. May exacerbate uremic GI and hematologic symptoms ^(Ref).

Peritoneal dialysis: Avoid use. May exacerbate uremic GI and hematologic symptoms ^(Ref).

CRRT: Avoid use ^(Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): Avoid use ^(Ref).

Dosing: Altered Liver Function: Adult

Avoid use in severe liver disease.

Dosing: Pediatric

Dosage guidance:

Dosing: Two immediate-release formulations are available: tablets and liquid-filled capsules (Vazalore). Immediate-release capsules cannot be divided and may not be an appropriate dosage form for some pediatric patients. All pediatric dosing for immediate-release formulations unless otherwise specified. Doses are typically rounded to a convenient amount (eg, $\frac{1}{4}$ of 81 mg tablet).

Analgesic

Analgesic: Oral, rectal: **Note:** Do not use aspirin in pediatric patients <18 years who have or who are recovering from chickenpox or flu symptoms (eg, viral illness) due to the association with Reye syndrome ^(Ref):

Infants, Children, and Adolescents weighing <50 kg: Limited data available: 10 to 15 mg/kg/dose every 4 to 6 hours; maximum daily dose: 90 mg/kg/**day** or 4,000 mg/**day** whichever is less ^(Ref).

Children ≥12 years and Adolescents weighing ≥50 kg: 325 to 650 mg every 4 to 6 hours; maximum daily dose: 4,000 mg/**day** ^(Ref).

Anti-inflammatory

Anti-inflammatory: Limited data available: Infants, Children, and Adolescents: Oral: Initial: 60 to 90 mg/kg/**day** in divided doses; usual maintenance: 80 to 100 mg/kg/**day** divided every 6 to 8 hours; monitor serum concentrations ^(Ref). **Note:** Although included in some product labeling, the use of salicylates including aspirin for the treatment of juvenile idiopathic arthritis has been replaced by other pharmacologic agents with improved efficacy and adverse effect profiles (eg, non-salicylate nonsteroidal anti-inflammatory drugs, disease-modifying antirheumatic drugs, glucocorticoids, biologic modifiers) ^(Ref).

Antiplatelet effect

Antiplatelet effect: Limited data available: Infants, Children, and Adolescents: Oral: Adequate pediatric studies have not been performed; pediatric dosage is derived from adult studies. Usual adult maximum daily dose for antiplatelet effects is 325 mg/**day**.

Acute ischemic stroke (AIS):

Noncardioembolic: 1 to 5 mg/kg/dose once daily for ≥2 years; patients with recurrent AIS or TIAs should be transitioned to clopidogrel, LMWH, or warfarin ^(Ref).

Secondary to Moyamoya and non-Moyamoya vasculopathy: 1 to 5 mg/kg/dose once daily; **Note:** In non-Moyamoya vasculopathy, continue aspirin for 3 months, with subsequent use guided by repeat cerebrovascular imaging ^(Ref).

Prosthetic heart valve:

Bioprosthetic aortic valve (with normal sinus rhythm): 1 to 5 mg/kg/dose once daily for 3 months ^(Ref).

Mechanical aortic and/or mitral valve: 1 to 5 mg/kg/dose once daily combined with vitamin K antagonist (eg, warfarin) is recommended as first-line antithrombotic therapy ^(Ref). Alternative regimens: 6 to 20 mg/kg/dose once daily in combination with dipyridamole ^(Ref).

Shunts: Blalock-Taussig; Glenn; postoperative; primary prophylaxis: 1 to 5 mg/kg/dose once daily ^(Ref).

Norwood, Fontan surgery, postoperative; primary prophylaxis: 1 to 5 mg/kg/dose once daily ^(Ref).

Transcatheter Atrial Septal Defect (ASD) or Ventricular Septal Defect (VSD) devices, postprocedure prophylaxis: 1 to 5 mg/kg/dose once daily starting one to several days prior to implantation and continued for at least 6 months. For older children and adolescents, after device closure of ASD, an additional anticoagulant may be given with aspirin for 3 to 6 months, but the aspirin should continue for at least 6 months ^(Ref).

Ventricular assist device (VAD) placement: 1 to 5 mg/kg/dose once daily initiated within 72 hours of VAD placement; should be used with heparin (initiated between 8 to 48 hours following implantation) and with or without dipyridamole ^(Ref).

Kawasaki disease, treatment

Kawasaki disease, treatment: Limited data available: **Note:** Patients with Kawasaki disease and presenting with influenza or viral illness should not receive aspirin; acetaminophen is suggested as an alternate antipyretic in these patients and an alternate antiplatelet agent is suggested for a minimum of 2 weeks ^(Ref).

Initial therapy (acute phase): **Note:** Optimal dose not established. Use in combination with IV immune globulin.

Infants and Children:

Moderate dose: Oral: 30 to 50 mg/kg/**day** divided every 6 hours until fever resolves for at least 48 to 72 hours ^(Ref).

High dose: Oral: 80 to 100 mg/kg/**day** divided every 6 hours until fever resolves for at least 48 to 72 hours ^(Ref).

Subsequent therapy (low-dose; antiplatelet effects): Infants and Children: Oral: 3 to 5 mg/kg/**day** once daily; initiate at least 48 to 72 hours after resolution of fever and continue for 6 to 8 weeks in patients without coronary artery abnormalities. In patients with coronary artery abnormalities ranging from dilation to large/giant aneurysms, low-dose aspirin should be continued (with or without dual antiplatelet therapy, warfarin or low molecular weight heparin); duration of aspirin therapy is based upon severity of coronary artery involvement; recommend consulting published treatment and institutional guidelines, and expert medical consultation for additional guidance ^(Ref).

Multisystem inflammatory syndrome in children associated with SARS-CoV-2

Multisystem inflammatory syndrome in children (MIS-C) associated with SARS-CoV-2 (antiplatelet/antithrombotic effects): Limited data available:

Note: Recommended for all MIS-C patients without active bleeding, significant bleeding risk, or platelet count $\leq 80,000$ cells/mm³ ^(Ref). Use is based on limited experience in patients with MIS-C, emerging data in adults with COVID-19, and extrapolation from Kawasaki disease and myocarditis ^(Ref).

Infants, Children, and Adolescents: Oral: 3 to 5 mg/kg/**day** once daily; maximum daily dose: 81 mg/**day**; continue until normalization of platelet count, confirmed normal coronary arteries at ≥ 4 weeks after diagnosis, and ejection fraction $>35\%$ ^(Ref).

Rheumatic fever

Rheumatic fever: Limited data available: Infants, Children, and Adolescents: Oral: Initial: 100 mg/kg/**day** divided into 4 to 5 doses; if response inadequate, may increase dose to 125 mg/kg/**day**; continue for 2 weeks; then decrease dose to 60 to 70 mg/kg/**day** in divided doses for an additional 3 to 6 weeks ^(Ref).

Migratory polyarthritis, with carditis without cardiomegaly or congestive heart failure: Oral: Initial: 50 to 70 mg/kg/**day** in 4 divided doses for 3 to 5 days, followed by 50 mg/kg/**day** in 4 divided doses for 2 to 3 weeks, followed by 25 mg/kg/**day** in 4 divided doses for 2 to 4 weeks ^(Ref); escalation to doses up to 80 to 100 mg/kg/**day** in 4 or 5 divided doses has been described for arthritis management ^(Ref).

Carditis and more than minimal cardiomegaly or congestive heart failure: **Note:** Aspirin should be initiated at the beginning of prednisone taper regimen to prevent rebound inflammation: Oral: 50 mg/kg/**day** in 4 divided doses for 6 weeks ^(Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Altered Kidney Function: Pediatric

Infants, Children, and Adolescents: There are no recommendations in the manufacturer's labeling; however, the following adjustments have been recommended ^(Ref):

GFR $\geq$ 10 mL/minute/1.73 m²: No dosage adjustment necessary.

GFR <10 mL/minute/1.73 m²: Avoid use.

Intermittent hemodialysis: Dialyzable: 50% to 100% (concentration dependent; higher salicylate concentrations are more readily dialyzable) ^(Ref); administer daily dose after dialysis session on dialysis days ^(Ref).

Peritoneal dialysis: Avoid use.

CRRT: No dosage adjustment necessary; monitor serum concentrations.

Dosing: Altered Liver Function: Pediatric

All ages: Avoid use in severe liver disease.

Calculations

- [Creatinine Clearance by Cockcroft-Gault](#)
- [Creatinine Clearance by Cockcroft-Gault \(SI units\)](#)
- [Creatinine Clearance by Cockcroft-Gault with IBW](#)
- [Creatinine Clearance by Cockcroft-Gault with IBW \(SI units\)](#)

Use: Labeled Indications

Oral:

Immediate release:

Analgesic, antipyretic, and anti-inflammatory: For the temporary relief of headache, pain, and fever caused by colds, muscle aches and pains, menstrual pain, toothache pain, and minor aches and pains of arthritis.

Revascularization procedures: For use in patients who have undergone revascularization procedures (ie, coronary artery bypass graft, percutaneous transluminal coronary angioplasty, or carotid endarterectomy).

Vascular indications, including ischemic stroke, transient ischemic attack, acute coronary syndromes (ST-elevation myocardial infarction or non-ST-elevation acute coronary syndromes [non-ST-elevation myocardial infarction or unstable angina]), secondary prevention after acute coronary syndromes, and management of stable ischemic heart disease: To reduce the combined risk of death and nonfatal stroke in patients who have had ischemic stroke or transient ischemia of the brain due to fibrin platelet emboli; to reduce the risk of vascular mortality in patients with a suspected acute myocardial infarction (MI); to reduce the combined risk of death and nonfatal MI in patients with a previous MI or unstable angina; to reduce the combined risk of MI and sudden death in patients with stable ischemic heart disease.

ER capsules:

Ischemic stroke or transient ischemic attack: To reduce the risk of death and recurrent stroke in patients who have had an ischemic stroke or transient ischemic attack.

Stable ischemic heart disease: To reduce the risk of death and MI in patients with stable ischemic heart disease.

Limitations of use: Do not use ER capsules in situations for which a rapid onset of action is required (such as acute treatment of MI or before percutaneous coronary intervention); use IR formulations instead.

Injection [International product]: Analgesic and antipyretic: Treatment of severe pain and relief of fever symptoms in adult patients unable to take oral NSAIDs.

* See [Uses in AHFS Essentials](#) for additional information.

Use: Off-Label: Adult

Atherosclerotic cardiovascular disease, primary prevention Level of Evidence [G]

Based on the American College of Cardiology/American Heart Association (ACC/AHA) guideline on the primary prevention of cardiovascular disease and the American Diabetes Association standards of care in diabetes, aspirin may be used for the primary prevention of cardiovascular disease in select patients after weighing the cardiovascular disease risk versus benefits ^(Ref).

Carotid artery atherosclerosis, asymptomatic or symptomatic Level of Evidence [G]

Based on the [American College of Chest Physicians \(ACCP\) guidelines for antithrombotic therapy and prevention of thrombosis \(9th edition\)](#), daily aspirin is suggested in patients with asymptomatic or symptomatic carotid artery atherosclerosis based on a slight reduction in total mortality observed when aspirin is taken over 10 years (regardless of cardiovascular risk profile) ^(Ref). The [AHA/American Stroke Association guidelines for the primary prevention of stroke](#) recommend daily aspirin for patients with asymptomatic or symptomatic carotid atherosclerosis to reduce the risk of a first stroke ^(Ref).

Carotid artery stenting Level of Evidence [C, G]

A randomized, controlled trial with blinded end point adjudication evaluated percutaneous carotid artery stenting versus carotid endarterectomy in patients with carotid artery stenosis. In this trial, aspirin in combination with clopidogrel was used for patients who underwent carotid artery stenting, which suggests that this antiplatelet combination is effective ^(Ref). In a prospective, single-arm multicenter trial, aspirin in combination with clopidogrel was also effective for patients who received carotid artery stenting by a transcarotid approach ^(Ref).

Based on the ASA/ACCF/AHA guideline on the management of patients with extracranial carotid and vertebral artery disease, dual antiplatelet therapy with aspirin and clopidogrel is effective at reducing ischemic complications after carotid artery stenting ^(Ref).

Colorectal cancer risk reduction, primary prevention Level of Evidence [B]

Two meta-analyses support the long-term (≥5 year) use of aspirin for primary prevention of colorectal cancer ^(Ref).

Migraine, acute treatment Level of Evidence [B, G]

In a meta-analysis of trials evaluating the treatment of acute migraine, aspirin was shown to be safe and effective ^(Ref). Based on the [2018 American Headache Society position statement on integrating new migraine treatments into clinical practice](#), aspirin is effective and recommended for the acute treatment of migraines of mild to moderate pain severity.

Percutaneous coronary intervention for stable ischemic heart disease Level of Evidence [G]

Based on ACC/AHA guidelines for coronary artery revascularization, aspirin in combination with a P2Y₁₂ inhibitor is effective for reducing ischemic complications after percutaneous coronary intervention (PCI). Aspirin is recommended for use prior to PCI and indefinitely afterwards (^{Ref}).

Pericarditis, acute or recurrent Level of Evidence [C, G]

Data from a prospective, open-label clinical trial support the use of high-dose aspirin treatment in patients with acute low-risk pericarditis (^{Ref}).

Based on the European Society of Cardiology guidelines for the diagnosis and management of pericardial diseases, aspirin is effective and recommended for the treatment of acute or recurrent pericarditis (^{Ref}).

Peripheral artery disease Level of Evidence [G]

Based on the ACC/AHA guidelines for the management of lower extremity peripheral arterial disease (PAD), aspirin is recommended to reduce the risk of vascular events (ie, myocardial infarction, stroke, vascular death) in patients with PAD, with or without a revascularization procedure (^{Ref}).

Based on the ACCP guidelines for antithrombotic therapy in PAD (9th edition), aspirin is recommended for patients with symptomatic or asymptomatic PAD to prevent cardiovascular events (^{Ref}).

Polycythemia vera, prevention of thrombosis Level of Evidence [B, G]

Aspirin may be used in all patients with polycythemia vera without a history of major bleeding or gastric intolerance. Aspirin has been studied in >1,000 patients and is recommended in multiple polycythemia vera guidelines (^{Ref}). Clinicians must consider the potential risk for bleeding and monitor for signs and symptoms of bleeding.

Preeclampsia prevention Level of Evidence [G]

Based on the American College of Obstetricians and Gynecologists committee opinion on low-dose aspirin use during pregnancy, low-dose aspirin is effective and recommended for use in patients at risk of preeclampsia (^{Ref}).

Surgical prosthetic heart valve replacement, thromboprophylaxis Level of Evidence [G]

Based on ACC/AHA guidelines for the management of patients with valvular heart disease, in patients who receive a surgically placed bioprosthetic aortic or mitral valve and are at low bleeding risk, aspirin is not necessary immediately after surgery. In these patients, monotherapy with warfarin is recommended for 3 to 6 months, followed by discontinuation of warfarin and transition to long-term aspirin monotherapy thereafter. In patients who receive a surgically placed bioprosthetic aortic or mitral valve and are at elevated risk of bleeding, aspirin monotherapy is recommended within 24 hours after surgery and long term (ie, no initial treatment with warfarin). In patients who receive a surgically placed mechanical aortic or mitral valve, aspirin is typically not recommended; only monotherapy with warfarin is recommended, unless there is another indication for aspirin. An exception is patients who receive a surgically placed mechanical On-X aortic valve. Aspirin is recommended in combination with warfarin in these patients (^{Ref}).

Transcatheter aortic valve replacement, thromboprophylaxis Level of Evidence [B, G]

Data from multiple clinical trials support the use of aspirin, with or without clopidogrel as an antithrombotic strategy for transcatheter aortic valve replacement (TAVR) ^(Ref).

Based on the ACC/AHA guideline for the management of patients with valvular heart disease, long-term aspirin is effective and recommended in patients with TAVR to reduce the risk of thromboembolic events and mortality. Clopidogrel may be used for 3 to 6 months after TAVR in addition to long-term aspirin. ^(Ref)

Transcatheter mitral valve repair with MitraClip device, thromboprophylaxis Level of Evidence [C]

Data from a limited number of patients in one clinical trial suggest that aspirin may be an effective component of the thromboprophylaxis regimen for transcatheter mitral valve repair when using the MitraClip device ^(Ref).

Venous thromboembolism prevention, indefinite therapy Level of Evidence [A, G]

Data from randomized, double-blind, placebo-controlled trials have demonstrated that low-dose aspirin reduces the overall risk of venous thromboembolism (VTE) recurrence in patients with a first unprovoked VTE (deep vein thrombosis [DVT] or pulmonary embolism [PE]) following completion and discontinuation of oral anticoagulant therapy ^(Ref). However, data from a double-blind, randomized, placebo-controlled trial did not demonstrate a significant reduction in the rate of VTE recurrence, although it did show a significant reduction in the rate of major vascular events ^(Ref).

Based on the ACCP guidelines for antithrombotic therapy for VTE disease, aspirin may be considered for extended treatment, after 3 months of therapeutic anticoagulation, to reduce VTE recurrence in patients who are stopping anticoagulant therapy following an unprovoked DVT or PE. Aspirin is *not* considered a reasonable alternative to anticoagulation for extended treatment because aspirin is expected to be much less effective at preventing recurrent VTE; however, in patients who have decided to stop anticoagulation (therapeutic dose or reduced intensity [eg, reduced dose apixaban or rivaroxaban]), aspirin may be considered over no aspirin therapy ^(Ref).

Venous thromboembolism prophylaxis for total hip or knee arthroplasty Level of Evidence [A]

Data from a large multicenter, randomized, double-blind, controlled trial support the use of aspirin as part of a hybrid VTE prophylaxis strategy after total hip arthroplasty (THA) or total knee arthroplasty (TKA). In this trial, patients undergoing THA or TKA received rivaroxaban for VTE prophylaxis for the first 5 postoperative days then switched to aspirin or continued rivaroxaban for an additional 30 days (35 days total) for THA or an additional 9 days (14 days total) for TKA. Aspirin was found to be noninferior to rivaroxaban at preventing symptomatic VTE, with no difference in bleeding ^(Ref). Data from another multicenter, randomized, double-blind, controlled trial support the use of aspirin as part of a hybrid VTE prophylaxis strategy after THA. Patients received dalteparin for the first 10 postoperative days and were then randomized to continue dalteparin or switch to aspirin for an additional 28 days (38 days total). Although the trial was stopped early due to slow enrollment, aspirin was found to be noninferior at preventing symptomatic VTE with no difference in bleeding ^(Ref).

Venous thromboembolism prophylaxis in lower-risk patients with multiple myeloma receiving immunomodulatory therapy Level of Evidence [B, G]

Data from a multicenter, randomized, open-label phase 3 sub-study support the use of aspirin as thromboprophylaxis in patients without a history of thromboembolism who were receiving thalidomide-based therapy for the treatment of newly diagnosed multiple myeloma ^(Ref). Data from a multicenter, randomized, open-label phase 3 sub-study support the use of aspirin as thromboprophylaxis in patients without a history of DVT or arterial thromboembolism who were

receiving lenalidomide-based therapy for the treatment of newly diagnosed multiple myeloma (^{Ref}). Thromboprophylaxis with aspirin was routinely used in a phase 2 study of patients with previously untreated multiple myeloma who were receiving lenalidomide and dexamethasone (^{Ref}).

According to updated guidelines from the American Society of Clinical Oncology for venous thromboembolism prophylaxis and treatment in patients with cancer, aspirin is an option for pharmacologic thromboprophylaxis in lower-risk patients with multiple myeloma receiving thalidomide- or lenalidomide-based regimens (^{Ref}). According to updated guidelines from the American Society of Hematology for management of venous thromboembolism: prevention and treatment in patients with cancer, low-dose aspirin is suggested for patients with multiple myeloma receiving lenalidomide, thalidomide, or pomalidomide-based regimens (^{Ref}).

Level of Evidence Definitions

Level of Evidence Scale

- A** - Consistent evidence from well-performed randomized, controlled trials or overwhelming evidence of some other form (eg, results of the introduction of penicillin treatment) to support the off-label use. Further research is unlikely to change confidence in the estimate of benefit.
- B** - Evidence from randomized, controlled trials with important limitations (inconsistent results, methodological flaws, indirect or imprecise), or very strong evidence of some other research design. Further research (if performed) is likely to have an impact on confidence in the estimate of benefit and risk and may change the estimate.
- C** - Evidence from observational studies (eg, retrospective case series/reports providing significant impact on patient care), unsystematic clinical experience, or from potentially flawed randomized, controlled trials (eg, when limited options exist for condition). Any estimate of effect is uncertain.
- G** - Use has been substantiated by inclusion in at least one evidence-based or consensus-based clinical practice guideline.

Clinical Practice Guidelines

Coronary Artery Bypass Graft Surgery:

ACC/AHA, "2021 ACC/AHA Guideline for Coronary Artery Revascularization," [December 2021](#)

AHA Scientific Statement, "Secondary Prevention After Coronary Artery Bypass Graft Surgery," [February 2015](#)

GI Bleed:

ACG/CAG, "Clinical Practice Guideline: Management of Anticoagulants and Antiplatelets During Acute Gastrointestinal Bleeding and Periendoscopic Period," [March 2022](#)

Hematology/Oncology:

BSH, "Guideline for the Diagnosis and Management of Polycythemia Vera," [November 2018](#)

ELN, "Philadelphia-Negative Classical Myeloproliferative Neoplasms: Revised Management Recommendations from European LeukemiaNet," [2018](#)

ESMO, "Philadelphia Chromosome-Negative Chronic Myeloproliferative Neoplasms: ESMO Clinical Practice Guidelines for Diagnosis, Treatment and Follow-up," [2015](#)

National Comprehensive Cancer Network® (NCCN), "NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)," [Myeloproliferative Neoplasms](#)

Ischemic Heart Disease:

AHA/ACC, "Guideline for the Management of Patients With Chronic Coronary Disease: a Report of the American Heart Association/American College of Cardiology Joint Committee on Clinical Practice Guidelines," [July 2023](#)

ACC/AHA "Guideline Focused Update on Duration of Dual Antiplatelet Therapy in Patients with Coronary Artery Disease," [March 2016](#)

AHA/ACC, "2014 AHA/ACC Guideline for the Management of Patients with Non-ST-Elevation Acute Coronary Syndromes," [September 2014](#).

ACCF/AHA, "2013 ACCF/AHA Guideline for the Management of ST-Elevation Myocardial Infarction," [December 2012](#)

ESC, "Guidelines for the Management of Acute Coronary Syndromes in Patients Presenting Without Persistent ST-Segment Elevation," [August 2020](#)

Medication Safety:

"STOPP/START Criteria for Potentially Inappropriate Prescribing in Older People: Version 3," [May 2023](#)

Pericarditis:

ESC, "2015 Guidelines for the Diagnosis and Management of Pericardial Diseases," [November 2015](#)

Peripheral Arterial Disease:

ACC/AHA, "2024 Guideline for the Management of Lower Extremity Peripheral Artery Disease," [May 2024](#)

ACCP, "2012 Guideline on antithrombotic therapy in peripheral artery disease," [February 2012](#)

ACCF/AHA, "2011 Focused Update of the Guideline for the Management of Patients with Peripheral Artery Disease (Updating the 2005 Guideline)," [September 2011](#)

CCS, "Canadian Cardiovascular Society 2022 Guidelines for Peripheral Arterial Disease," [2022](#)

Preeclampsia:

USPSTF, "Aspirin Use to Prevent Preeclampsia and Related Morbidity and Mortality," [2021](#)

Prevention:

ACC/AHA, "2019 Guideline on the primary prevention of cardiovascular disease," [March 2019](#)

ADA, "Standards of Care in Diabetes - 2023," [January 2023](#)

"ASCO Hereditary Colorectal Cancer Syndromes Clinical Practice Guideline [Endorsement]," December 2014

KDIGO, "KDIGO 2020 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease," [October 2020](#)

USPSTF, "Aspirin Use to Prevent Cardiovascular Disease: US Preventive Services Task Force Recommendation Statement," [April 2022](#)

Stroke:

AHA/ASA, "Guideline for the Prevention of Stroke in Patients With Stroke and Transient Ischemic Attack," [2021](#)

AHA/ASA, "Guidelines for the Early Management of Patients With Acute Ischemic Stroke: 2019 Update to the 2018 Guidelines for the Early Management of Acute Ischemic Stroke," [2019](#)

AHA/ASA, "Guidelines for the Early Management of Patients With Acute Ischemic Stroke," [2018](#)

AHA/ASA, "Guidelines for the Early Management of Patients With Acute Ischemic Stroke," [February 2013](#)

AHA/ASA, "Guidelines for the Prevention of Stroke in Women," [2014](#)

AHA/ASA, "Guidelines for Prevention of Stroke in Patients with Stroke and Transient Ischemic Attack," [May 2014](#).

AHA/ASA, "Guidelines for the Primary Prevention of Stroke," [December 2014](#)

ASA/ACCF/AHA, "Guideline on the management of patients with extracranial carotid and vertebral artery disease," [January 2011](#)

Surgical and Procedural Management:

ACC/AHA, "2014 ACC/AHA Guideline on Perioperative Cardiovascular Evaluation and Management of Patients Undergoing Noncardiac Surgery," [August 2014](#)

ACCP, "Perioperative Management of Antithrombotic Therapy," [August 2022](#)

ESC, "Guidelines on Cardiovascular Assessment and Management of Patients Undergoing Non-Cardiac Surgery," [August 2022](#)

STS, "2012 Update to the Society of Thoracic Surgeons Guideline on Use of Antiplatelet Drugs in Patients Having Cardiac and Noncardiac Operations," [November 2012](#)

Valvular Heart Disease:

ACC/AHA, "ACC/AHA Guideline for the Management of Patients with Valvular Heart Disease," [December 2020](#)

Venous Thromboembolism:

ACCP, "Antithrombotic Therapy for VTE Disease: CHEST Guideline and Expert Panel Report," [February 2016](#)

ACCP, "Antithrombotic Therapy for VTE Disease: Second update of the CHEST Guideline and Expert Panel Report," [August 2021](#)

ASCO, "Venous Thromboembolism Prophylaxis and Treatment in Patients With Cancer: ASCO Clinical Practice Guideline Update," [August 2019](#)

ASH, "2021 Guidelines for Management of Venous Thromboembolism: Prevention and Treatment in Patients with Cancer," [February 2021](#)

ESC, "Guidelines for the diagnosis and management of acute pulmonary embolism," [August 2019](#)

Other:

AHA, "Prevention and Treatment of Thrombosis in Pediatric and Congenital Heart Disease," [December 2013](#)

Administration: IM

International product: Inject deeply into muscle.

Administration: IV

International product: May be administered IV undiluted or by IV infusion using compatible solutions (NS or D5W).

Administration: Oral

IR tablets: Administer with food or a full glass of water to minimize GI distress. In situations for which a rapid onset of action is required (eg, acute treatment of myocardial infarction), have patient chew IR tablet.

Preeclampsia (prevention): Administration as an evening dose may be more beneficial than administration in the morning ^(Ref).

ER capsules: Do not cut, crush, or chew. Administer with a full glass of water at the same time each day. Do not administer 2 hours before or 1 hour after alcohol consumption.

Bariatric surgery: Some institutions may have specific protocols that conflict with these recommendations; refer to institutional protocols as appropriate. Switch to IR formulation.

Administration: Rectal

Remove suppository from plastic packet and insert into rectum as far as possible.

Administration: Pediatric

Oral: Administer with water, food, or milk to decrease GI upset.

Immediate release:

Capsule: Swallow whole; do not cut, crush, or chew. Administer with a full glass of water.

Tablets: Do not crush or chew enteric-coated tablets; these preparations should be swallowed whole. In situations for which rapid onset of action is required, patient should chew immediate-release tablet.

Extended release: Do not cut, crush, or chew. Administer with a full glass of water at the same time each day.

Rectal: Remove suppository from plastic packet and insert into rectum as far as possible.

Storage/Stability

Store oral dosage forms (caplets, tablets, capsules) at room temperature; protect from moisture; see product-specific labeling for details. Keep suppositories in refrigerator; do not freeze. Hydrolysis of aspirin occurs upon exposure to water or moist air, resulting in salicylate and acetate, which possess a vinegar-like odor. Do not use if a strong odor is present.

Injection [International product]: Store below 25°C (77°F) and protect from light prior to reconstitution.

Preparation for Administration: Adult

Injection [International product]: After reconstitution, may dilute further with a compatible solution (eg, D5W, NS) to administer via IV infusion.

Medication Safety Issues

Sound-alike/look-alike issues:

Aspirin may be confused with Afrin

Ascriptin may be confused with Aricept

Ecotrin may be confused with Edocrin, Epogen

Halfprin may be confused with Haltran

ZORprin may be confused with Zylprim

International issues:

Anacin 81 [Puerto Rico] may be confused with Anacin brand name for lidocaine/prilocaine [Korea]; Anacin-3 brand name for acetaminophen [Taiwan]; Anacin New brand name for acetaminophen/caffeine [India]

Cartia [multiple international markets] may be confused with Cartia XT brand name for diltiazem [US]

Older Adult: High-Risk Medication:

Beers Criteria: Aspirin is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients ≥ 65 years of age for the primary prevention of cardiovascular disease due to increased risk for major bleeding. A potential for harm and lack of benefit when initiated as primary prevention in older adults is suggested by studies. In older adults already taking for primary prevention, deprescribing should be considered. In addition, aspirin, when used chronically at doses more than 325 mg/day, is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients ≥ 65 years of age (unless alternative agents ineffective and patient can receive concomitant gastroprotective agent) due to increased risk of GI bleeding and peptic ulcer disease in older adults in high-risk category (eg, > 75 years of age or receiving concomitant oral/parenteral corticosteroids, anticoagulants, or antiplatelet agents) (Beers Criteria [AGS 2023]).

Aspirin is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥ 65 years of age) at risk of major bleeding (eg, uncontrolled hypertension, recent significant bleeding, bleeding diathesis) or for long-term use at doses > 100 mg/day. Use is also potentially inappropriate for primary prevention of cardiovascular disease, stroke prevention in patients with chronic atrial fibrillation, and long-term use for relief of symptoms in osteoarthritis or gout. Some disease states of concern include gastric antral vascular ectasia (GAVE), peptic ulcer disease, renal impairment, gastrointestinal bleeding, and severe hypertension (O'Mahony 2023).

Pediatric patients: High-risk medication:

KIDs List: Salicylates, when used in pediatric patients <18 years of age with suspicion of viral illness (influenza, chickenpox), are identified on the Key Potentially Inappropriate Drugs in Pediatrics (KIDs) list and should be used with caution due to risk of Reye syndrome (weak recommendation; very low quality of evidence) (PPA [Meyers 2020]).

Contraindications

Hypersensitivity to NSAIDs; patients with asthma, rhinitis, and nasal polyps; use in children or teenagers for viral infections, with or without fever.

Warnings/Precautions***Disease-related concerns:***

- Bariatric surgery:
 - Altered absorption and efficacy: Altered absorption and efficacy may occur. The T_{max} of salicylic acid after gastric bypass was observed to be significantly shorter after surgery. C_{max} and AUC_{0-24} were also significantly higher (Mitrov-Winkelmolen 2016). In a mixed surgery population (80% gastric bypass, 20% sleeve), aspirin-induced platelet reactivity was significantly reduced and correlated to the extent of weight loss after surgery (Norgard 2017).
 - Gastric ulceration: Evaluate the risk vs benefit of aspirin after surgery; if aspirin therapy is continued (eg, cardiovascular indications), use the lowest possible dose with concurrent administration of proton pump inhibitor (PPI); risk of gastric ulceration after gastric bypass and sleeve gastrectomy may be increased. A population-based study of over 20,000 patients identifying risk factors for marginal ulceration after gastric bypass suggests limited doses of aspirin may not increase risk whereas higher doses may (Sverden 2016). Another study identified that following gastric bypass, there is no significant difference in marginal ulceration rate between those given low-dose aspirin and those not on aspirin; however, long-term PPI therapy (>90 days) was found to significantly reduce marginal ulceration rate in both groups (Kang 2017).
- Bleeding disorders: Use with caution in patients with platelet and bleeding disorders.
- Dehydration: Use with caution in patients with dehydration.
- Ethanol use: Heavy ethanol use (>3 drinks/day) can increase bleeding risks.
- Gastrointestinal disease: Use with caution in patients with erosive gastritis. Avoid use in patients with active peptic ulcer disease.
- Hepatic impairment: Avoid use in severe hepatic failure.
- Renal impairment: Use high dosages (eg, analgesic or anti-inflammatory uses) with caution (NKF [Henrich 1996]; Whelton 2000). Low-dose aspirin (eg, 75 to 162 mg daily) may be safely used in patients with any degree of renal impairment (KDOQI 2005; KDOQI 2007).

Concurrent drug therapy issues:

- Thrombolytics: In the treatment of acute ischemic stroke, avoid aspirin for 24 hours following administration of a thrombolytic; administration within 24 hours increases the risk of hemorrhagic transformation (AHA/ASA [Powers 2019]; Jauch 2013).

Special populations:

- GI bleed patients: An individualized and multidisciplinary approach should be used to manage patients with an acute GI bleed who are on antiplatelet medications. Aspirin for primary prevention of cardiovascular events should be avoided in most patients with GI bleed who do not have high risk factors for cardiovascular events. However, aspirin for secondary cardiovascular prevention should *not* be discontinued in patients with established cardiovascular disease, even in the setting of a GI bleed. If held in the setting of a GI bleed, aspirin for secondary cardiovascular prevention should be resumed on the day hemostasis is confirmed by endoscopy (ACG/CAG [Abraham 2022]).

- Pediatric: When used for self-medication (OTC labeling): Children and teenagers who have or are recovering from chickenpox or flu-like symptoms should not use this product. Changes in behavior (along with nausea and vomiting) may be an early sign of Reye syndrome; patients should be instructed to contact their healthcare provider if these occur.
- Surgical patients: Patients who have recently undergone percutaneous coronary intervention with stenting or balloon angioplasty should continue antiplatelet therapy until it is safe to temporarily hold treatment. In patients undergoing CABG, aspirin may be continued until the time of surgery. Elective surgery for these patients should be delayed. Aspirin can typically be continued perioperatively in patients who undergo elective non-cardiac surgery. If aspirin interruption is necessary, discontinue therapy ≤ 7 days before surgery, and resume as soon as possible (eg, ≤ 24 hours) after surgery based on bleeding and thrombotic risks. Patient-specific situations should be discussed with cardiologist (ACC/AHA [Fleisher 2014]; ACC/AHA [Lawton 2022]; ACCP [Douketis 2022]; AHA/ACC/SCAI/ACS/ADA [Grines 2007]).

Dosage form specific issues:

- Polysorbate 80: Some dosage forms may contain polysorbate 80 (also known as Tweens). Hypersensitivity reactions, usually a delayed reaction, have been reported following exposure to pharmaceutical products containing polysorbate 80 in certain individuals (Isaksson 2002; Lucente 2000; Shelley 1995). Thrombocytopenia, ascites, pulmonary deterioration, and renal and hepatic failure have been reported in premature neonates after receiving parenteral products containing polysorbate 80 (Alade 1986; CDC 1984). See manufacturer's labeling.

Other warnings/precautions:

- Resistance: Aspirin resistance is defined as measurable, persistent platelet activation that occurs in patients prescribed a therapeutic dose of aspirin. Clinical aspirin resistance, the recurrence of some vascular event despite a regular therapeutic dose of aspirin, is considered aspirin treatment failure. Proposed mechanisms of aspirin resistance include poor adherence with therapy, poor absorption, inadequate dosage, drug interactions, increased isoprostane activity, platelet hypersensitivity to agonists, increased COX-2 activity, COX-1 polymorphism, and platelet alloantigen 2 polymorphism of platelet glycoprotein IIIa. Estimates of biochemical aspirin resistance range from 5.5% to 60% depending on the population studied and the assays used (Gasparyan 2008). Patients with aspirin resistance may have a higher risk of cardiovascular events compared to those who are aspirin sensitive (Gum 2003). Aspirin resistance is likely dose-related but may be influenced by dynamic factors yet to be identified; further research is required.

* See [Cautions in AHFS Essentials](#) for additional information.

Older Adult Considerations

Older adult patients are at high risk for adverse effects from nonsteroidal anti-inflammatory drugs (NSAIDs). Older adults can develop an asymptomatic peptic ulcer and/or hemorrhage (Savage 2005). Using the lowest effective dose for the shortest period possible is recommended. Tinnitus may be a difficult and unreliable indication of toxicity due to age-related hearing loss or eighth cranial nerve damage (Curhan 2022). CNS adverse effects such as confusion, agitation, and hallucination are generally seen in overdose or high-dose situations, but older adults may demonstrate these adverse effects at lower doses than younger adults (Savage 2005). NSAIDs seem to be a significant risk factor for falls, which can lead to morbidity and mortality in older adults. Therefore, risks and benefits should be balanced carefully in individual patients (Wongrakpanich 2018).

Low-dose aspirin increased incident anemia and decline in ferritin in otherwise healthy older adults, independent of major bleeding. Periodic monitoring of hemoglobin should be considered in older persons on aspirin (McQuilten 2023).

Warnings: Additional Pediatric Considerations

Do not use aspirin in pediatric patients <18 years of age (APS 2016) who have or who are recovering from chickenpox or flu symptoms (due to the association with Reye syndrome); when using aspirin, changes in behavior (along with nausea and vomiting) may be an early sign of Reye syndrome; instruct patients and caregivers to contact their health care provider if these symptoms occur; patients should be kept current on their influenza and varicella immunizations. Although Reye syndrome has been observed in patients receiving prolonged, high-dose aspirin therapy after Kawasaki disease presentation; it has not been observed in pediatric patients receiving low (antiplatelet) dosing regimens. Patients with Kawasaki disease and presenting with influenza or viral illness should not receive aspirin; acetaminophen is suggested as an antipyretic in these patients, and an alternate antiplatelet agent is suggested for a minimum of 2 weeks (AHA [McCrindle 2017]).

Reproductive Considerations

Low-dose aspirin has been evaluated to improve live birth rates in patients with antiphospholipid syndrome (APS) diagnosed with recurrent pregnancy loss (Hamulyák 2020; Hamulyák 2021). Recommendations for when to initiate treatment differ between guidelines. Some guidelines initiate aspirin prior to conception, while others prefer use of other agents (heparin or low-molecular-weight heparin) in patients planning to become pregnant (ACR [Sammaritano 2020]); ESHRE [Bender Atik 2018]; EULAR [Andreoli 2017]). Use of low-dose aspirin prior to conception in patients with a history of recurrent pregnancy loss but who are not diagnosed with APS has also been evaluated (Naimi 2021).

Pregnancy Considerations

Salicylate is present in umbilical cord and newborn serum following maternal use of aspirin prior to delivery (Garrettson 1975; Levy 1975; Palmisano 1969); salicylic acid and other metabolites can also be detected in the newborn urine following in utero exposure (Garrettson 1975).

Fetal outcomes are influenced by maternal dose; low dose aspirin (≤ 150 mg/day) is not associated with the same risks as higher doses and has a positive effect on some pregnancy outcomes (ACOG 2018; Choi 2021; Turner 2020). Adverse effects reported in the fetus following maternal use of high dose aspirin include mortality, intrauterine growth restriction, salicylate intoxication, bleeding abnormalities, and neonatal acidosis. Use of aspirin close to delivery may cause premature closure of the ductus arteriosus. Adverse effects reported in the mother include anemia, hemorrhage, prolonged gestation, and prolonged labor (Corby 1978; Østensen 1998).

Except when used in lower doses for pregnancy-related conditions, maternal use of aspirin should be avoided beginning 20 weeks gestation (FDA Safety Communication 2020).

Due to pregnancy-induced physiologic changes, some pharmacokinetic properties of aspirin may be altered (Rymark 1994; Shanmugalingam 2019).

Low-dose aspirin is recommended to prevent preeclampsia in patients at risk. The definition of risk factors varies by guideline and dosing varies by country due to dosage form availability (ACOG 2018; FIGO [Poon 2019]; SOGC [Magee 2022]; USPSTF [Davidson 2021]). The American College of Obstetricians and Gynecologists and United States Preventive Services Task Force recommend the use of low-dose aspirin for patients with ≥ 1 of the following high-risk factors: history of preeclampsia, multifetal gestation, chronic hypertension, type 1 or type 2 diabetes mellitus, renal disease, autoimmune disease (eg, systemic lupus erythematosus, antiphospholipid syndrome), or combinations of multiple moderate risk factors. Low-dose aspirin is also recommended for patients with ≥ 2 of the following moderate risk factors: nulliparity, BMI >30 , family history of preeclampsia, Black race (due to social, rather than biological, factors), lower income, age ≥ 35 years, personal history factors (eg, low birth weight or small for GA, previous adverse pregnancy outcome, >10 -year pregnancy interval), or in vitro conception. Low-dose aspirin may also be considered in patients with ≥ 1 of the following moderate risk factors: Black race (due to social, rather than biological, factors), lower income (ACOG 2018; ACOG 2021; USPSTF [Davidson 2021]).

In addition to preventing preeclampsia, low-dose aspirin is also recommended to improve other pregnancy outcomes and decrease the risk of thrombosis in patients with a positive antiphospholipid antibody (aPL) test. The addition of heparin or low-molecular-weight heparin may also be needed in patients with obstetric or thrombotic antiphospholipid syndrome (APS). Low-dose aspirin is also recommended for pregnant patients with systemic lupus erythematosus (SLE) (ACR [Sammaritano 2020]; EULAR [Andreoli 2017]; SMFM [Silver 2022]).

Low-dose aspirin, in addition to anticoagulation, may be used to prevent thrombosis in pregnant patients with a mechanical heart valve (ACC/AHA [Otto 2021]).

Aspirin is not recommended for the treatment of migraine headaches during pregnancy (ACOG 2022).

Breastfeeding Considerations

Salicylic acid is present in breast milk following maternal use of aspirin (Bailey 1982; Findlay 1981; Jamali 1981).

Actual breast milk concentrations of salicylic acid vary by maternal dose. In a study of six breastfeeding women, 1 to 8 months' postpartum, the highest breast milk concentration of salicylic acid was 0.0001654 mcg/mL in six women, following a maternal dose of aspirin 81 mg/day (duration of treatment not presented). Authors of this study calculated the relative infant dose (RID) of aspirin to be <1% of the weight-adjusted maternal dose (Datta 2017). In general, breastfeeding is considered acceptable when the RID is <10% (Anderson 2016; Ito 2000). Significantly higher breast milk concentrations (48.1 mcg/mL) were reported following maternal administration of aspirin 1,500 mg as a single dose (Jamali 1981).

The reported time to peak milk concentration of salicylic acid is variable (2 to 9 hours), milk concentrations decline slowly, and do not correlate strongly to maternal serum concentration (Bailey 1982; Bar-Oz 2003; Datta 2017; Findlay 1981; Jamali 1981). Salicylate is measurable in the serum (Unsworth 1987) and urine (Clark 1981) of breastfed infants following maternal use of oral aspirin. Higher salicylate concentrations may be present in breast milk following multiple maternal doses. In addition, salicylate concentrations may be higher than reported as metabolite concentrations of salicylic acid were not evaluated in most studies; the longer elimination half-life in infants compared to adults should also be considered (Bar-Oz 2003; Spigset 2000).

Metabolic acidosis was reported in a 16-day old breastfed full-term infant following maternal doses of aspirin 3.9 g/day (Clark 1981). Thrombocytopenic purpura was also reported in one infant following salicylate exposure via breast milk (maternal dose not specified) (Spigset 2000; Terragna 1967). There were no cases of diarrhea, drowsiness, or irritability noted in breastfed infants in the study which included 15 mother-infant pairs following aspirin exposure (dose, duration, and relationship to breastfeeding not provided) (Ito 1993).

Low-dose aspirin may be used in breastfeeding patients (ACOG 2022; ESC [Regitz-Zagrosek 2018]); however, standard doses of aspirin should be avoided due to the possible risk of neonatal metabolic acidosis, hemolysis, and the theoretical risk of Reye syndrome (ACOG 2022). Nonopioid analgesics are preferred for breastfeeding patients who require pain control peripartum or for surgery outside of the postpartum period. Low doses of aspirin (75 to 162 mg/day) may be used; however, other agents are preferred if higher doses are needed (ABM [Martin 2018]; ABM [Reece-Stremtan 2017]; Sachs 2013). When used for vascular indications, breastfeeding may be continued during low-dose aspirin therapy (ESC [Regitz-Zagrosek 2018]; WHO 2002). The WHO considers occasional doses of aspirin to be compatible with breastfeeding patients for the treatment of rheumatic and musculoskeletal diseases or migraine but recommends avoiding long-term therapy and consider monitoring the infant for adverse effects (hemolysis, prolonged bleeding, metabolic acidosis) (WHO 2002). High doses of aspirin in breastfeeding patients for the treatment of rheumatic and musculoskeletal diseases or migraine are not recommended (ACOG 2022; ACR [Sammaritano 2020]).

Briggs' Drugs in Pregnancy & Lactation

- [Aspirin](#)

Adverse Reactions (Significant): Considerations

GI effects

Aspirin use is associated with a 2- to 4-fold increase in upper gastrointestinal (UGI) events, such as symptomatic or complicated **gastrointestinal ulcers** and GI mucosal damage (^{Ref}). Lower GI (LGI) events have also been reported with chronic aspirin use (^{Ref}). Symptoms can range from mild (**dyspepsia**) to severe (peptic ulcer disease, **gastrointestinal hemorrhage**). The use of enteric-coated preparations of aspirin does not decrease the risk of UGI events (^{Ref}).

Mechanism: Dose-related; related to the pharmacologic action; damage to the GI mucosa is typically caused by aspirin's effects on the epithelial and microvascular tissue of the GI tract via inhibition of cyclooxygenase-1 and subsequent prostaglandin depletion, which is gastro-protective (^{Ref}). Additionally, aspirin accumulates in gastric mucosal cells, altering the permeability, resulting in an increased risk of ulceration (^{Ref}).

Onset: Varied; topical gastric mucosa toxicity can be seen on endoscopy within minutes of aspirin administration (^{Ref}). Other studies reported GI ulcers after at least 3 months of use (^{Ref}). Risk of GI events is generally seen when beginning aspirin therapy; however, long-term administration of aspirin is associated with a greater risk of GI events (^{Ref}).

Risk factors:

- Dose >100 mg (^{Ref})
- Bleeding ulcers (^{Ref})

- Peptic ulcer disease ^(Ref)
- Concurrent use of nonsteroidal anti-inflammatory drugs ^(Ref)
- Concurrent corticosteroids ^(Ref)
- Concurrent dual antiplatelet and anticoagulant therapy ^(Ref)
- *Helicobacter pylori* infection ^(Ref)
- Smoking ^(Ref)
- Excessive alcohol consumption ^(Ref)
- Older age (>70 years) ^(Ref)

Hypersensitivity reactions (immediate and delayed)

Hypersensitivity reactions (immediate and delayed) involving the skin (eg, **angioedema**, **urticaria**), airways (eg, dyspnea, rhinorrhea), and/or other organs ^(Ref) have been reported. Clinical phenotypes of aspirin/nonsteroidal anti-inflammatory drug (NSAID) hypersensitivity reactions include aspirin/NSAID-exacerbated respiratory disease (AERD and NERD), aspirin/NSAID-induced urticaria/angioedema (NIUA), aspirin/NSAID-exacerbated cutaneous disease (NICD) and single NSAID-induced urticaria/angioedema or **anaphylaxis** ^(Ref). Delayed hypersensitivity reactions, including **drug rash with eosinophilia and systemic symptoms** (DRESS), have also been rarely associated with aspirin ^(Ref).

Mechanism: Immediate reactions: Non-dose-related; most reactions (ie, AERD, NERD, NECD, NIUA) are nonimmunologic related to inhibition of cyclooxygenase-1 (COX-1) with subsequent activation of mast cells and eosinophils causing release of inflammatory mediators including cysteinyl-leukotrienes (cysLTs) ^(Ref). Some immediate reactions are IgE-mediated ^(Ref); although, these are rare with aspirin and more common with NSAIDs, where they can be associated with an immediate reaction specific to the specific NSAID or class of NSAID. Delayed hypersensitivity reactions, including DRESS, are T-cell-mediated ^(Ref).

Onset: Immediate hypersensitivity reactions: Rapid; occur within 1 hour of administration but may occur up to 6 hours after exposure ^(Ref). Delayed hypersensitivity reactions, including DRESS: Varied, by definition occur after 6 hours of exposure but most commonly occur 2 to 8 weeks after initiation ^(Ref).

Risk factors:

- Presence of chronic rhinosinusitis with nasal polyps, family history of AERD/NERD, and/or severe asthma may increase the risk of AERD/NERD ^(Ref). The prevalence of NERD in adult patients with asthma is ~10% to 20% ^(Ref).
- Chronic urticaria increases the risk of NECD ^(Ref). Aspirin-induced reactions are less frequent and less intense when chronic urticaria is in remission or under control ^(Ref). Approximately 12% to 30% of patients with chronic idiopathic urticaria develop exacerbations of their disease with use of aspirin and other COX-1 inhibitors ^(Ref).
- Cross-reactivity between aspirin and NSAIDs (with predominant COX-1 inhibition) have been described in patients with a history of NERD, NECD, and NIUA ^(Ref). Patients who develop a reaction (immediate or delayed) to only aspirin or a single NSAID are often able to tolerate chemically unrelated NSAIDs ^(Ref).
- Pharmacological cross-reactivity between aspirin and acetaminophen, a weak COX-1 inhibitor, and between aspirin and nonselective COX-2 inhibitors (eg, meloxicam, nimesulide) may occur ^(Ref). Due to the weak COX-1 inhibitory activity, AERD reactions associated with acetaminophen typically only occur with higher doses such as >4 g/day. Although selective COX-2 inhibitors (eg, celecoxib, etoricoxib) are generally tolerated in patients with AERD or NERD ^(Ref), pharmacological cross-reactions may occur, especially in patients with histories of urticaria/angioedema ^(Ref).
- Although earlier reports suggested that tartrazine and aspirin/NSAIDs may cross react, more recent clinical evidence has indicated that tartrazine is well-tolerated in patients with AERD/NERD ^(Ref)
- Despite structural similarity between aspirin and 5-aminosalicylic acid (ASA) (mesalamine), there is lack of evidence demonstrating occurrence of AERD in association with 5-ASA ^(Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. As with all drugs which may affect hemostasis, bleeding is associated with aspirin. Hemorrhage may occur at virtually any site. Risk is dependent on multiple variables including dosage, concurrent use of multiple agents which alter hemostasis, and patient susceptibility. Many adverse reactions with aspirin are dose related and are rare at low dosages. Other serious reactions are idiosyncratic, related to allergy or individual sensitivity. Accurate estimation of frequencies is not possible.

Frequency not defined:

Cardiovascular: Cardiac arrhythmia, hypotension, tachycardia

Endocrine & metabolic: Dehydration, hyperglycemia, hyperkalemia, hypoglycemia (children), increased thirst, metabolic acidosis

Gastrointestinal: Abdominal pain, dyspepsia, gastrointestinal perforation, gastrointestinal ulcer, heartburn, nausea, vomiting

Genitourinary: Postpartum hemorrhage, post-term pregnancy, prolonged labor, proteinuria, stillborn infant

Hematologic & oncologic: Disorder of hemostatic components of blood, disseminated intravascular coagulation, hemorrhage, prolonged bleeding time, prolonged prothrombin time, thrombocytopenia

Hepatic: Hepatitis, increased liver enzymes

Nervous system: Agitation, brain edema, coma, confusion, dizziness, headache, hypothermia, lethargy, seizure

Renal: Increased blood urea nitrogen, increased serum creatinine, interstitial nephritis, renal failure syndrome, renal insufficiency, renal papillary necrosis

Respiratory: Hyperventilation, laryngeal edema, pulmonary edema, respiratory alkalosis, tachypnea

Miscellaneous: Fever, low birth weight

Postmarketing:

Dermatologic: Urticaria (Kowalski 2013)

Endocrine & metabolic: Hyperuricemia (doses ≤ 325 mg/day) (Caspi 2000; Segal 2006)

Gastrointestinal: Gastrointestinal hemorrhage (Garcia Rodriguez 2019), pancreatitis (Hoyte 2012)

Hypersensitivity: Anaphylaxis (Kowalski 2013), angioedema (Kowalski 2013)

Immunologic: Drug reaction with eosinophilia and systemic symptoms (Kawakami 2009)

Nervous system: Intracranial hemorrhage (Huang 2019), Reye's syndrome (McGovern 2001)

Neuromuscular & skeletal: Rhabdomyolysis (Leventhal 1989)

Ophthalmic: Macular degeneration (age-related) (Li 2015)

Otic: Hearing loss (Koren 2009), tinnitus (Koren 2009)

Respiratory: Asthma (Kowalski 2013), bronchospasm (Kowalski 2013)

* See [Cautions in AHFS Essentials](#) for additional information.

Allergy and Idiosyncratic Reactions

- [Salicylate Allergy/Sensitivity](#)

Toxicology

- [Salicylates](#)

Metabolism/Transport Effects

Substrate of CYP2C9 (minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

[Open Interactions](#)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Open Interactions" button above.

Miscellaneous Antiplatelets: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Abrocitinib: Aspirin may enhance the antiplatelet effect of Abrocitinib. Management: Do not use aspirin at doses greater than 81 mg/day with abrocitinib during the first 3 months of abrocitinib therapy. The abrocitinib prescribing information lists this combination as contraindicated. *Risk D: Consider therapy modification*

Acalabrutinib: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Aducanumab: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Agents with Blood Glucose Lowering Effects: Salicylates may enhance the hypoglycemic effect of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor therapy*

Ajmaline: Salicylates may enhance the adverse/toxic effect of Ajmaline. Specifically, the risk for cholestasis may be increased. *Risk C: Monitor therapy*

Alcohol (Ethyl): May enhance the adverse/toxic effect of Aspirin. Specifically, alcohol may increase the bleeding risk of aspirin. Alcohol (Ethyl) may diminish the therapeutic effect of Aspirin. Specifically, alcohol may interfere with the controlled release mechanism of extended release aspirin. *Risk C: Monitor therapy*

Alendronate: Aspirin may enhance the adverse/toxic effect of Alendronate. Specifically, the incidence of upper gastrointestinal adverse events may be increased *Risk C: Monitor therapy*

Ammonium Chloride: May increase the serum concentration of Salicylates. *Risk C: Monitor therapy*

Anagrelide: May enhance the antiplatelet effect of Aspirin. *Risk C: Monitor therapy*

Angiotensin-Converting Enzyme Inhibitors: Salicylates may enhance the nephrotoxic effect of Angiotensin-Converting Enzyme Inhibitors. Salicylates may diminish the therapeutic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Anticoagulants (Miscellaneous Agents): Aspirin may enhance the anticoagulant effect of Anticoagulants (Miscellaneous Agents). *Risk C: Monitor therapy*

Antiplatelet Agents (P2Y12 Inhibitors): Therapeutic Antiplatelets may enhance the antiplatelet effect of Antiplatelet Agents (P2Y12 Inhibitors). *Risk C: Monitor therapy*

Benzbromarone: Salicylates may diminish the therapeutic effect of Benzbromarone. *Risk C: Monitor therapy*

Calcium Channel Blockers (Nondihydropyridine): May enhance the antiplatelet effect of Aspirin. *Risk C: Monitor therapy*

Caplacizumab: May enhance the antiplatelet effect of Therapeutic Antiplatelets. Management: Avoid this combination if possible. If coadministration is required, monitor closely for bleeding. Interrupt caplacizumab if clinically significant bleeding occurs and administer von Willebrand factor concentrate to rapidly correct hemostasis, if needed. *Risk D: Consider therapy modification*

Carbonic Anhydrase Inhibitors: Salicylates may enhance the adverse/toxic effect of Carbonic Anhydrase Inhibitors. Salicylate toxicity might be enhanced by this same combination. Management: Avoid these combinations when possible. Dichlorphenamide use with high-dose aspirin as contraindicated. If another combination is used, monitor patients closely for adverse effects. Tachypnea, anorexia, lethargy, and coma have been reported. *Risk D: Consider therapy modification*

ClomPRAMINE: May enhance the antiplatelet effect of Aspirin. *Risk C: Monitor therapy*

Collagenase (Systemic): Aspirin may enhance the adverse/toxic effect of Collagenase (Systemic). Specifically, the risk of injection site bruising and or bleeding may be increased. *Risk C: Monitor therapy*

Corticosteroids (Systemic): Salicylates may enhance the adverse/toxic effect of Corticosteroids (Systemic). These specifically include gastrointestinal ulceration and bleeding. Corticosteroids (Systemic) may decrease the serum concentration of Salicylates. Withdrawal of corticosteroids may result in salicylate toxicity. *Risk C: Monitor therapy*

Dasatinib: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Deoxycholic Acid: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Desirudin: Therapeutic Antiplatelets may enhance the anticoagulant effect of Desirudin. *Risk C: Monitor therapy*

Dexibuprofen: Aspirin may enhance the adverse/toxic effect of Dexibuprofen. Dexibuprofen may diminish the cardioprotective effect of Aspirin. *Risk X: Avoid combination*

Dexketoprofen: Salicylates may enhance the adverse/toxic effect of Dexketoprofen. Dexketoprofen may diminish the therapeutic effect of Salicylates. Salicylates may decrease the serum concentration of Dexketoprofen. Management: The use of high-dose salicylates (3 g/day or more in adults) together with dexketoprofen is inadvisable. Consider administering dexketoprofen 30-120 min after or at least 8 hrs before cardioprotective doses of aspirin to minimize any possible interaction. *Risk X: Avoid combination*

Dipyrrone: May diminish the antiplatelet effect of Aspirin. Management: Use caution and consider avoiding use of dipyrrone in patients treated with aspirin for the treatment or prevention of cardiovascular events or stroke. *Risk D: Consider therapy modification*

Direct Oral Anticoagulants (DOACs): Aspirin may enhance the anticoagulant effect of Direct Oral Anticoagulants (DOACs). *Risk C: Monitor therapy*

Donanemab: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Fondaparinux: Aspirin may enhance the anticoagulant effect of Fondaparinux. Management: Discontinue aspirin prior to fondaparinux therapy, if possible. If co-administration is required use caution and monitor for bleeding. *Risk D: Consider therapy modification*

Ginkgo Biloba: May enhance the anticoagulant effect of Salicylates. Management: Consider alternatives to this combination of agents. Monitor for signs and symptoms of bleeding (especially intracranial bleeding) if salicylates are used in combination with ginkgo biloba. *Risk D: Consider therapy modification*

Glycoprotein IIb/IIIa Inhibitors: Therapeutic Antiplatelets may enhance the anticoagulant effect of Glycoprotein IIb/IIIa Inhibitors. *Risk C: Monitor therapy*

Heparin: Aspirin may enhance the anticoagulant effect of Heparin. *Risk C: Monitor therapy*

Heparins (Low Molecular Weight): Aspirin may enhance the anticoagulant effect of Heparins (Low Molecular Weight). *Risk C: Monitor therapy*

Herbal Products with Anticoagulant/Antiplatelet Effects (eg, Alfalfa, Anise, Bilberry): May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Hyaluronidase: Salicylates may diminish the therapeutic effect of Hyaluronidase. *Risk C: Monitor therapy*

Ibritumomab Tiuxetan: Therapeutic Antiplatelets may enhance the antiplatelet effect of Ibritumomab Tiuxetan. *Risk C: Monitor therapy*

Ibrutinib: Therapeutic Antiplatelets may enhance the adverse/toxic effect of Ibrutinib. Specifically, the risks of bleeding and hemorrhage may be increased. *Risk C: Monitor therapy*

Icosapent Ethyl: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Influenza Virus Vaccine (Live/Attenuated): May enhance the adverse/toxic effect of Salicylates. Specifically, Reye's syndrome may develop. *Risk X: Avoid combination*

Inotersen: Therapeutic Antiplatelets may enhance the adverse/toxic effect of Inotersen. Specifically, the risk of bleeding may be increased. *Risk C: Monitor therapy*

Ketorolac (Nasal): May enhance the adverse/toxic effect of Aspirin. An increased risk of bleeding may be associated with use of this combination. Ketorolac (Nasal) may diminish the cardioprotective effect of Aspirin. Management: Concurrent use of nasal ketorolac with analgesic doses of aspirin is generally not recommended. If using low-dose, cardioprotective aspirin with nasal ketorolac, monitor the patient closely for evidence of adverse GI effects. *Risk D: Consider therapy modification*

Ketorolac (Systemic): May enhance the adverse/toxic effect of Aspirin. An increased risk of bleeding may be associated with use of this combination. Ketorolac (Systemic) may diminish the cardioprotective effect of Aspirin. *Risk X: Avoid combination*

Lecanemab: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Limaprost: May enhance the adverse/toxic effect of Therapeutic Antiplatelets. Specifically, the risk of bleeding may be increased. *Risk C: Monitor therapy*

Lipid Emulsion (Fish Oil Based): May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Loop Diuretics: Salicylates may diminish the therapeutic effect of Loop Diuretics. Loop Diuretics may increase the serum concentration of Salicylates. *Risk C: Monitor therapy*

Macimorelin: Aspirin may diminish the diagnostic effect of Macimorelin. *Risk X: Avoid combination*

Methotrexate: Salicylates may increase the serum concentration of Methotrexate. Salicylate doses used for prophylaxis of cardiovascular events are not likely to be of concern. Management: Consider avoiding coadministration of methotrexate and salicylates. If coadministration cannot be avoided, monitor for increased toxic effects of methotrexate. Salicylate doses used for prophylaxis of cardiovascular events are not likely to be of concern. *Risk D: Consider therapy modification*

Multivitamins/Fluoride (with ADE): May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Multivitamins/Minerals (with ADEK, Folate, Iron): May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Multivitamins/Minerals (with AE, No Iron): May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Nicorandil: Aspirin may enhance the adverse/toxic effect of Nicorandil. Specifically, the risk of gastrointestinal ulceration and hemorrhage may be increased. *Risk C: Monitor therapy*

Nonsteroidal Anti-Inflammatory Agents (COX-2 Selective): Aspirin may enhance the adverse/toxic effect of Nonsteroidal Anti-Inflammatory Agents (COX-2 Selective). Specifically, the risk of gastrointestinal adverse effects may be increased. Management: Concurrent use of aspirin at doses beyond cardioprotective levels is not recommended. While concurrent use of low-dose aspirin with a COX-2 inhibitor is permissible, patients should be monitored closely for signs/symptoms of GI ulceration/bleeding. *Risk D: Consider therapy modification*

Nonsteroidal Anti-Inflammatory Agents (Nonselective): May enhance the adverse/toxic effect of Salicylates. An increased risk of bleeding may be associated with use of this combination. Nonsteroidal Anti-Inflammatory Agents (Nonselective) may diminish the cardioprotective effect of Salicylates. Salicylates may decrease the serum concentration of Nonsteroidal Anti-Inflammatory Agents (Nonselective). Management: Nonselective NSAIDs may reduce aspirin's cardioprotective effects. Administer ibuprofen 30-120 minutes after immediate-release aspirin, 2 to 4 hours after extended-release aspirin, or 8 hours before aspirin. *Risk D: Consider therapy modification*

Nonsteroidal Anti-Inflammatory Agents (Topical): May enhance the adverse/toxic effect of Salicylates. Specifically, the risk of gastrointestinal (GI) toxicity is increased. Management: Coadministration of salicylates and topical NSAIDs is not recommended. If salicylates and topical NSAIDs are coadministered, ensure the benefits outweigh the risks and monitor for increased NSAID toxicities. *Risk D: Consider therapy modification*

Obinutuzumab: Therapeutic Antiplatelets may enhance the adverse/toxic effect of Obinutuzumab. Specifically, the risk of bleeding may be increased. Management: Consider avoiding coadministration of obinutuzumab and therapeutic antiplatelets, especially during the first cycle of obinutuzumab therapy. *Risk D: Consider therapy modification*

Omacetaxine: Aspirin may enhance the adverse/toxic effect of Omacetaxine. Specifically, the risk for bleeding-related events may be increased. Management: Avoid concurrent use of aspirin with omacetaxine in patients with a platelet count of less than 50,000/uL. *Risk X: Avoid combination*

Omega-3 Fatty Acids: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Pentosan Polysulfate Sodium: Therapeutic Antiplatelets may enhance the adverse/toxic effect of Pentosan Polysulfate Sodium. Specifically, the risk of hemorrhage may be increased. *Risk C: Monitor therapy*

Pirtobrutinib: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Potassium Phosphate: May increase the serum concentration of Salicylates. *Risk C: Monitor therapy*

PRALAtrexate: Salicylates may increase the serum concentration of PRALAtrexate. Salicylate doses used for prophylaxis of cardiovascular events are unlikely to be of concern. Management: Consider avoiding concomitant use of salicylates and pralatrexate. If coadministered, monitor for increased pralatrexate adverse effects. Salicylate doses used for prophylaxis of cardiovascular events are not likely to be of concern. *Risk D: Consider therapy modification*

Probenecid: Salicylates may diminish the therapeutic effect of Probenecid. *Risk X: Avoid combination*

Salicylates: May enhance the anticoagulant effect of other Salicylates. *Risk C: Monitor therapy*

Selective Serotonin Reuptake Inhibitors: May enhance the antiplatelet effect of Aspirin. *Risk C: Monitor therapy*

Selumetinib: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Serotonin/Norepinephrine Reuptake Inhibitors: May enhance the antiplatelet effect of Aspirin. *Risk C: Monitor therapy*

Sincalide: Drugs that Affect Gallbladder Function may diminish the therapeutic effect of Sincalide. Management: Consider discontinuing drugs that may affect gallbladder motility prior to the use of sincalide to stimulate gallbladder contraction. *Risk D: Consider therapy modification*

Spironolactone: Aspirin may diminish the therapeutic effect of Spironolactone. *Risk C: Monitor therapy*

Sucroferric Oxyhydroxide: May decrease the serum concentration of Aspirin. Management: Administer aspirin at least 1 hour before administration of sucroferric oxyhydroxide. *Risk D: Consider therapy modification*

Sulfinpyrazone: Salicylates may decrease the serum concentration of Sulfinpyrazone. *Risk X: Avoid combination*

Talniflumate: Aspirin may enhance the adverse/toxic effect of Talniflumate. Management: When possible, consider alternatives to this combination. Concurrent use is generally not recommended. *Risk D: Consider therapy modification*

Thiopental: Aspirin may decrease the protein binding of Thiopental. *Risk C: Monitor therapy*

Thrombolytic Agents: Therapeutic Antiplatelets may enhance the adverse/toxic effect of Thrombolytic Agents. Specifically, the risk of bleeding may be increased. *Risk C: Monitor therapy*

Ticagrelor: Aspirin may enhance the antiplatelet effect of Ticagrelor. Aspirin may diminish the therapeutic effect of Ticagrelor. More specifically, the benefits of ticagrelor relative to clopidogrel may be diminished in adult patients receiving daily aspirin doses greater than 100-150 mg daily. Management: Avoid maintenance aspirin doses greater than 150 mg/day in patients receiving ticagrelor. After any initial dose, only low-dose aspirin (75 to 100 mg/day) is recommended. *Risk D: Consider therapy modification*

Tipranavir: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Valproate Products: Salicylates may increase the serum concentration of Valproate Products. *Risk C: Monitor therapy*

Varicella Virus-Containing Vaccines: Salicylates may enhance the adverse/toxic effect of Varicella Virus-Containing Vaccines. Specifically, the risk for Reye's syndrome may increase. *Risk X: Avoid combination*

Vitamin E (Systemic): May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Vitamin K Antagonists (eg, warfarin): Aspirin may enhance the anticoagulant effect of Vitamin K Antagonists. *Risk C: Monitor therapy*

Volanesorsen: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Zanubrutinib: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Food Interactions

Food may decrease the rate but not the extent of oral absorption. Benedictine liqueur, prunes, raisins, tea, and gherkins have a potential to cause salicylate accumulation. Fresh fruits containing vitamin C may displace drug from binding sites, resulting in increased urinary excretion of aspirin. Curry powder, paprika, licorice; may cause salicylate accumulation. These foods contain 6 mg salicylate/100 g. An ordinary American diet contains 10-200 mg/day of salicylate. Management: Administer with food or large volume of water or milk to minimize GI upset. Limit curry powder, paprika, licorice.

Test Interactions

False-negative results for glucose oxidase urinary glucose tests (Clinistix); false-positives using the cupric sulfate method (Clinitest); also, interferes with Gerhardt test, VMA determination; 5-HIAA, xylose tolerance test and T₃ and T₄; may lead to false-positive aldosterone/renin ratio (ARR) (Funder 2016)

Genes of Interest

- [CYP2C9 - Aspirin](#)
- [G6PD - Aspirin](#)

Monitoring Parameters

Monitor for signs and symptoms of drug reaction with eosinophilia and systemic symptoms (eg, fever, rash, lymphadenopathy, eosinophilia in association with other organ system involvement such as hepatitis, nephritis, hematological abnormalities, myocarditis, myositis; early symptoms of hypersensitivity reaction may occur without rash).

Reference Range

Timing of serum samples: Peak levels usually occur 2 hours after ingestion. Salicylate serum concentrations correlate with the pharmacological actions and adverse effects observed. The serum salicylate concentration and the corresponding clinical correlations are as follows: See table.

Serum Salicylate Concentration	Desired Effects	Adverse Effects / Intoxication
~100 mcg/mL (SI: ~724 micromole/L)	Antiplatelet Antipyresis Analgesia	GI intolerance and bleeding, hypersensitivity, hemostatic defects
150 to 300 mcg/mL (SI: 1,086 to 2,172 micromole/L)	Anti-inflammatory	Mild salicylism
250 to 400 mcg/mL (SI: 1,810 to 2,896 micromole/L)	Treatment of rheumatic fever	Nausea/vomiting, hyperventilation, salicylism, flushing, sweating, thirst, headache, diarrhea, and tachycardia
>400 to 500 mcg/mL (SI: >2,896 to 3,620 micromole/L)		Respiratory alkalosis, hemorrhage, excitement, confusion, asterixis, pulmonary edema, convulsions, tetany, metabolic acidosis, fever, coma, cardiovascular collapse, renal and respiratory failure

Nursing Practice Points

Check ordered labs and report abnormalities. Monitor and instruct patients to report any signs of bleeding, signs of abdominal ulcer, or hypersensitivity reactions. Educate patient to avoid alcohol consumption within 1 hour before and 2 hours after aspirin administration. Educate pregnant patient about the correct way to perform fetal kick counts and to monitor fetal movement. Report any concerns of decreased fetal movement to physician.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Capsule, Oral:

Vazalore: 81 mg, 325 mg [contains carrageenan, fd&c blue #1 (brilliant blue), soybean oil]

Capsule Extended Release 24 Hour, Oral:

Durlaza: 162.5 mg [DSC]

Packet, Oral:

Stanback Headache Powders: 650-200-32 MG (2 ea [DSC], 6 ea [DSC], 24 ea [DSC], 50 ea [DSC])

Suppository, Rectal:

Generic: 60 mg (12 ea, 100 ea, 1000 ea); 300 mg (12 ea); 600 mg (12 ea [DSC])

Tablet, Oral:

Aspirin Adult: 325 mg [DSC] [contains corn starch]

Aspirin Maximum Strength: 500 mg [DSC]

Bayer Aspirin: 325 mg [contains corn starch]

FT Aspirin: 325 mg [contains corn starch]

GoodSense Aspirin: 325 mg [contains corn starch]

GoodSense Aspirin: 325 mg [gluten free; contains corn starch]

Medi-First Aspirin: 325 mg [contains corn starch]

Medique Aspirin: 325 mg [contains corn starch]

Generic: 325 mg

Tablet, Oral, (buffered):

Ascriptin: 325 mg [contains methylparaben, saccharin sodium]

Bufferin: 325 mg [DSC]

Bufferin: 325 mg [contains benzoic acid, fd&c blue #1 (brilliant blue)]

Bufferin Extra Strength: 500 mg [contains benzoic acid, fd&c blue #1 (brilliant blue)]

Tri-Buffered Aspirin: 325 mg

Tablet Chewable, Oral:

Aspirin Childrens: 81 mg [orange flavor]

Aspirin Childrens: 81 mg [contains fd&c yellow #6 (sunset yellow)]

Aspirin Low Dose: 81 mg [orange flavor]

Aspirin Low Dose: 81 mg [contains corn starch, fd&c yellow #6 (sunset yellow), saccharin sodium]

Aspirin Low Dose: 81 mg [contains corn starch, fd&c yellow #6 (sunset yellow), saccharin sodium; orange flavor]

Aspirin Low Dose: 81 mg [contains corn starch, fd&c yellow #6(sunset yellow)alumin lake; orange flavor]

Bayer Low Dose: 81 mg [caffeine free, sodium free; contains corn starch, fd&c yellow #6(sunset yellow)alumin lake, saccharin sodium]

Childrens Aspirin Low Strength: 81 mg [DSC] [caffeine free]

FT Aspirin: 81 mg [gluten free; contains corn starch, fd&c yellow #6(sunset yellow)alumin lake, saccharin sodium; orange flavor]

GoodSense Aspirin: 81 mg [gluten free; contains fd&c red #40(allura red ac)aluminum lake, saccharin sodium; cherry flavor]

GoodSense Aspirin: 81 mg [gluten free; contains fd&c yellow #6(sunset yellow)alumin lake, saccharin sodium]

GoodSense Aspirin: 81 mg [gluten free; contains fd&c yellow #6(sunset yellow)alumin lake, saccharin sodium; orange flavor]

GoodSense Aspirin Adult Low St: 81 mg [DSC] [gluten free; contains fd&c yellow #6(sunset yellow)alumin lake, saccharin sodium; orange flavor]

St Joseph Low Dose: 81 mg [contains corn starch, fd&c yellow #6(sunset yellow)alumin lake; orange flavor]

Generic: 81 mg

Tablet Delayed Release, Oral:

Aspirin 81: 81 mg [DSC] [contains fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

Aspirin 81: 81 mg [DSC] [contains fd&c yellow #6(sunset yellow)alumin lake, quinoline yellow (d&c yellow #10)]

Aspirin Adult Low Dose: 81 mg [contains fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

Aspirin Adult Low Dose: 81 mg [DSC] [contains quinoline (d&c yellow #10) aluminum lake]

Aspirin Adult Low Dose: 81 mg [contains quinoline (d&c yellow #10) aluminum lake, soybeans (glycine soja)]

Aspirin Adult Low Strength: 81 mg [contains corn starch, quinoline yellow (d&c yellow #10)]

Aspirin EC Adult Low Dose: 81 mg [contains fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

Aspirin EC Low Strength: 81 mg [contains corn starch, quinoline (d&c yellow #10) aluminum lake]

Aspirin Low Dose: 81 mg [contains corn starch, fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

Aspirin Low Dose: 81 mg [contains fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

Aspirin Low Dose: 81 mg [contains fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

Aspirin Low Dose: 81 mg [contains quinoline (d&c yellow #10) aluminum lake]

Bayer Aspirin: 325 mg [contains corn starch, fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

Bayer Aspirin EC Low Dose: 81 mg [contains corn starch, fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

Ecotrin Maximum Strength: 500 mg [DSC]

EcPirin: 325 mg [DSC] [contains fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

FT Aspirin Low Dose: 81 mg [contains corn starch, fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

FT Aspirin Low Dose: 81 mg [contains fd&c red #40(allura red ac)aluminum lake, fd&c yellow #6(sunset yellow)alumin lake]

FT Aspirin Low Dose: 81 mg [contains quinoline (d&c yellow #10) aluminum lake]

FT Aspirin Low Dose: 81 mg [contains quinoline yellow (d&c yellow #10)]

FT Enteric Coated Aspirin: 325 mg [contains corn starch, fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

GoodSense Aspirin: 325 mg [DSC] [contains corn starch, fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

GoodSense Aspirin Low Dose: 81 mg [contains fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

GoodSense Aspirin Low Dose: 81 mg [contains quinoline (d&c yellow #10) aluminum lake]

Miniprin Low Dose: 81 mg [DSC] [contains quinoline (d&c yellow #10) aluminum lake]

St Joseph Aspirin: 81 mg [contains fd&c red #40 (allura red ac dye), fd&c yellow #6 (sunset yellow); orange flavor]

St Joseph Low Dose: 81 mg [gluten free; contains fd&c red #40 (allura red ac dye), fd&c yellow #6 (sunset yellow); orange flavor]

Generic: 81 mg, 325 mg

Dosage Forms: International

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. **Note:** Availability of specific dosage forms may vary by region/country.

Additional dosage forms not available in the United States and/or Canada:

Solution Reconstituted, Injection: Lysine Aspirin: 900 mg (equivalent to aspirin 500 mg)

Generic Available (US)

May be product dependent

Pricing: US

Capsules (Vazalore Oral)

81 mg (per each): \$0.93

325 mg (per each): \$0.90

Chewable (Aspirin Oral)

81 mg (per each): \$0.02 - \$0.06

Chewable (St Joseph Low Dose Oral)

81 mg (per each): \$0.05

Suppository (Aspirin Rectal)

300 mg (per each): \$1.46

Tablet, EC (Aspirin Oral)

81 mg (per each): \$0.01 - \$0.03

325 mg (per each): \$0.02 - \$0.05

Tablet, EC (Bayer Aspirin EC Low Dose Oral)

81 mg (per each): \$0.04

Tablet, EC (Bayer Aspirin Oral)

325 mg (per each): \$0.08

Tablets (Ascriptin Oral)

325 mg (per each): \$0.08

Tablets (Aspirin Oral)

325 mg (per each): \$0.01 - \$0.03

Tablets (Bayer Aspirin Oral)

325 mg (per each): \$0.14

Tablets (Bufferin Extra Strength Oral)

500 mg (per each): \$0.10

Tablets (Bufferin Oral)

325 mg (per each): \$0.07

Tablets (Medi-First Aspirin Oral)

325 mg (per each): \$0.04

Tablets (Medique Aspirin Oral)

325 mg (per each): \$0.04

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Mechanism of Action

Irreversibly inhibits cyclooxygenase-1 and 2 (COX-1 and 2) enzymes, via acetylation, which results in decreased formation of prostaglandin precursors; irreversibly inhibits formation of prostaglandin derivative, thromboxane A₂, via acetylation of platelet cyclooxygenase, thus inhibiting platelet aggregation; has antipyretic, analgesic, and anti-inflammatory properties

Pharmacokinetics (Adult Data Unless Noted)

Onset: Immediate release: Platelet inhibition: Nonenteric-coated: <1 hour; enteric-coated: 3 to 4 hours (Eikelboom 2012). **Note:** Chewing nonenteric-coated or enteric-coated tablets results in inhibition of platelet aggregation within 20 minutes (Eikelboom 2012; Feldman 1999; Sai 2011).

Duration: Immediate release: 4 to 6 hours; however, platelet inhibitory effects last the lifetime of the platelet (~10 days) due to its irreversible inhibition of platelet COX-1 (Eikelboom 2012).

Absorption: Immediate release: Rapidly absorbed in stomach and upper intestine (Eikelboom 2012); Extended-release capsule: Rate of absorption is dependent upon food, alcohol, and gastric pH.

Distribution: V_d: 10 L; readily into most body fluids and tissues; hydrolyzed to salicylate (active) by esterases in the GI mucosa, red blood cells, synovial fluid and blood

Protein binding: Concentration dependent; as salicylate concentration increases, protein binding decreases: ~90% to 94% (to albumin) at concentrations ≤80 mcg/mL (Rosenberg 1981; Juurlink 2015); ~30% with concentrations seen in overdose (Juurlink 2015).

Metabolism: Hydrolyzed to salicylate (active) by esterases in GI mucosa, red blood cells, synovial fluid, and blood; metabolism of salicylate occurs primarily by hepatic conjugation; metabolic pathways are saturable

Bioavailability: Immediate release: 50% to 75% reaches systemic circulation

Half-life elimination: Parent drug: Plasma concentration: 15 to 20 minutes; Salicylates (dose dependent): 3 hours at lower doses (300 to 600 mg), 5 to 6 hours (after 1 g), 10 hours with higher doses

Time to peak, serum: Immediate release: ~1 to 2 hours (nonenteric-coated), 3 to 4 hours (enteric-coated) (Eikelboom 2012);
Extended-release capsule: ~2 hours. **Note:** Chewing nonenteric-coated tablets results in a time to peak concentration of 20 minutes (Feldman 1999). Chewing enteric-coated tablets results in a time to peak concentration of 2 hours (Sai 2011).

Excretion: Urine (75% as salicyluric acid, 10% as salicylic acid)

Dental Use

Treatment of postoperative pain

* See [Uses in AHFS Essentials](#) for additional information.

Dental: Local Anesthetic/Vasoconstrictor Precautions

No information available to require special precautions

Dental Health Professional Considerations

The Food and Drug Administration (FDA), has issued a letter updating information and considerations regarding the use of ibuprofen (400 mg doses) in patients who are taking low dose aspirin (81 mg, immediate release; not enteric coated) for cardioprotection and stroke prevention. Ibuprofen, at these doses, may interfere with aspirin's antiplatelet effect depending upon when it is administered. Patients initiated on aspirin first (for ~1 week) then ibuprofen (400 mg 3 times/day for 10 days) seem to maintain aspirin's platelet effect (Cryer, 2005). Ibuprofen has the greatest impact on aspirin if administered less than 8 hours before aspirin (Catella-Lawson, 2001).

Patients may require counseling about the appropriate timing of ibuprofen dosing in relationship to aspirin therapy. With occasional use of ibuprofen, a clinically-significant interaction with aspirin is unlikely. To avoid interference during chronic dosing, a single dose of ibuprofen should be taken 30 to 120 minutes after aspirin ingestion or at least 8 hours should elapse after ibuprofen dosing before giving aspirin (Catella-Lawson, 2001; FDA, 2006).

The clinical implications of the interaction are unclear. There have not been any clinical endpoint studies conducted at this time. Avoidance of this interaction is potentially important because aspirin's vascular protection could be decreased or negated.

Other nonselective NSAIDs may have potential for a similar interaction with aspirin. Such has been described with naproxen (Capone, 2005). Acetaminophen does not appear to interfere with the antiplatelet effect of aspirin. Other clinical scenarios (use of smaller ibuprofen doses, other aspirin products, other doses of aspirin) have not been evaluated.

Additional information is available at:

<http://www.fda.gov/Drugs/DrugSafety/PostmarketDrugSafetyInformationforPatientsandProviders/ucm125222.htm>

Dental: Effects on Dental Treatment

Key adverse event(s) related to dental treatment: As with all drugs which may affect hemostasis, bleeding is associated with aspirin. Hemorrhage may occur at virtually any site; risk is dependent on multiple variables including dosage, concurrent use of multiple agents which alter hemostasis, and patient susceptibility. Many adverse effects of aspirin are dose related, and are rare at low dosages. Other serious reactions are idiosyncratic, related to allergy or individual sensitivity (see Dental Health Professional Considerations).

Aspirin as sole antiplatelet agent: Patients taking aspirin for ischemic stroke prevention are safe to continue it during dental procedures (Armstrong, 2013).

Concurrent aspirin use with other antiplatelet agents: Aspirin in combination with clopidogrel (Plavix), prasugrel (Effient), or ticagrelor (Brilinta) is the primary prevention strategy against stent thrombosis after placement of drug-eluting metal stents in coronary patients. Premature discontinuation of combination antiplatelet therapy (ie, dual antiplatelet therapy) strongly increases the risk of a catastrophic event of stent thrombosis leading to myocardial infarction and/or death, so says a science advisory issued in January 2007 from the American Heart Association in collaboration with the American Dental Association and other professional healthcare organizations. The advisory stresses a 12-month therapy of dual antiplatelet therapy after placement of a

drug-eluting stent in order to prevent thrombosis at the stent site. Any elective surgery should be postponed for 1 year after stent implantation, and if surgery must be performed, consideration should be given to continuing the antiplatelet therapy during the perioperative period in high-risk patients with drug-eluting stents.

This advisory was issued from a science panel made up of representatives from the American Heart Association (AHA), the American College of Cardiology, the Society for Cardiovascular Angiography and Interventions, the American College of Surgeons, the American Dental Association (ADA), and the American College of Physicians (Grines, 2007).

Dental: Effects on Bleeding

Aspirin irreversibly inhibits platelet aggregation which can prolong bleeding. Upon discontinuation, normal platelet function returns only when new platelets are released (~7 to 10 days). However, in the case of dental surgery, there is no scientific evidence to support discontinuation of aspirin. This was recently supported by the American Academy of Neurology in patients with ischemic cerebrovascular disease (Armstrong, 2013). A recent study compared blood loss after a single tooth extraction in coronary artery disease patients who were either on aspirin (100 mg daily) or off aspirin for the extraction. The mean volume of bleeding was not statistically different between the groups. Local hemostatic measures were sufficient to control bleeding and there were no reported episodes of hemorrhaging intra- or postoperatively (Medeiros, 2011).

Dental Usual Dosing

Postoperative pain:

Analgesic and antipyretic: Oral, rectal:

Children: 10 to 15 mg/kg/dose every 4 to 6 hours, up to a total of 4 g/day

Adults: 325 to 650 mg every 4 to 6 hours up to 4 g/day

Anti-inflammatory: Oral: Initial:

Children: 60 to 90 mg/kg/day in divided doses; usual maintenance: 80 to 100 mg/kg/day divided every 6 to 8 hours; monitor serum concentrations

Adults: 2.4 to 3.6 g/day in divided doses; usual maintenance: 3.6 to 5.4 g/day; monitor serum concentrations

Related Information

- [Beers Criteria 2023: Potentially Inappropriate Medication Use in Older Adults ≥65 Years of Age](#)
- [KIDs List](#)
- [Oral Antiplatelet Comparison Chart](#)
- [Oral Medications That Should Not Be Crushed or Altered](#)

Index Terms

Acetylsalicylic Acid; ASA; Baby Aspirin; DL-Lysine Acetylsalicylate; Lysine Aspirin

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References: International

Lag injection (aspirin) [product information]. Hsinchu County, Taiwan: Taiyu Chemical Pharmaceutical Factory Co Ltd; December 2015.

Brand Names: US

Ascriptin [OTC]; Aspirin 81 [OTC] [DSC]; Aspirin Adult Low Dose [OTC]; Aspirin Adult Low Strength [OTC]; Aspirin Adult [OTC] [DSC]; Aspirin Childrens [OTC]; Aspirin EC Adult Low Dose [OTC]; Aspirin EC Low Strength [OTC]; Aspirin Low Dose [OTC]; Aspiatab Maximum Strength [OTC] [DSC]; Bayer Aspirin EC Low Dose [OTC]; Bayer Aspirin [OTC]; Bayer Low Dose [OTC]; Bufferin Extra Strength [OTC]; Bufferin [OTC]; Childrens Aspirin Low Strength [OTC] [DSC]; Durlaza [DSC]; Ecotrin Maximum Strength [OTC] [DSC]; EcPirin [OTC] [DSC]; FT Aspirin Low Dose [OTC]; FT Aspirin [OTC]; FT Enteric Coated Aspirin [OTC]; GoodSense Aspirin Adult Low St [OTC] [DSC]; GoodSense Aspirin Low Dose [OTC]; GoodSense Aspirin [OTC]; Medi-First Aspirin [OTC]; Medique Aspirin [OTC]; Miniprin Low Dose [OTC] [DSC]; St Joseph Aspirin [OTC]; St Joseph Low Dose [OTC]; Stanback Headache Powders [OTC] [DSC]; Tri-Buffered Aspirin [OTC]; Vazalore [OTC]

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Therapeutic Cases for AI Trial

1. 59-year-old male patient with a history of hypertension for 10 years, DM II for 5 years, and myocardial infarction 2 years ago. His home medication consists of bisoprolol 5mg Q24 – enalapril 10mg Q24 – atorvastatin 80mg Q24 and SL NTG PRN. Mention the missing medication needed to protect the patient's heart.

ANSWER: Aspirin 100mg Q24 for secondary prevention.

Therapeutic Cases for AI Trial

2. A 50-year-old male patient newly diagnosed with hypertension (average BP reading 137/80) with albuminuria. He is a known case of migraine headache and needs prophylactic. What would be your recommendation for treating this patient's hypertension and migraine?

ANSWER: Lisinopril 10mg Q24

Lisinopril (Lexi-Drugs Multinational)

Monograph

Images

Adult Patient Education

Pediatric Patient Education

ALERT: US Boxed Warning

Fetal toxicity:

When pregnancy is detected, discontinue lisinopril as soon as possible. Drugs that act directly on the renin-angiotensin system can cause injury and death to the developing fetus.

Drug Shortages: US

One or more forms of this drug may be in short supply or unavailable. Refer to the following for additional information:

FDA: https://www.accessdata.fda.gov/scripts/drugshortages/dsp_ActiveIngredientDetails.cfm?AI=Mefloquine+Hydrochloride+Tablet&st=c&tab=tabs-4&panels=0

Brand Names: International

Find brand name(s) by country (for United States and Canada see separate Brand Names sections)

🔍 Enter a Country or Country Code

For country code abbreviations ( [show table](#))

International Nonproprietary Names (INN)

Lisinopril [English]; Lisinopril [French]; Lisinopril [Spanish]; Lisinoprilum [Latin]; Лизиноприл [Russian]; ليزينوبريل [Arabic]; 赖诺普利 [Chinese]

Brazilian Nonproprietary Names (DCB)

Lisinopril

Japanese Accepted Name (JAN)

リシノプリル水和物

Anatomic Therapeutic Chemical (ATC) Classification

- C09AA03

Pharmacologic Category

Angiotensin-Converting Enzyme (ACE) Inhibitor; Antihypertensive

Dosing: Adult

Acute coronary syndrome

Acute coronary syndrome:

Non-ST-elevation acute coronary syndrome (off-label use):

Note: Initiate in stable patients prior to hospital discharge as a component of an appropriate medical regimen, which may include antiplatelet agent(s), a beta-blocker, and a statin. Continue indefinitely for patients with concurrent diabetes, left ventricular ejection fraction $\leq 40\%$, hypertension, or stable chronic kidney disease (^{Ref}). Dosing recommendations based on general dosing range in manufacturer's labeling.

Oral: Initial: 2.5 to 10 mg once daily (depending on initial blood pressure); titrate slowly based on tolerability and response up to 40 mg/day.

ST-elevation myocardial infarction:

Note: In hemodynamically stable patients with large anterior ST-elevation myocardial infarction, consider starting within 24 hours of presentation as a component of an appropriate medical regimen, which may include antiplatelet agent(s), a beta-blocker, and a statin. Continue therapy indefinitely ^(Ref). Dosing recommendations based on general dosing range in manufacturer's labeling.

Oral: Initial: 2.5 to 5 mg once daily initiated within 24 hours of presentation; titrate slowly up to 10 mg/day or higher as tolerated under close monitoring to avoid hypotension; maximum: 40 mg/day ^(Ref).

Heart failure with reduced ejection fraction

Heart failure with reduced ejection fraction:

Note: If tolerated, an angiotensin II receptor-neprilysin inhibitor is generally preferred over an angiotensin-converting enzyme inhibitor ^(Ref).

Oral: Initial: 2.5 to 5 mg once daily; increase dose (eg, double but by no more than 10 mg increments) as tolerated every ≥ 1 to 2 weeks to a target dose of 20 to 40 mg once daily ^(Ref). In hospitalized patients, may titrate more rapidly as tolerated ^(Ref).

Hypertension, chronic

Hypertension, chronic:

Note: For patients who warrant combination therapy (BP >20/10 mm Hg above goal or suboptimal response to initial monotherapy), may use with another appropriate agent (eg, long-acting dihydropyridine calcium channel blocker or thiazide diuretic) ^(Ref).

Oral: Initial: 5 to 10 mg once daily; evaluate response after ~2 to 4 weeks and titrate dose (eg, increase the daily dose by doubling) as needed, up to 40 mg once daily; if additional blood pressure control is needed, consider combination therapy. Patients with severe asymptomatic hypertension and no signs of acute end organ damage should be evaluated for medication titration within 1 week ^(Ref).

Migraine, prevention

Migraine, prevention (alternative agent) (off-label use):

Note: Among second-line agents for migraine prevention, consider use in patients with comorbid hypertension ^(Ref). An adequate trial for assessment of effect is considered to be at least 2 to 3 months at a therapeutic dose ^(Ref).

Oral: Initial: 10 mg once daily; may increase dose after 1 week to 20 mg daily based on response and tolerability ^(Ref).

Posttransplant erythrocytosis, kidney transplant recipients

Posttransplant erythrocytosis, kidney transplant recipients (off-label use):

Note: For patients with a hemoglobin concentration >17 g/dL ^(Ref).

Oral: Initial: 2.5 or 5 mg once daily; if inadequate response seen within 4 weeks, may titrate up to 40 mg/day based on hemoglobin and blood pressure response; if hemoglobin remains >17 g/dL after an additional 4 weeks, consider additional therapy (eg, phlebotomy) ^(Ref).

Proteinuric chronic kidney disease, diabetic or nondiabetic

Proteinuric chronic kidney disease, diabetic or nondiabetic (off-label use):

Oral: Initial: 2.5 to 10 mg once daily depending on baseline BP. Titrate gradually (eg, by doubling the dose every 2 to 4 weeks) up to the maximally tolerated dose, not to exceed 40 mg/day. If proteinuria target is not met despite optimized dosage, consider additional therapies (eg, sodium-glucose cotransporter-2 inhibitor) (^{Ref}).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

* See [Dosage and Administration in AHFS Essentials](#) for additional information.

Dosing: Older Adult

Refer to adult dosing. In the management of hypertension, consider lower initial doses (eg, 2.5 to 5 mg once daily) and titrate to response (^{Ref}).

Dosing: Altered Kidney Function: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Kidney impairment prior to treatment initiation:

Altered kidney function:

CrCl ≥30 mL/minute: No dosage adjustment necessary.

CrCl 10 to <30 mL/minute: Initial: 2.5 mg once daily (may consider beginning 5 mg once daily in some patients [eg, hypertension]); titrate slowly based on tolerability and response with close monitoring, not to exceed usual maximum dose of up to 40 mg/day (^{Ref}).

CrCl <10 mL/minute: Consider alternative therapy; risk of adverse effects or complications (eg, hyperkalemia, kidney failure) are increased (^{Ref}). If necessary, begin 2.5 mg once daily; titrate slowly based on tolerability and response with close monitoring, not to exceed usual maximum dose of up to 40 mg/day (^{Ref}).

Hemodialysis, intermittent (thrice weekly): Dialyzable (50%) (^{Ref}):

Daily dosing: Initial: 2.5 mg once daily; when scheduled dose falls on a dialysis day, administer post hemodialysis. Titrate slowly based on tolerability and response with close monitoring (^{Ref}), not to exceed usual maximum dose of up to 40 mg/day (^{Ref}).

Three times weekly (post dialysis) dosing (hypertension only): Initial: 10 mg 3 times weekly administered post hemodialysis on dialysis days; titrate slowly based on tolerability and response with close monitoring to a maximum of 40 mg 3 times weekly administered post hemodialysis on dialysis days (^{Ref}).

Peritoneal dialysis: Initial: 2.5 mg once daily; titrate slowly based on tolerability and response with close monitoring, not to exceed usual maximum dose of up to 40 mg/day (^{Ref}).

Alterations in kidney function during treatment: Small, transient increases in serum creatinine are likely to occur within 4 weeks following initiation of therapy or an increase in dose. If serum creatinine increases by >30%, review for possible etiologies (eg, acute kidney injury, volume depletion, concomitant medications, renal artery stenosis) before determining if dose reduction or discontinuation of lisinopril therapy should be considered (^{Ref}).

Dosing: Altered Liver Function: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: Use of angiotensin-converting enzyme inhibitors in patients with cirrhosis and ascites should be avoided as use can further diminish renal blood flow and precipitate hepatorenal syndrome (^{Ref}). Lisinopril does not require liver metabolism for activation (^{Ref}).

Liver impairment prior to treatment initiation:

Initial or dose adjustment in patients with preexisting liver cirrhosis:

Child-Turcotte-Pugh class A to C: No dosage adjustment necessary (^{Ref}); avoid use in patients with ascites (^{Ref}).

Dosing: Adjustment for Toxicity: Adult

Hyperkalemia: Assess for alternative or contributing factors (eg, diet, concomitant medications) of increased potassium before determining if dose reduction or discontinuation of lisinopril therapy should be considered ^(Ref).

Dosing: Pediatric**Dosage guidance:**

Safety: Compounded and commercially available oral solutions have multiple concentrations; precautions should be taken to verify and avoid confusion between the different concentrations; dose should be clearly presented as mg.

Hypertension**Hypertension:**

Children <6 years: Limited data available: Oral: Initial: 0.07 mg/kg/dose once daily; maximum initial daily dose: 5 mg/**day**; titrate dose based on response; maximum daily dose: 0.6 mg/kg/**day** not to exceed 40 mg/**day** ^(Ref).

Children ≥6 years and Adolescents: Oral: Initial: 0.07 mg/kg/dose once daily; titrate dose based on response; maximum initial daily dose: 5 mg/**day**; maximum daily dose: 0.61 mg/kg/**day** not to exceed 40 mg/**day** ^(Ref).

Proteinuria

Proteinuria (eg, mild IgA nephropathy, focal segmental glomerulosclerosis [FSGS]): Limited data available:

Children ≥2 years and Adolescents: Oral: Initial: 0.1 to 0.2 mg/kg/dose once daily; increase at 1- to 2-week intervals to target dose of 0.4 mg/kg/dose once daily ^(Ref); maximum dose: 40 mg/**day**. Dosing based on a prospective study (total patients: n=138; pediatric patients: n=93, age range: 2 to 17 years) in patients with steroid-resistant FSGS comparing mycophenolate to cyclosporine in which most patients received lisinopril (n=118) for protein sparing effects. Lisinopril was initiated at 0.1 mg/kg/dose and increased every 2 weeks to target dose of 0.4 mg/kg/dose (maximum dose: 40 mg/dose) ^(Ref). In pediatric patients with mild IgA nephropathy (n=40, mean age: 11.4 years; range: 4.4 to 15.4 years), a similar protocol was used, with an initial dose of 0.2 mg/kg/dose increased after 7 days to 0.4 mg/kg/dose (maximum dose: 20 mg/dose) ^(Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Altered Kidney Function: Pediatric

Children ≥6 years and Adolescents:

CrCl ≥30 mL/minute/1.73 m²: No dosage adjustment needed.

CrCl <30 mL/minute/1.73 m²: Use is not recommended.

Dosing: Altered Liver Function: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution.

Calculations

- [Child Pugh Classification](#)
- [Child Pugh Classification \(SI units\)](#)
- [Creatinine Clearance by Cockcroft-Gault](#)
- [Creatinine Clearance by Cockcroft-Gault \(SI units\)](#)
- [Creatinine Clearance by Cockcroft-Gault with IBW](#)
- [Creatinine Clearance by Cockcroft-Gault with IBW \(SI units\)](#)
- [Creatinine Clearance by Jelliffe](#)
- [Glomerular Filtration Rate by MDRD](#)
- [Glomerular Filtration Rate by Schwartz](#)

Use: Labeled Indications

Heart failure with reduced ejection fraction: Adjunctive therapy to reduce signs and symptoms of systolic heart failure.

Hypertension, chronic: Management of hypertension in adult and pediatric patients ≥ 6 years of age.

ST-elevation myocardial infarction: Treatment of acute MI within 24 hours in hemodynamically stable patients to improve survival.

* See [Uses in AHFS Essentials](#) for additional information.

Use: Off-Label: Adult

Migraine, prevention Level of Evidence [B, G]

Data from small controlled and noncontrolled studies support the use of lisinopril for the prevention of migraine (^{Ref}).

Based on the American Headache Society consensus statement update on integrating new migraine treatments into clinical practice, lisinopril is probably effective and should be considered for migraine prevention (^{Ref}). Based on the American Academy of Neurology/American Headache Society evidence-based guideline on pharmacologic treatment for episodic migraine prevention in adults, lisinopril is considered possibly effective and may be considered for migraine prevention (^{Ref}). [Access Full Off-Label Monograph](#)

Non-ST-elevation acute coronary syndrome Level of Evidence [G]

Based on the 2014 [American College of Cardiology/American Heart Association \(ACC/AHA\) guidelines for the management of patients with non-ST-elevation acute coronary syndromes \(NSTEMI-ACS\)](#), an angiotensin-converting enzyme (ACE) inhibitor should be initiated and continued indefinitely after NSTEMI-ACS in patients with a left ventricular ejection fraction (LVEF) $\leq 40\%$ and in those with hypertension, diabetes mellitus, or stable chronic kidney disease (CKD) unless contraindicated. Use of an ACE inhibitor may also be useful in all other patients with cardiac or other vascular disease.

Posttransplant erythrocytosis, kidney transplant recipients Level of Evidence [C]

Data from a limited number of small clinical trials suggest that lisinopril may be beneficial for the treatment of kidney transplant recipients who have posttransplant erythrocytosis (^{Ref}).

Proteinuric chronic kidney disease, diabetic or nondiabetic Level of Evidence [G]

Based on the Kidney Disease Improving Global Outcomes guidelines for the evaluation and management of CKD, for the management of glomerular disease, and for diabetes management in CKD, the use of an ACE inhibitor (eg, lisinopril) or an angiotensin receptor blocker is recommended in patients with proteinuric diabetic or nondiabetic (eg, associated with IgA nephropathy or focal segmental glomerulosclerosis) CKD to prevent progression of CKD (^{Ref}).

Level of Evidence Definitions

Level of Evidence Scale

A - Consistent evidence from well-performed randomized, controlled trials or overwhelming evidence of some other form (eg, results of the introduction of penicillin treatment) to support the off-label use. Further research is unlikely to change confidence in the estimate of benefit.

- B** - Evidence from randomized, controlled trials with important limitations (inconsistent results, methodological flaws, indirect or imprecise), or very strong evidence of some other research design. Further research (if performed) is likely to have an impact on confidence in the estimate of benefit and risk and may change the estimate.
- C** - Evidence from observational studies (eg, retrospective case series/reports providing significant impact on patient care), unsystematic clinical experience, or from potentially flawed randomized, controlled trials (eg, when limited options exist for condition). Any estimate of effect is uncertain.
- G** - Use has been substantiated by inclusion in at least one evidence-based or consensus-based clinical practice guideline.

Class and Related Monographs

Angiotensin-Converting Enzyme Inhibitors

Clinical Practice Guidelines

Atrial Fibrillation:

AHA/ACC/HRS, "2014 AHA/ACC/HRS Guideline for the Management of Patients with Atrial Fibrillation," March 2014

Chronic Kidney Disease:

"KDIGO 2012 Clinical Practice Guidelines for the Evaluation and Management of Chronic Kidney Disease," January 2013

Coronary Artery Bypass Graft Surgery:

ACCF/AHA, "2011 ACCF/AHA Guideline for Coronary Artery Bypass Graft Surgery," November 2011

AHA Scientific Statement, "Secondary Prevention After Coronary Artery Bypass Graft Surgery," February 2015

Heart Failure:

AHA/ACC/HFSA, "Guideline for the Management of Heart Failure," April 2022

ESC, "Guidelines for the Diagnosis and Treatment of Acute and Chronic Heart Failure," September 2021

Hypertension:

"2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults," November 2017.

"ACC/AHA Expert Consensus Document on Hypertension in the Elderly," 2011

KDIGO, "Clinical Practice Guideline for the Management of Blood Pressure in Chronic Kidney Disease," 2021

Ischemic Heart Disease:

AHA/ACC, "2014 AHA/ACC Guideline for the Management of Patients with Non-ST-Elevation Acute Coronary Syndromes," September 2014.

ACC/AHA, "2013 ACCF/AHA Guideline for the Management of ST-Elevation Myocardial Infarction," December 2012

ESC, "Guidelines for the Management of Acute Coronary Syndromes in Patients Presenting Without Persistent ST-Segment Elevation," August 2020

Ischemic Stroke:

AHA/ASA, "2014 Guidelines for the Prevention of Stroke in Patients with Stroke and Transient Ischemic Attack," May 2014

AHA/ASA, "Guideline for the Prevention of Stroke in Patients with Stroke and Transient Ischemic Attack," 2021

ESO, "Guideline on Pharmacological Interventions for Long-term Secondary Prevention after Ischaemic Stroke or Transient Ischaemic Attack," 2022

Liver Ascites, Cirrhosis:

AASLD "Diagnosis, Evaluation, and Management of Ascites, Spontaneous Bacterial Peritonitis and Hepatorenal Syndrome, Guidance" [August 2021](#)

AASLD "Management of Adult Patients with Ascites Due to Cirrhosis," [2012](#)

Medication Safety:

"STOPP/START Criteria for Potentially Inappropriate Prescribing in Older People: Version 3," [May 2023](#)

Migraine:

AHS, "The American Headache Society Consensus Statement: Update on Integrating New Migraine Treatments into Clinical Practice," [2021](#)

CHS, "Canadian Headache Society Guideline for Migraine Prophylaxis," [2012](#)

EHF/WHO, "Aids to Management of Headache Disorders in Primary Care," [2019](#)

Peripheral Arterial Disease:

AHA/ACC, "2016 Guideline on the Management of Patients with Lower Extremity Peripheral Artery Disease," [November 2016](#)

CCS, "Canadian Cardiovascular Society 2022 Guidelines for Peripheral Arterial Disease," [2022](#)

Prevention:

"AHA/ACCF Secondary Prevention and Risk Reduction Therapy for Patients with Coronary and Other Atherosclerotic Vascular Disease: 2011 Update," [November 2011](#)

Surgical and Procedural Management:

ACC/AHA, "2014 ACC/AHA Guideline on Perioperative Cardiovascular Evaluation and Management of Patients Undergoing Noncardiac Surgery," [August 2014](#)

ESC, "Guidelines on Cardiovascular Assessment and Management of Patients Undergoing Non-Cardiac Surgery," [August 2022](#)

Valvular Heart Disease:

AHA/ACC, "2014 AHA/ACC Guideline for the Management of Patients with Valvular Heart Disease," [March 2014](#)

Administration: Oral

Administer as a single daily dose and without regard to meals.

Administration: Enteral Feeding Tube

The following recommendations are based upon the best available evidence and clinical expertise. Senior editorial team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Pegg A. Guenter PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN.

Oral solution (commercially available):

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes: Dilute dose with at least an equivalent volume of purified water prior to administering to reduce osmolality and viscosity; draw up diluted solution into enteral dosing syringe and administer via feeding tube ^(Ref).

General guidance: Hold enteral nutrition (EN) during lisinopril administration ^(Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration ^(Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose ^(Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart EN ^(Ref).

Oral tablet:

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes (≥8 French): Crush tablet(s) into a fine powder and disperse in 10 to 20 mL purified water immediately prior to administration; draw up mixture into enteral dosing syringe and administer via feeding tube (^{Ref}).

Dosage form information: Additional flushes may be required; tablets may be poorly soluble in water (^{Ref}).

General guidance: Hold EN during lisinopril administration (^{Ref}). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration (^{Ref}). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose; extra rinsing may be required (^{Ref}). Flush feeding tube with an appropriate volume of purified water (eg, 30 mL) and restart EN (^{Ref}).

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Administration: Pediatric

The following feeding tube recommendations are based upon the best available evidence and clinical expertise. Senior editor panel: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter, PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang, MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN.

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Oral: May be administered without regard to food.

Oral solution (commercially available): Administer with an accurate measuring device (calibrated oral syringe or measuring cup); do not use a household teaspoon or tablespoon to measure dose (overdosage may occur).

Administration via feeding tube:

Gastric (eg, NG, G-tube) or post-pyloric tubes (eg, J-tube): Dilute dose with at least an equivalent volume of purified water prior to administering to reduce osmolality and viscosity; draw up diluted solution into enteral dosing syringe and administer via feeding tube (^{Ref}).

General guidance: Hold enteral nutrition during lisinopril administration (^{Ref}). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures (^{Ref}). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (^{Ref}). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition (^{Ref}).

Tablets:

Administration via feeding tube:

Gastric (eg, NG, G-tube) or post-pyloric tubes (eg, J-tube) (≥8 French): Crush tablet(s) into a fine powder and disperse in 10 to 20 mL purified water immediately prior to administration; draw up mixture into enteral dosing syringe and administer via feeding tube (^{Ref}).

Dosage form information: Additional flushes may be required; tablets may be poorly soluble in water (^{Ref}).

General guidance: Hold enteral nutrition during lisinopril administration (^{Ref}). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures (^{Ref}). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (^{Ref}). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition (^{Ref}).

Dietary Considerations

Use potassium-containing salt substitutes cautiously in patients with diabetes, patients with kidney impairment, or those maintained on potassium supplements or potassium-sparing diuretics.

Storage/Stability

Oral solution: Store at 20°C to 25°C (68°F to 77°F). Protect from freezing and excessive heat.

Tablet: Store at 15°C to 30°C (59°F to 86°F). Protect from moisture.

Extemporaneously Prepared

Note: A lisinopril oral solution (1 mg/mL) is commercially available (Qbrelis).

1 mg/mL Oral Suspension

A 1 mg/mL lisinopril oral suspension may be made with tablets and a 1:1 mixture of Ora-Plus and Ora-Sweet. Crush ten 10 mg tablets in a mortar and reduce to a fine powder. Add small portions of the vehicle and mix to a uniform paste; mix while adding the vehicle in incremental proportions to **almost** 100 mL; transfer to a graduated cylinder; rinse mortar with vehicle, and add quantity of vehicle sufficient to make 100 mL. Store in amber plastic prescription bottles; label "shake well." Stable for 13 weeks at room temperature or refrigerated (Nahata 2004).

Nahata MC and Morosco RS. Stability of lisinopril in two liquid dosage forms. *Ann Pharmacother*. 2004;38(3):396-399.[[PubMed 14742834](#)]

A 1 mg/mL lisinopril oral suspension may be made with tablets and a mixture of sodium citrate/citric acid oral solution or Cytra-2 diluent and Ora-Sweet SF. Place ten 20 mg tablets into an 8 ounce amber polyethylene terephthalate (PET) bottle and then add 10 mL purified water and shake for at least 1 minute. Gradually add 30 mL of sodium citrate/citric acid oral solution or Cytra-2 diluent and 160 mL of Ora-Sweet SF to the bottle and gently shake after each addition to disperse the contents. Store resulting suspension at ≤25°C (77°F) for up to 4 weeks. Label bottle "shake well" (Prinivil prescribing information 2016; Thompson 2003).

Prinivil (lisinopril) [prescribing information]. Whitehouse Station, NJ: Merck & Co., Inc; October 2016.

Thompson KC, Zhao Z, Mazakas JM, et al. Characterization of an extemporaneous liquid formulation of lisinopril. *Am J Health Syst Pharm*. 2003;60(1):69-74.[[PubMed 12533979](#)]

A 1 mg/mL lisinopril oral suspension also be made with tablets, methylcellulose 1% with parabens, and simple syrup NF. Crush ten 10 mg tablets in a mortar and reduce to a fine powder. Add 7.7 mL of methylcellulose gel and mix to a uniform paste; mix while adding the simple syrup in incremental proportions to **almost** 100 mL; transfer to a graduated cylinder; rinse mortar with vehicle, and add quantity of vehicle sufficient to make 100 mL. Store in amber plastic prescription bottles; label "shake well." Stable for 13 weeks refrigerated or 8 weeks at room temperature (Nahata 2004).

Nahata MC and Morosco RS. Stability of lisinopril in two liquid dosage forms. *Ann Pharmacother*. 2004;38(3):396-399.[[PubMed 14742834](#)]

2 mg/mL Syrup

A 2 mg/mL lisinopril syrup may be made with powder (Sigma Chemical Company, St. Louis, MO) and simple syrup. Dissolve 1 g of lisinopril powder in 30 mL of distilled water. Mix while adding simple syrup in incremental proportions in a quantity sufficient to make 500 mL. Label "shake well" and "refrigerate." Stable for 30 days when stored in amber plastic prescription bottles at room temperature or refrigerated. **Note:** Although no visual evidence of microbial growth was observed, the authors recommend refrigeration to inhibit microbial growth (Webster 1997).

Webster AA, English BA, and Rose DJ. The stability of lisinopril as an extemporaneous syrup. *Intr J Pharmaceut Compound*. 1997;1:352-353.

Medication Safety Issues

Sound-alike/look-alike issues:

Lisinopril may be confused with fosinopril, Lioresal, Lipitor, RisperDAL

Prinivil may be confused with Plendil, Pravachol, Prevacid, PriLOSEC, Proventil

Zestril may be confused with Desyrel, Restoril, Vistaril, Zegerid, Zerit, Zetia, Zostrix, ZyPREXA

Older Adult: High-Risk Medication:

Lisinopril is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥65 years of age) with hyperkalemia or severe symptomatic aortic stenosis (O'Mahony 2023).

International issues:

Acepril [Malaysia] may be confused with Accupril which is a brand name for quinapril [US]

Acepril: Brand name for lisinopril [Malaysia], but also the brand name for captopril [Great Britain]; enalapril [Hungary, Switzerland]

Contraindications

Hypersensitivity to lisinopril, other ACE inhibitors, or any component of the formulation; angioedema related to previous treatment with an ACE inhibitor; idiopathic or hereditary angioedema; concomitant use with aliskiren in patients with diabetes mellitus; coadministration with or within 36 hours of switching to or from a neprilysin inhibitor (eg, sacubitril).

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Canadian labeling: Additional contraindications (not in US labeling): Women who are pregnant, intend to become pregnant, or of childbearing potential and not using adequate contraception; breastfeeding; concomitant use with aliskiren, angiotensin receptor blockers (ARBs), or other ACE inhibitors in patients with moderate to severe kidney impairment (GFR <60 mL/minute/1.73 m²), hyperkalemia (>5 mmol/L), or with heart failure who are hypotensive; concomitant use with ARBs or other ACE inhibitors in diabetic patients with end organ damage; pediatric patients <6 years; pediatric patients 6 to 16 years of age with severe kidney impairment (GFR <60 mL/minute/1.73 m²).

Warnings/Precautions**Concerns related to adverse effects:**

- Hematologic effects: Patients with kidney impairment are at high risk of developing neutropenia. Patients with both kidney impairment and collagen vascular disease (eg, systemic lupus erythematosus) are at an even higher risk of developing neutropenia. Periodically monitor CBC with differential in these patients.
- Hypersensitivity reactions: Anaphylactic/anaphylactoid reactions can occur with angiotensin-converting enzyme (ACE) inhibitors. Severe anaphylactoid reactions may be seen during hemodialysis (eg, CVVHD) with high-flux dialysis membranes (eg, AN69), and rarely, during low density lipoprotein apheresis with dextran sulfate cellulose. Rare cases of anaphylactoid reactions have been reported in patients undergoing sensitization treatment with hymenoptera (bee, wasp) venom while receiving ACE inhibitors.
- Hypotension/syncope: Symptomatic hypotension with or without syncope can occur with ACE inhibitors (usually with the first several doses). Effects are most often observed in volume-depleted patients; correct volume depletion prior to initiation. Close monitoring of patients is required especially within the first few weeks of initial dosing and with dosing increases; blood pressure must be lowered at a rate appropriate for the patient's clinical condition. Although dose reduction may be necessary, hypotension is not a reason for discontinuation of future ACE inhibitor use especially in patients with heart failure where a reduction in systolic blood pressure is a desirable observation. Avoid use in hemodynamically unstable patients after acute myocardial infarction (MI).

Disease-related concerns:

- Aortic stenosis: Use with caution in patients with severe aortic stenosis; may reduce coronary perfusion resulting in ischemia.
- Ascites: Generally, avoid use in patients with ascites due to cirrhosis or refractory ascites; if use cannot be avoided in patients with ascites due to cirrhosis, monitor BP and kidney function carefully to avoid rapid development of kidney failure (AASLD [Biggins 2021]; AASLD [Runyon 2013]).

- Cardiovascular disease: Initiation of therapy in patients with ischemic heart disease or cerebrovascular disease warrants close observation due to the potential consequences posed by falling blood pressure (eg, MI, stroke). Fluid replacement, if needed, may restore blood pressure; therapy may then be resumed. Discontinue therapy in patients whose hypotension recurs. In a retrospective cohort study of patients ≥ 65 years of age with MI and impaired left ventricular function, administration of an ACE inhibitor was associated with a survival benefit, including patients with serum creatinine concentrations >3 mg/dL (265 micromol/L) (Frances 2000).
- Collagen vascular disease: Use with caution in patients with collagen vascular disease especially with concomitant kidney impairment; may be at increased risk for hematologic toxicity.
- Hypertrophic cardiomyopathy with left ventricular outflow tract obstruction: Use with caution in patients with hypertrophic cardiomyopathy and left ventricular outflow tract obstruction since reduction in afterload may worsen symptoms associated with this condition (AHA/ACC [Ommen 2020]).
- Kidney impairment: Use with caution in preexisting kidney insufficiency; dosage adjustment may be needed. Avoid rapid dosage escalation, which may lead to further kidney impairment.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Oral solution: Some dosage forms may contain sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982). Some data suggest that benzoate displaces bilirubin from protein-binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol derivative with caution in neonates. See manufacturer's labeling.

Special populations:

- Race/Ethnicity: In Black patients, the BP-lowering effects of ACE inhibitors may be less pronounced. The exact mechanism is not known; differences in the renin-angiotensin-aldosterone system, low renin levels, and salt sensitivity more commonly found in Black patients may contribute (Brewster 2013; Helmer 2018).
- Surgical patients: In patients on chronic ACE inhibitor therapy, intraoperative hypotension may occur with induction and maintenance of general anesthesia; use with caution before, during, or immediately after major surgery. Cardiopulmonary bypass, intraoperative blood loss, or vasodilating anesthesia increases endogenous renin release. Use of ACE inhibitors perioperatively will blunt angiotensin II formation and may result in hypotension. However, discontinuation of therapy prior to surgery is controversial. If continued preoperatively, avoidance of hypotensive agents during surgery is prudent (Hillis 2011). Based on current research and clinical guidelines in patients undergoing noncardiac surgery, continuing ACE inhibitors is reasonable in the perioperative period. If ACE inhibitors are held before surgery, it is reasonable to restart postoperatively as soon as clinically feasible (ACC/AHA [Fleisher 2014]).

* See [Cautions in AHFS Essentials](#) for additional information.

Older Adult Considerations

The AHA/ACC/ASH 2015 scientific statement on the treatment of hypertension in patients with coronary artery disease (CAD) warns to use caution and avoid decreases in diastolic blood pressure (DBP) <60 mm Hg especially in patients >60 years of age since reduced coronary perfusion may occur. When lowering systolic blood pressure in older hypertensive patients with wide pulse pressures, very low DBP values (<60 mm Hg) may result. In patients with obstructive CAD, clinicians should lower blood pressure slowly and carefully monitor for any untoward signs or symptoms, especially those resulting from myocardial ischemia and worsening heart failure (AHA/ACC/ASH [Rosendorff 2015]).

Reproductive Considerations

Avoid use of angiotensin-converting enzyme (ACE) inhibitor therapy in patients who may become pregnant and who are not using effective contraception (ADA 2023).

Medications considered acceptable for the treatment of chronic hypertension during pregnancy may generally be used in patients trying to conceive. Angiotensin-converting enzyme (ACE) inhibitors are fetotoxic. Transition patients prior to conception to an agent preferred for use during pregnancy unless treatment with an ACE inhibitor is absolutely necessary (ACC/AHA [Whelton

2018]; ACOG 2019; NICE 2019).

When ACE inhibitors are used for the treatment of proteinuric chronic kidney disease in patients who could become pregnant, discontinue use at the first positive pregnancy test (ADA 2023; Fakhouri 2023).

ACE inhibitors are not recommended for the treatment of heart failure in patients planning to become pregnant (AHA/ACC/HFSA [Heidenreich 2022]).

Lisinopril may be effective for prevention of migraines. In general, preventive treatment for migraine in patients trying to become pregnant should be avoided. Options for patients planning a pregnancy should be considered as part of a shared decision-making process. Nonpharmacologic interventions should be considered initially. When needed, preventive treatment should be individualized considering the available safety data and needs of the patient should pregnancy occur. A gradual discontinuation of preventive medications is generally preferred when the decision is made to stop treatment prior to conception (ACOG 2022; AHS [Ailani 2021]). When lisinopril is used for migraine prevention, treatment should be discontinued prior to attempting to conceive (ACOG 2022).

Pregnancy Considerations

Lisinopril crosses the placenta (Bhatt-Mehta 1993; Filler 2003).

Drugs that act on the renin-angiotensin system can cause injury and death to the developing fetus. Exposure to an angiotensin-converting enzyme (ACE) inhibitor during the first trimester of pregnancy may be associated with an increased risk of fetal malformations (ACOG 2019; ESC [Regitz-Zagrosek 2018]). Following exposure during the second or third trimesters, drugs that act on the renin-angiotensin system are associated with oligohydramnios. Oligohydramnios, due to decreased fetal renal function, may lead to fetal lung hypoplasia and skeletal malformations. Oligohydramnios may not appear until after an irreversible fetal injury has occurred. ACE inhibitor use during pregnancy is also associated with anuria, hypotension, renal failure, skull hypoplasia, and death in the fetus/neonate. Monitor infants exposed to an ACE inhibitor in utero for hyperkalemia, hypotension, and oliguria. Exchange transfusions or dialysis may be required to reverse hypotension or improve renal function.

Chronic maternal hypertension is also associated with adverse events in the fetus/infant. Chronic maternal hypertension may increase the risk of birth defects, low birth weight, premature delivery, stillbirth, and neonatal death. Actual fetal/neonatal risks may be related to the duration and severity of maternal hypertension. Untreated chronic hypertension may also increase the risks of adverse maternal outcomes, including gestational diabetes, pre-eclampsia, delivery complications, stroke and myocardial infarction (ACOG 2019).

Discontinue ACE inhibitors as soon as possible once pregnancy is detected. Agents other than ACE inhibitors are recommended for the treatment of chronic hypertension during pregnancy (ACOG 2019; ESC [Cífková 2020]; SOGC [Magee 2022]). Consider the use of ACE inhibitors only for pregnant patients with hypertension refractory to other medications (ACOG 2019). Closely monitor pregnant patients on ACE inhibitors with serial ultrasounds.

ACE inhibitors are not recommended for the treatment of heart failure or proteinuric chronic kidney disease during pregnancy (AHA/ACC/HFSA [Heidenreich 2022]; ESC [Regitz-Zagrosek 2018]; Fakhouri 2023).

In general, preventive treatment for migraine should be avoided during pregnancy. Options for pregnant patients should be considered as part of a shared decision-making process. Nonpharmacologic interventions should be considered initially. When needed, preventive treatment should be individualized considering the available safety data, the potential for adverse maternal and fetal events, and needs of the patient (ACOG 2022; AHS [Ailani 2021]). Lisinopril is not recommended for migraine prevention during pregnancy (ACOG 2022).

Breastfeeding Considerations

It is not known if lisinopril is present in breast milk.

Due to the potential for serious adverse reactions in the breastfed infant, the manufacturer recommends a decision be made whether to discontinue breastfeeding or to discontinue lisinopril.

In general, preventive treatment for migraine in lactating patients should be avoided. When needed, therapy should be individualized considering the available safety data and needs of the patient (AHS [Ailani 2021]). When postpartum treatment with an ACE inhibitor is needed, consider use of an agent other than lisinopril (ESC [Cífková 2020]). Avoid breastfeeding if high maternal doses of an ACE inhibitor are needed (ACOG 2019).

Briggs' Drugs in Pregnancy & Lactation

- [Lisinopril](#)

Adverse Reactions (Significant): Considerations

Acute kidney injury

Use may be associated with **increased blood urea nitrogen** and **increased serum creatinine**, resulting in oliguria and **acute kidney injury** (AKI). Increases in serum creatinine are expected and usually stabilize within 20% to 30% of baseline; higher increases may indicate high efferent tone (such as with hypovolemia, congestive heart failure, or renal artery stenosis) ^(Ref).

Mechanism: Related to pharmacologic action; inhibits efferent arteriolar vasoconstriction, lowering glomerular filtration pressure, which can lead to a reduction in glomerular filtration rate (GFR). Kidney hypoperfusion from hypotension may also occur ^(Ref).

Onset: Intermediate; increases in serum creatinine generally occur within 2 weeks of initiation and stabilize within 2 to 4 weeks ^(Ref). However, more immediate increases may occur in patients with other risk factors for AKI ^(Ref).

Risk factors:

- Patients with low renal blood flow whose GFR is dependent on efferent arteriolar vasoconstriction by angiotensin II ^(Ref):
 - Low effective circulating volume (sodium or volume depletion)
 - Congestive heart failure
 - Hypotension or shock
 - Renal artery stenosis
- High-dose initiation (especially with concomitant diuretic use) ^(Ref)
- Older patients ^(Ref)
- Preexisting kidney impairment ^(Ref)
- Concurrent diuretic and/or nonsteroidal anti-inflammatory drug use ^(Ref)

Angioedema

Angioedema may occur rarely; edema may manifest in the head and neck (potentially compromising the airway) or the intestine (presenting with abdominal pain). Use is contraindicated in patients with idiopathic or hereditary angioedema or previous angioedema associated with any angiotensin-converting enzyme inhibitors or neprilysin inhibitors ^(Ref).

Mechanism: Related to pharmacologic action (ie, increased bradykinin and Substance P, vascular permeability, and vasodilation) ^(Ref).

Onset: Varied; may occur at any time during treatment. Most cases occur within the first week of therapy but may also occur after years of therapy ^(Ref).

Risk factors:

- Black patients (estimated 4- to 5-fold higher risk); the mechanism for this is not completely understood but may be related to genetic variants ^(Ref)
- Females ^(Ref)

- Smoking history ^(Ref)
- Previous history of angioedema ^(Ref)
- Age >65 years ^(Ref)
- Seasonal allergies ^(Ref)
- Concurrent use of mechanistic target of rapamycin (mTOR) inhibitors (eg, everolimus) ^(Ref)
- Concurrent use of neprilysin inhibitor (contraindicated)

Cough

A dry, hacking, nonproductive **cough** that is typically associated with tickling or scratching in the throat may occur with angiotensin converting enzyme inhibitors (ACEI), including lisinopril in adult and pediatric patients ^(Ref). Recurrence is likely with rechallenge ^(Ref). Resolution of cough typically occurs 1 to 4 weeks after ACEI discontinuation but may persist for up to 3 months ^(Ref).

Mechanism: Various proposed mechanisms. May be related to the pharmacologic action (ie, increased bradykinin and substance P, resulting in accumulation in the lungs and bronchoconstriction) ^(Ref).

Onset: Variable; within hours to 4 weeks after initiation but can be delayed for up to 6 months ^(Ref).

Risk factors:

- Females ^(Ref)
- Possibly certain genetic variants (some of which may be independent of the bradykinin pathway) ^(Ref)

Hyperkalemia

Hyperkalemia (elevated serum potassium) may occur with angiotensin converting enzyme inhibitors (ACEI), including lisinopril.

Mechanism: Related to pharmacologic action; inhibits formation of circulating angiotensin II, which leads to efferent arteriole vasodilation and subsequent lowering of GFR, which lowers potassium elimination. Additionally, interferes with generation and release of aldosterone from the adrenal cortex, leading to an impairment of potassium excretion from the kidney ^(Ref).

Risk factors:

- Disease states associated with hyperkalemia (congestive heart failure, diabetes mellitus, chronic kidney disease) ^(Ref)
- Concurrent use of medications which cause hyperkalemia (ACEI, ARBs, spironolactone, NSAIDs, beta blockers, heparin, tacrolimus, cyclosporine) ^(Ref)
- Acute kidney injury (elevated BUN/serum creatinine) ^(Ref)
- High dietary intake of potassium or concurrent use of potassium supplements (including potassium-containing salt substitutes) ^(Ref)
- Baseline elevated potassium (≥ 5 mmol/L) ^(Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%:

Cardiovascular: Hypotension (4% to 11%)

Nervous system: Dizziness (5% to 19%)

1% to 10%:

Cardiovascular: Flushing, orthostatic hypotension, syncope (5% to 7%), vasculitis

Dermatologic: Alopecia, diaphoresis, erythema of skin, pruritus, skin photosensitivity, skin rash, Stevens-Johnson syndrome, toxic epidermal necrolysis, urticaria

Endocrine & metabolic: Diabetes mellitus, hyperkalemia (2% to 6%; incidence varies in literature and may be impacted by comorbidities) ^(Ref), SIADH

Gastrointestinal: Constipation, diarrhea, dysgeusia, flatulence, pancreatitis, xerostomia

Genitourinary: Impotence

Hematologic & oncologic: Bone marrow depression, eosinophilia, hemolytic anemia, increased erythrocyte sedimentation rate, leukocytosis, leukopenia, neutropenia, positive ANA titer, pseudolymphoma (cutaneous), thrombocytopenia

Nervous system: Altered sense of smell, asthenia, fatigue, paresthesia, vertigo

Neuromuscular & skeletal: Arthralgia, arthritis, gout, myalgia

Ophthalmic: Blurred vision, diplopia, photophobia, vision loss

Otic: Tinnitus

Renal: Increased blood urea nitrogen ($\leq 2\%$; transient), increased serum creatinine ($\leq 10\%$; transient), renal insufficiency (in patients with acute myocardial infarction: 1% to 2%)

Respiratory: Cough (4%; literature suggests incidence may be as high as 20%) ^(Ref) ([table 1](#))

Miscellaneous: Fever

Postmarketing:

Dermatologic: Psoriasis ^(Ref)

Endocrine & metabolic: Hyponatremia ^(Ref)

Hematologic & oncologic: Aplastic anemia ^(Ref), pancytopenia ^(Ref)

Hepatic: Cholestatic hepatitis ^(Ref), fulminant hepatitis ^(Ref), hepatocellular hepatitis ^(Ref)

Hypersensitivity: Angioedema ^(Ref)

Nervous system: Confusion, mood changes (including depressive symptoms), visual hallucination ^(Ref)

Renal: Acute kidney injury ^(Ref)

* See [Cautions in AHFS Essentials](#) for additional information.

Allergy and Idiosyncratic Reactions

- [ACE Inhibitor Allergy/Hypersensitivity](#)

Toxicology

- [Angiotensin-Converting Enzyme \(ACE\) Inhibitors](#)

Metabolism/Transport Effects

None known.

Drug Interactions

[Open Interactions](#)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Open Interactions" button above.

Alfuzosin: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Aliskiren: May enhance the hyperkalemic effect of Angiotensin-Converting Enzyme Inhibitors. Aliskiren may enhance the hypotensive effect of Angiotensin-Converting Enzyme Inhibitors. Aliskiren may enhance the nephrotoxic effect of Angiotensin-Converting Enzyme Inhibitors. Management: Aliskiren use with ACEIs or ARBs in patients with diabetes is contraindicated. Combined use in other patients should be avoided, particularly when CrCl is less than 60 mL/min. If combined, monitor potassium, creatinine, and blood pressure closely. *Risk D: Consider therapy modification*

- Allopurinol: Angiotensin-Converting Enzyme Inhibitors may enhance the potential for allergic or hypersensitivity reactions to Allopurinol. *Risk C: Monitor therapy*
- Alteplase: Angiotensin-Converting Enzyme Inhibitors may enhance the adverse/toxic effect of Alteplase. Specifically, the risk for angioedema may be increased. *Risk C: Monitor therapy*
- Amifostine: Blood Pressure Lowering Agents may enhance the hypotensive effect of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider therapy modification*
- Amphetamines: May diminish the antihypertensive effect of Antihypertensive Agents. *Risk C: Monitor therapy*
- Angiotensin II: Angiotensin-Converting Enzyme Inhibitors may enhance the therapeutic effect of Angiotensin II. *Risk C: Monitor therapy*
- Angiotensin II Receptor Blockers: May enhance the adverse/toxic effect of Angiotensin-Converting Enzyme Inhibitors. Angiotensin II Receptor Blockers may increase the serum concentration of Angiotensin-Converting Enzyme Inhibitors. Management: Use of telmisartan and ramipril is not recommended. It is not clear if any other combination of an ACE inhibitor and an ARB would be any safer. Consider alternatives when possible. Monitor blood pressure, renal function, and potassium if combined. *Risk D: Consider therapy modification*
- Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may enhance the hypotensive effect of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor therapy*
- Aprotinin: May diminish the antihypertensive effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*
- Arginine: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*
- AzaTHIOprine: Angiotensin-Converting Enzyme Inhibitors may enhance the myelosuppressive effect of AzaTHIOprine. *Risk C: Monitor therapy*
- Barbiturates: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*
- Benperidol: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*
- Brigatinib: May diminish the antihypertensive effect of Antihypertensive Agents. Brigatinib may enhance the bradycardic effect of Antihypertensive Agents. *Risk C: Monitor therapy*
- Brimonidine (Topical): May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*
- Bromperidol: May diminish the hypotensive effect of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may enhance the hypotensive effect of Bromperidol. *Risk X: Avoid combination*
- Dapoxetine: May enhance the orthostatic hypotensive effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*
- Dexmethylphenidate: May diminish the therapeutic effect of Antihypertensive Agents. *Risk C: Monitor therapy*
- Diazoxide: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*
- Dipeptidyl Peptidase-IV Inhibitors: May enhance the adverse/toxic effect of Angiotensin-Converting Enzyme Inhibitors. Specifically, the risk of angioedema may be increased. *Risk C: Monitor therapy*
- Drospirenone-Containing Products: May enhance the hyperkalemic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*
- DULoxetine: Blood Pressure Lowering Agents may enhance the hypotensive effect of DULoxetine. *Risk C: Monitor therapy*
- Eplerenone: May enhance the hyperkalemic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*
- Everolimus: May enhance the adverse/toxic effect of Angiotensin-Converting Enzyme Inhibitors. Specifically, the risk of angioedema may be increased. *Risk C: Monitor therapy*
- Ferric Gluconate: Angiotensin-Converting Enzyme Inhibitors may enhance the adverse/toxic effect of Ferric Gluconate. *Risk C: Monitor therapy*
- Ferric Hydroxide Polymaltose Complex: Angiotensin-Converting Enzyme Inhibitors may enhance the adverse/toxic effect of Ferric Hydroxide Polymaltose Complex. Specifically, the risk for angioedema or allergic reactions may be increased. *Risk C: Monitor therapy*

Finerenone: Angiotensin-Converting Enzyme Inhibitors may enhance the hyperkalemic effect of Finerenone. *Risk C: Monitor therapy*

Flunarizine: May enhance the therapeutic effect of Antihypertensive Agents. *Risk C: Monitor therapy*

Gelatin (Succinylated): Angiotensin-Converting Enzyme Inhibitors may enhance the adverse/toxic effect of Gelatin (Succinylated). Specifically, the risk of a paradoxical hypotensive reaction may be increased. *Risk C: Monitor therapy*

Grass Pollen Allergen Extract (5 Grass Extract): Angiotensin-Converting Enzyme Inhibitors may enhance the adverse/toxic effect of Grass Pollen Allergen Extract (5 Grass Extract). Specifically, ACE inhibitors may increase the risk of severe allergic reaction to Grass Pollen Allergen Extract (5 Grass Extract). *Risk X: Avoid combination*

Green Tea: May decrease the serum concentration of Lisinopril. *Risk C: Monitor therapy*

Heparin: May enhance the hyperkalemic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Heparins (Low Molecular Weight): May enhance the hyperkalemic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Herbal Products with Blood Pressure Increasing Effects: May diminish the antihypertensive effect of Antihypertensive Agents. *Risk C: Monitor therapy*

Herbal Products with Blood Pressure Lowering Effects: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may enhance the hypotensive effect of Hypotension-Associated Agents. *Risk C: Monitor therapy*

Icatibant: May diminish the antihypertensive effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Iloperidone: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Indoramin: May enhance the hypotensive effect of Antihypertensive Agents. *Risk C: Monitor therapy*

Iron Dextran Complex: Angiotensin-Converting Enzyme Inhibitors may enhance the adverse/toxic effect of Iron Dextran Complex. Specifically, patients receiving an ACE inhibitor may be at an increased risk for anaphylactic-type reactions. *Risk C: Monitor therapy*

Isocarboxazid: May enhance the antihypertensive effect of Antihypertensive Agents. *Risk X: Avoid combination*

Lanthanum: May decrease the serum concentration of Angiotensin-Converting Enzyme Inhibitors. Management: Administer angiotensin-converting enzyme (ACE) inhibitors at least two hours before or after lanthanum. *Risk D: Consider therapy modification*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may enhance the hypotensive effect of Levodopa-Foslevodopa. *Risk C: Monitor therapy*

Lithium: Angiotensin-Converting Enzyme Inhibitors may increase the serum concentration of Lithium. Management: Lithium dosage reductions will likely be needed following the addition of an ACE inhibitor. Monitor for increased concentrations/toxic effects of lithium if an ACE inhibitor is initiated/dose increased, or if switching between ACE inhibitors. *Risk D: Consider therapy modification*

Loop Diuretics: May enhance the hypotensive effect of Angiotensin-Converting Enzyme Inhibitors. Loop Diuretics may enhance the nephrotoxic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Lormetazepam: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Methylphenidate: May diminish the antihypertensive effect of Antihypertensive Agents. *Risk C: Monitor therapy*

Molsidomine: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Naftopidil: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Nicergoline: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Nicorandil: May enhance the hyperkalemic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Nicorandil: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Nitroprusside: Blood Pressure Lowering Agents may enhance the hypotensive effect of Nitroprusside. *Risk C: Monitor therapy*

Nonsteroidal Anti-Inflammatory Agents: Angiotensin-Converting Enzyme Inhibitors may enhance the adverse/toxic effect of Nonsteroidal Anti-Inflammatory Agents. Specifically, the combination may result in a significant decrease in renal function. Nonsteroidal Anti-Inflammatory Agents may diminish the antihypertensive effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Nonsteroidal Anti-Inflammatory Agents (Topical): May diminish the therapeutic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Obinutuzumab: May enhance the hypotensive effect of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider therapy modification*

Ocreotide: May increase the serum concentration of Lisinopril. *Risk C: Monitor therapy*

Pentoxifylline: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Perazine: May enhance the hypotensive effect of Antihypertensive Agents. *Risk C: Monitor therapy*

Pholcodine: Blood Pressure Lowering Agents may enhance the hypotensive effect of Pholcodine. *Risk C: Monitor therapy*

Phosphodiesterase 5 Inhibitors: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Polyethylene Glycol-Electrolyte Solution: Angiotensin-Converting Enzyme Inhibitors may enhance the nephrotoxic effect of Polyethylene Glycol-Electrolyte Solution. *Risk C: Monitor therapy*

Potassium Salts: May enhance the hyperkalemic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Potassium-Sparing Diuretics: May enhance the hyperkalemic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Prazosin: Antihypertensive Agents may enhance the hypotensive effect of Prazosin. *Risk C: Monitor therapy*

Pregabalin: Angiotensin-Converting Enzyme Inhibitors may enhance the adverse/toxic effect of Pregabalin. Specifically, the risk of angioedema may be increased. *Risk C: Monitor therapy*

Prostacyclin Analogues: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Quinagolide: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Racecadotril: May enhance the adverse/toxic effect of Angiotensin-Converting Enzyme Inhibitors. Specifically, the risk for angioedema may be increased with this combination. *Risk C: Monitor therapy*

Ranolazine: May enhance the adverse/toxic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Sacubitril: Angiotensin-Converting Enzyme Inhibitors may enhance the adverse/toxic effect of Sacubitril. Specifically, the risk of angioedema may be increased with this combination. *Risk X: Avoid combination*

Salicylates: May enhance the nephrotoxic effect of Angiotensin-Converting Enzyme Inhibitors. Salicylates may diminish the therapeutic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Silodosin: May enhance the hypotensive effect of Blood Pressure Lowering Agents. *Risk C: Monitor therapy*

Sirolimus Products: May enhance the adverse/toxic effect of Angiotensin-Converting Enzyme Inhibitors. Specifically, the risk for angioedema may be increased. *Risk C: Monitor therapy*

Sodium Phosphates: Angiotensin-Converting Enzyme Inhibitors may enhance the nephrotoxic effect of Sodium Phosphates. Specifically, the risk of acute phosphate nephropathy may be enhanced. *Risk C: Monitor therapy*

Sparsentan: May enhance the adverse/toxic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk X: Avoid combination*

Tacrolimus (Systemic): Angiotensin-Converting Enzyme Inhibitors may enhance the hyperkalemic effect of Tacrolimus (Systemic). *Risk C: Monitor therapy*

Temsirolimus: May enhance the adverse/toxic effect of Angiotensin-Converting Enzyme Inhibitors. Specifically, the risk of angioedema may be increased. *Risk C: Monitor therapy*

Terazosin: Antihypertensive Agents may enhance the hypotensive effect of Terazosin. *Risk C: Monitor therapy*

Thiazide and Thiazide-Like Diuretics: May enhance the hypotensive effect of Angiotensin-Converting Enzyme Inhibitors. Thiazide and Thiazide-Like Diuretics may enhance the nephrotoxic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Tolvaptan: May enhance the hyperkalemic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Trimethoprim: May enhance the hyperkalemic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Urapidil: May interact via an unknown mechanism with Angiotensin-Converting Enzyme Inhibitors. Management: Avoid concomitant use of urapidil and angiotensin-converting enzyme (ACE) inhibitors. *Risk D: Consider therapy modification*

Urokinase: May enhance the adverse/toxic effect of Angiotensin-Converting Enzyme Inhibitors. Specifically, the risk of angioedema may be increased. *Risk C: Monitor therapy*

Test Interactions

May lead to false-negative aldosterone/renin ratio (ARR) (Funder 2016)

Monitoring Parameters

Blood pressure; BUN; serum creatinine; electrolytes (eg, potassium [especially in patients on concomitant potassium-sparing diuretics, potassium supplements, and/or potassium-containing salts]); consider baseline LFTs (if preexisting liver impairment); monitor for jaundice or signs of liver failure; if patient has collagen vascular disease and/or kidney impairment, periodically monitor CBC with differential. If angioedema is suspected, assess risk of airway obstruction (eg, involvement of tongue, glottis, larynx, and/or history of airway surgery).

Nursing Practice Points

Boxed warning information: This drug has a boxed warning.

Nursing care and monitoring:

- Monitor patient for:
 - Hypotension/orthostatic hypotension, flushing

Additional patient education: Teach patient:

- To notify doctor right away if pregnancy is planned or suspected.
- About use, risk of falls, and to avoid operating machinery including motor vehicles.
- About the importance of staying well hydrated.

Note: This information does not cover all factors that could impact the safe and effective use of the medication. Refer to the full drug monograph for more details.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Oral:

Qbrelis: 1 mg/mL (150 mL) [contains sodium benzoate]

Tablet, Oral:

Prinivil: 20 mg [DSC]

Zestril: 2.5 mg

Zestril: 5 mg [scored]

Zestril: 10 mg, 20 mg, 30 mg, 40 mg

Generic: 2.5 mg, 5 mg, 10 mg, 20 mg, 30 mg, 40 mg

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Tablet, Oral:

Zestril: 5 mg, 10 mg, 20 mg [contains corn starch]

Generic: 5 mg, 10 mg, 20 mg

Generic Available (US)

May be product dependent

Pricing: US

Solution (Qbrelis Oral)

1 mg/mL (per mL): \$5.01

Tablets (Lisinopril Oral)

2.5 mg (per each): \$0.01 - \$0.65

5 mg (per each): \$0.02 - \$0.97

10 mg (per each): \$0.01 - \$1.00

20 mg (per each): \$0.03 - \$1.08

30 mg (per each): \$0.04 - \$1.51

40 mg (per each): \$0.04 - \$1.57

Tablets (Zestril Oral)

2.5 mg (per each): \$15.95

5 mg (per each): \$15.95

10 mg (per each): \$15.95

20 mg (per each): \$15.95

30 mg (per each): \$15.95

40 mg (per each): \$15.95

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Mechanism of Action

Competitive inhibitor of angiotensin-converting enzyme (ACE); prevents conversion of angiotensin I to angiotensin II, a potent vasoconstrictor; results in lower levels of angiotensin II which causes an increase in plasma renin activity and a reduction in aldosterone secretion; a CNS mechanism may also be involved in hypotensive effect as angiotensin II increases adrenergic outflow from CNS; vasoactive kallikreins may be decreased in conversion to active hormones by ACE inhibitors, thus reducing blood pressure

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: 1 hour; Peak effect: Hypotensive: Oral: ~6 hours

Duration: 24 hours

Metabolism: Not metabolized

Bioavailability:

Pediatric patients:

2 to 15 years: 20% to 36% (Hogg 2007)

6 to 16 years: ~28%

Adults: ~25% (range: 6% to 60%); decreased to 16% with NYHA Class II-IV heart failure

Half-life elimination: 12 hours

Time to peak:

Pediatric patients 6 months to 15 years: Median (range): 5 to 6 hours (Hogg 2007)

Adults: ~7 hours

Excretion: Primarily urine (as unchanged drug)

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Decreased elimination when glomerular filtration rate is <30 mL/minute. With greater impairment, peak and trough lisinopril levels increase, time to peak concentration increases, and time to attain steady state is prolonged.

Dental: Local Anesthetic/Vasoconstrictor Precautions

No information available to require special precautions

Dental: Effects on Dental Treatment

Key adverse event(s) related to dental treatment: Cough, Stevens-Johnson syndrome, dysgeusia, and xerostomia (normal salivary flow resumes upon discontinuation); other risk factors for xerostomia include use of oral desiccants (alcohol, coffee, cannabis, tobacco), urinary incontinence and depression medications, and poor hydration. Orthostatic hypotension may occur; use caution with sudden changes in position during and after dental treatment.

An angiotensin-converting enzyme inhibitor cough is a dry, hacking, nonproductive cough that can potentially interfere with longer dental procedures if patient has this side effect.

Dental: Effects on Bleeding

No information available to require special precautions

Related Information

- [Angiotensin Agents](#)

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Brand Names: US

Prinivil [DSC]; Qbrelis; Zestril

Brand Names: Canada

APO-Lisinopril; Auro-Lisinopril; Lisinopril (Type Z); PRO-Lisinopril-10; PRO-Lisinopril-20; PRO-Lisinopril-5; SANDOZ Lisinopril [DSC]; TARO-Lisinopril; TEVA-Lisinopril (Type P) [DSC]; TEVA-Lisinopril (Type Z); Zestril

Last Updated 12/12/24

Therapeutic Cases for AI Trial

3. A 35-year-old pregnant female was diagnosed with GDM during her 20-week pregnancy. Her gynecologist started her on insulin, but she has a fear of injections. She requested an oral route medication to manage her GDM. What would be the best option?
ANSWER: Metformin 500mg BID.

MetFORMIN (Lexi-Drugs Multinational)

Monograph

Images

Adult Patient Education

Pediatric Patient Education

ALERT: US Boxed Warning

Lactic acidosis

Postmarketing cases of metformin-associated lactic acidosis have resulted in death, hypothermia, hypotension, and resistant bradyarrhythmias. The onset of metformin-associated lactic acidosis is often subtle, accompanied only by nonspecific symptoms such as malaise, myalgias, respiratory distress, somnolence, and abdominal pain. Metformin-associated lactic acidosis was characterized by elevated blood lactate levels (>5 mmol/L), anion gap acidosis (without evidence of ketonuria or ketonemia), and increased lactate/pyruvate ratio; and metformin plasma levels generally >5 mcg/mL.

Risk factors for metformin-associated lactic acidosis include renal impairment, concomitant use of certain drugs (eg, carbonic anhydrase inhibitors such as topiramate), ≥ 65 years, having a radiological study with contrast, surgery and other procedures, hypoxic states (eg, acute congestive heart failure), excessive alcohol intake, and hepatic impairment.

Steps to reduce the risk of and manage metformin-associated lactic acidosis in these high-risk groups are provided in the full prescribing information.

If metformin-associated lactic acidosis is suspected, immediately discontinue metformin and institute general supportive measures in a hospital setting. Prompt hemodialysis is recommended.

Brand Names: International

Find brand name(s) by country (for United States and Canada see separate Brand Names sections)

 Enter a Country or Country Code

For country code abbreviations ( [show table](#))

International Nonproprietary Names (INN)

Metformin [English]; Metformina [Spanish]; Metformine [French]; Metforminum [Latin]; Метформин [Russian]; مينفورمين [Arabic]; 二甲双胍 [Chinese]

Brazilian Nonproprietary Names (DCB)

Metformina

Japanese Accepted Name (JAN)

メトホルミン塩酸塩

Anatomic Therapeutic Chemical (ATC) Classification

- A10BA02

Pharmacologic Category

[Antidiabetic Agent, Biguanide](#)

Dosing: Adult

Antipsychotic-induced weight gain, treatment

Antipsychotic-induced weight gain, treatment (off-label use):

Note: When used as an early intervention strategy, the goal is plateau of weight gain; however, reversal of weight gain may also be possible. With weight loss, goals of at least 5% of baseline body weight within 6 months are suggested (^{Ref}).

Immediate release: **Oral:** Initial: 500 mg once or twice daily; to minimize GI effects gradually increase dose based on response and tolerability in increments of 500 mg every 1 to 2 weeks up to a target dose of 1 g twice daily (^{Ref}).

Extended release: **Oral:** 500 mg once daily; to minimize GI effects gradually increase dose based on response and tolerability in increments of 500 mg every 1 to 2 weeks up to a target dose of 2 g once daily (^{Ref}).

Diabetes mellitus, type 2, prevention

Diabetes mellitus, type 2, prevention (off-label use):

Note: For select patients with prediabetes, particularly for those with BMI ≥ 35 kg/m², age <60 years, and patients with prior gestational diabetes mellitus, in whom lifestyle interventions fail to improve glycemic indices (^{Ref}).

Immediate release: **Oral:** Initial: 850 mg once daily for 1 month, then increase to 850 mg twice daily; unless GI adverse effects warrant a longer titration period (^{Ref}).

Diabetes mellitus, type 2, treatment

Diabetes mellitus, type 2, treatment:

Note: In patients in whom glycemic targets are not met despite diet, exercise, and metformin, combination therapy is necessary to achieve optimal results (^{Ref}).

Immediate release:

Initial: **Oral:** 500 mg once or twice daily **or** 850 mg once daily (^{Ref}).

Dosage adjustments: **Oral:** The dose should be increased gradually to minimize GI adverse effects. Titration strategies vary widely, but usually done in 500 mg or 850 mg increments every 7 days (range: 5 days to 1 month).

Usual maintenance dosage: **Oral:** 1 g twice daily **or** 850 mg twice daily (^{Ref}).

Maximum: **Oral:** 2.55 g/day. Modest additional benefit has been observed with doses up to ~2.5 g/day; however, GI adverse effects may limit use (^{Ref}). If doses >2 g/day are needed, consider administering in 3 divided doses to minimize GI adverse effects.

Extended release:

Initial: **Oral:** 500 mg to 1 g once daily

Dosage adjustments: **Oral:** The dose should be increased gradually to minimize GI adverse effects. Titration strategies vary widely, but usually done in 500 mg increments every 7 days (range: 7 days to 6 weeks).

Maximum: **Oral:** 2 g/day. If glycemic control is not achieved at maximum dose given once daily, may divide maximum dose and administer twice daily.

Gestational diabetes mellitus, treatment

Gestational diabetes mellitus, treatment (alternative agent) (off-label use): Immediate release: **Oral:** Initial: 500 mg once or twice daily; increase dosage to meet glycemic targets, typically over 1 to 2 weeks, up to a maximum of 2 to 2.5 g daily in 2 to 3 divided doses. If targets not achieved with metformin alone, insulin may be added (^{Ref}). **Note:** Insulin is the preferred medication for gestational diabetes as it does not cross the placenta to a measurable extent; all oral agents lack long-term safety data (^{Ref}).

Ovarian hyperstimulation syndrome prevention in patients with polycystic ovary syndrome undergoing in vitro fertilization/intracytoplasmic sperm injection

Ovarian hyperstimulation syndrome prevention in patients with polycystic ovary syndrome undergoing in vitro fertilization/intracytoplasmic sperm injection (alternative agent) (off-label use):

Note: For use prior to and/or during ovarian stimulation in patients undergoing a gonadotropin-releasing hormone agonist protocol ^(Ref).

Immediate release: **Oral:** Initial: 500 mg once daily; increase dose gradually as tolerated up to 2 g/day in divided doses ^(Ref). Dosage range studied in trials: 1 to 2.55 g/day in 2 or 3 divided doses; use of ER products may reduce GI adverse effects ^(Ref). Discontinue metformin with a positive pregnancy test ^(Ref).

Metformin conversion recommendations:

Conversion from IR to ER dosage forms: Patients receiving metformin immediate-release may be switched to metformin extended-release once daily at the same total daily dose, up to 2 g once daily. However, in patients who are doing well with immediate-release metformin, some experts recommend they continue using it, as there is little additional benefit documented with ER tablets ^(Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

* See [Dosage and Administration in AHFS Essentials](#) for additional information.

Dosing: Older Adult

Refer to adult dosing. The initial and maintenance dosing should be conservative, due to the potential for decreased renal function (monitor).

Dosing: Altered Kidney Function: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function:

eGFR ≥ 60 mL/minute/1.73 m²: No dosage adjustment necessary. Monitor renal function at least annually.

eGFR >45 to <60 mL/minute/1.73 m²: No dosage adjustment necessary. Metformin plasma concentrations may be higher compared to patients with an eGFR ≥ 60 mL/minute/1.73 m²; increase monitoring of renal function (eg, every 3 to 6 months) ^(Ref).

eGFR 30 to 45 mL/minute/1.73 m²:

Initiation of therapy: Use generally not recommended ^(Ref); however, initial therapy with 500 mg once daily with the evening meal titrated to 500 mg twice daily, if tolerated, with close monitoring of kidney function has been recommended by some experts ^(Ref).

Continuation of existing therapy: May continue at a reduced dose up to a maximum of 500 mg twice daily with close monitoring of kidney function ^(Ref).

eGFR <30 mL/minute/1.73 m²: Use is contraindicated.

Acute kidney injury during therapy: If acute kidney injury occurs or if risk factors are present (eg, severe vomiting or diarrhea), instruct patient to temporarily hold metformin ^(Ref).

Dosing: Altered Liver Function: Adult

The hepatic dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: Key pharmacokinetic changes in hepatic insufficiency: Metformin is not metabolized by the liver and is primarily eliminated unchanged by the kidney. The liver is the primary site of lactic acid metabolism and lactic acid-producing events, such as active alcohol consumption, dehydration, hypotension, sepsis, reduced cardiac function, or reduced kidney function, may enhance accumulation of lactic acid. Metformin increases lactic acid production in a dose-dependent manner and may result in overt lactic acidosis when coupled with liver dysfunction ^(Ref).

Hepatic impairment prior to treatment initiation:

Initial or dose titration in patients with preexisting liver cirrhosis:

Note: In patients with concurrent kidney impairment, eGFR ≤ 45 mL/minute/1.73 m² and Child-Turcotte-Pugh A or B, defer to dosing in altered kidney function in adults ^(Ref). Monitor kidney function frequently (eg, every 1 to 3 months) (expert opinion) during continuation of therapy ^(Ref).

Child-Turcotte-Pugh class A: No dosage adjustment necessary ^(Ref).

Child-Turcotte-Pugh class B: 500 mg once daily; may increase by ≤ 500 mg/day increments every 30 days based on tolerability and response (expert opinion); consider slower titration (eg, every 60 days) in patients on concurrent agents that cause diarrhea (eg, lactulose) or fluid loss (eg, diuretics) (expert opinion); maximum dose: 1.5 g/day ^(Ref).

Note: Do not initiate in patients at risk for lactic acid-producing events (eg, active alcohol consumption, dehydration, hypotension, sepsis, reduced cardiac function, reduced kidney function) ^(Ref).

Child-Turcotte-Pugh class C: Avoid use ^(Ref).

Dosage adjustment in patients with chronic, worsening hepatic function during treatment (eg, progression from Child-Turcotte-Pugh class A to B):

Note: In patients with concurrent kidney impairment, eGFR ≤ 45 mL/minute/1.73 m² and Child-Turcotte-Pugh A or B, defer to dosing in altered kidney function in adults ^(Ref). Monitor kidney function frequently (eg, every 1 to 3 months) ^(Ref) during continuation of therapy ^(Ref).

Progression from baseline to Child-Turcotte-Pugh class A: No dosage adjustment necessary ^(Ref).

Progression from Child-Turcotte-Pugh class A to B: No dosage adjustment necessary; however, a dose reduction may be required in patients at risk for lactic acid-producing events (eg, active alcohol consumption, dehydration, hypotension, sepsis, reduced cardiac function, reduced kidney function) ^(Ref).

Progression from Child-Turcotte-Pugh class B to C: If tolerating with appropriate clinical endpoints, use with caution; discontinue metformin therapy in patients at risk for or who experienced a lactic acid-producing event (eg, active alcohol consumption, dehydration, hypotension, sepsis, reduced cardiac function, reduced kidney function) ^(Ref).

Acute hepatotoxicity during treatment:

Acute worsening of hepatic function (eg, requiring hospitalization): Discontinue metformin during the acute event ^(Ref). Once the acute event has resolved, may resume metformin at pre-event doses (expert opinion). Permanent discontinuation should be considered for patients who recently experienced lactic acidosis, are at high risk of recurrent decompensation, or at risk of lactic acid-producing events ^(Ref).

Metformin-induced hepatotoxicity: Permanently discontinue metformin therapy. Although metformin is not metabolized by the liver, rare cases of metformin-induced hepatotoxicity have been reported within weeks of initiation, but concurrent use of other hepatotoxic medications is common. Most patients present with jaundice, fatigue, and elevated AST/ALT. Upon discontinuation of metformin, LFTs normalize over days to weeks and reoccurs upon rechallenge ^(Ref).

Dosing: Pediatric**Diabetes mellitus, type 2, treatment**

Diabetes mellitus, type 2, treatment: Note: Allow 1 to 2 weeks between dose titrations. Generally, clinically significant responses are not seen at doses less than 1,500 to 2,000 mg/day ^(Ref); however, a lower recommended starting dose with a gradual increase in dosage is recommended to minimize GI symptoms.

Immediate release: Children ≥ 10 years and Adolescents: Oral: Initial: 500 to 1,000 mg once daily **or** 500 mg twice daily; increase by 500 mg every 1 to 2 weeks as tolerated; maximum dose: 1,000 mg twice daily **or** 850 mg 3 times daily ^(Ref).

Extended release: Note: Fewer GI effects may be seen with extended-release products; however, no pediatric studies have compared extended-release products to standard metformin ^(Ref).

Children ≥ 10 years and Adolescents: Limited data available: Oral: Initial: 500 to 1,000 mg once daily for 7 days; may increase dose by 500 mg increments every 1 to 2 weeks as tolerated; maximum daily dose: 2,000 mg/day ^(Ref).

Note: If glycemic control is not achieved at maximum dose, may divide dose and administer twice daily ^(Ref).

Surgical patients ^(Ref):

Minor surgeries (surgery lasting <2 hours, with or without sedation or anesthesia, where rapid recovery is anticipated and patient is expected to be able to eat within 2 to 4 hours): Discontinue metformin the day of surgery; monitor glucose closely; administer rapid- or short-acting insulin if needed; resume metformin after oral intake is tolerated.

Major surgeries (any surgery under anesthesia that is more than minor, typically lasting >2 hours with a high likelihood of postoperative nausea, vomiting, and/or inability to adequately feed after procedure): Discontinue metformin the day of surgery; monitor glucose closely; administer rapid- or short-acting insulin if needed. Withhold metformin for 24 hours following surgery and until normal renal function has been confirmed.

Obesity; severe, adjunct therapy with lifestyle interventions

Obesity; severe, adjunct therapy with lifestyle interventions: Limited data available; data have shown modest efficacy; reported mean BMI reduction -0.86 to -1.16 kg/m², although reported efficacy endpoints/outcomes in trials are variable (Ref). Overall, due to limited efficacy, some experts do not consider metformin a weight loss treatment option for pediatric obesity (Ref). Optimal treatment duration not established; duration in trials varied from 3 to 12 months. Daily multivitamin supplement may be considered with therapy (Ref).

Immediate release: Dosing regimens variable: Children ≥6 years and Adolescents: Oral: Initial: 500 mg once or twice daily, may titrate upward at weekly intervals by 500 mg/day increments; reported final doses: 1,000 to 2,000 mg/day in 2 divided doses; a meta-analysis showed that metformin 2,000 mg/day intervention was most identified as most effective relative to lower doses or lifestyle modifications in adolescents without diabetes (Ref). Age-dependent efficacy findings are conflicting; some data suggest that metformin may have greater efficacy in prepubertal children and others have shown a greater effect in pubertal subjects (11 to 17 years) (Ref). A low dose (850 mg/day) over a prolonged period (24 months) showed positive effects on body composition (weight and BMI standard deviation scores) and other metabolic and inflammatory markers (eg, fat mass, liver fat, leptin, highly sensitive C-reactive protein) in a small, placebo-controlled trial of 22 subjects (metformin=13; age range: 6 to 13 years) (Ref).

Extended release: Adolescents: Oral: Initial: 500 mg once daily with dinner for 2 weeks; increase to 1,000 mg once daily for 2 weeks, and then 2,000 mg once daily; may slow titration if adverse GI effects (Ref).

Weight gain, atypical antipsychotic induced, treatment

Weight gain, atypical antipsychotic induced, treatment: Limited data available:

Immediate release: Note: In trials, all patients were diagnosed with autism spectrum disorder; trials used metformin oral solution and slowly titrated the dose to minimize GI effects and maximize tolerability. After 16 weeks of therapy, patients receiving metformin had a statistically significant decrease in BMI z-scores (primary efficacy outcome) as well as other secondary outcomes (eg, raw weight, BMI) compared to placebo. A 16-week, open-label extension study showed results achieved in the treatment group were maintained without further decreases (Ref).

6 to 9 years: Oral: Initial: 250 mg with evening meal for 1 week, then 250 mg twice daily for 1 week, then 500 mg twice daily.

10 to 17 years: Oral: Initial: 250 mg with evening meal for 1 week, then 250 mg twice daily for 1 week, then 500 mg twice daily for 1 week, then 850 mg twice daily.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Altered Kidney Function: Pediatric

Children ≥10 years and Adolescents: Oral: Immediate release, extended release:

eGFR >45 mL/minute/1.73 m²: No dosage adjustment necessary.

eGFR 30 to 45 mL/minute/1.73 m²:

Preexisting impairment: Initiation of therapy is not recommended.

During therapy:

If eGFR falls between 30 and <45 mL/minute/1.73 m²: Consider risk/benefit ratio for continuing therapy.

If eGFR falls to <30 mL/minute/1.73 m²: Discontinue therapy.

eGFR <30 mL/minute/1.73 m²: Use is contraindicated.

Dosing: Altered Liver Function: Pediatric

Children ≥10 years and Adolescents: Avoid metformin; liver disease is a risk factor for the development of lactic acidosis during metformin therapy.

Calculations

- [Child Pugh Classification](#)
- [Child Pugh Classification \(SI units\)](#)
- [Creatinine Clearance by Cockcroft-Gault](#)
- [Creatinine Clearance by Cockcroft-Gault \(SI units\)](#)
- [Creatinine Clearance by Cockcroft-Gault with IBW](#)
- [Creatinine Clearance by Cockcroft-Gault with IBW \(SI units\)](#)
- [Creatinine Clearance by Jelliffe](#)
- [Glomerular Filtration Rate by MDRD](#)
- [Glomerular Filtration Rate by Schwartz](#)

Use: Labeled Indications

Diabetes mellitus, type 2, treatment: Management of type 2 diabetes mellitus when hyperglycemia cannot be managed with diet and exercise alone.

* See [Uses in AHFS Essentials](#) for additional information.

Use: Off-Label: Adult

Antipsychotic-induced weight gain Level of Evidence [B, G]

Data from multiple meta-analyses of randomized controlled trials with varying degrees of heterogeneity (primarily in patients with schizophrenia and schizoaffective disorder) support the use of metformin in promoting modest weight loss and preventing weight gain associated with second-generation antipsychotics in adult patients (^{Ref}).

Based on British Association for Psychopharmacology guidelines on the management of weight gain, metabolic disturbances, and cardiovascular risk associated with psychosis and antipsychotic drug treatment and the VA/DoD clinical practice guideline for management of first-episode psychosis and schizophrenia, metformin should be considered to attenuate or reduce weight gain caused by antipsychotics after behavioral interventions have failed (^{Ref}) or parallel to nonpharmacologic strategies (^{Ref}). Based on the "Metformin in the management of antipsychotic-induced weight gain in adults with psychosis" guideline, metformin is strongly recommended as an alternative first-line option when lifestyle interventions available to patients are unacceptable or inappropriate. Metformin is also strongly recommended when non-pharmacological interventions are ineffective (^{Ref}). Based on the Canadian Network for Mood and Anxiety Treatments (CANMAT) recommendations for the management of patients with mood disorders and comorbid metabolic disorders, metformin is recommended as a second-line option after nonpharmacologic strategies for managing weight gain in patients with bipolar and major depressive disorders have failed, and is recognized as often being used as a secondary prevention strategy for antipsychotic-related weight gain (^{Ref}). [Access Full Off-Label Monograph](#)

Diabetes mellitus, type 2 (prevention) Level of Evidence [A, G]

Data from a multicenter, placebo-controlled clinical trial involving 3,234 patients without diabetes but at high risk for developing diabetes showed that metformin treatment reduced the incidence of diabetes compared to placebo (^{Ref}).

Based on the American Diabetes Association (ADA) Standards of Care in Diabetes, metformin is effective and may be considered for the prevention of type 2 diabetes in patients with prediabetes (ie, impaired glucose tolerance, impaired fasting glucose, or an HbA_{1c} 5.7% to 6.4%), especially for those with BMI ≥ 35 kg/m², those <60 years of age, and patients with prior gestational diabetes mellitus (^{Ref}).

Gestational diabetes mellitus (treatment) Level of Evidence [G]

Based on the ADA Standards of Care in Diabetes, metformin may be considered an option in the treatment of gestational diabetes mellitus in patients who cannot use insulin, based on short-term safety and efficacy studies. However, because additional long-term safety data are needed, and because of the failure of metformin to provide adequate glycemic control in some patients, insulin is the preferred treatment when pharmacologic therapy is needed (^{Ref}).

Ovarian hyperstimulation syndrome prevention in patients with polycystic ovary syndrome undergoing in vitro fertilization/intracytoplasmic sperm injection Level of Evidence [B, G]

Controlled trials and meta-analyses support the use of metformin to reduce the risk of ovarian hyperstimulation syndrome (OHSS) in patients with polycystic ovary syndrome (PCOS) receiving a gonadotropin during in vitro fertilization (IVF) or intracytoplasmic sperm injection (ICSI) (^{Ref}).

According to the International PCOS Network evidence-based guideline for the assessment and management of PCOS, metformin may be considered as an adjunctive agent to reduce the risk of OHSS and to improve pregnancy rate in patients with PCOS and anovulatory infertility undergoing IVF or ICSI (^{Ref}). [Access Full Off-Label Monograph](#)

Level of Evidence Definitions

Level of Evidence Scale

- A** - Consistent evidence from well-performed randomized, controlled trials or overwhelming evidence of some other form (eg, results of the introduction of penicillin treatment) to support the off-label use. Further research is unlikely to change confidence in the estimate of benefit.
- B** - Evidence from randomized, controlled trials with important limitations (inconsistent results, methodological flaws, indirect or imprecise), or very strong evidence of some other research design. Further research (if performed) is likely to have an impact on confidence in the estimate of benefit and risk and may change the estimate.
- C** - Evidence from observational studies (eg, retrospective case series/reports providing significant impact on patient care), unsystematic clinical experience, or from potentially flawed randomized, controlled trials (eg, when limited options exist for condition). Any estimate of effect is uncertain.
- G** - Use has been substantiated by inclusion in at least one evidence-based or consensus-based clinical practice guideline.

Clinical Practice Guidelines

Diabetes Mellitus:

AACE, "American Association of Clinical Endocrinology Consensus Statement: Comprehensive Type 2 Diabetes Management Algorithm – 2023 Update," [May 2023](#)

ADA, "Standards of Care in Diabetes - 2023," [January 2023](#)

American Diabetes Association and the European Association for the Study of Diabetes Consensus Report, "Management of Hyperglycemia in Type 2 Diabetes, 2018," [December 2018](#)

Diabetes Canada, "Clinical Practice Guidelines for the Prevention and Management of Diabetes in Canada," [2018](#)

KDIGO, "KDIGO 2020 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease," [October 2020](#)

Prevention:

"AHA/ACCF Secondary Prevention and Risk Reduction Therapy for Patients with Coronary and Other Atherosclerotic Vascular Disease: 2011 Update," [November 2011](#)

Metabolic Disorders:

APA, "Practice Guideline for the Treatment of Patients With Schizophrenia," [2020](#)

British Association of Psychopharmacology (BAP), "BAP Guidelines on the Management of Weight Gain, Metabolic Disturbances and Cardiovascular Risk Associated With Psychosis and Antipsychotic Drug Treatment," [August 2016](#)

Canadian Network for Mood and Anxiety Treatments (CANMAT), "Recommendations for the Management of Patients with Mood Disorders and Comorbid Metabolic Disorders," [February 2012](#)

"Metformin in the Management of Antipsychotic-Induced Weight Gain in Adults With Psychosis: Development of the First Evidence-Based Guideline Using GRADE Methodology," [February 2022](#)

VA/DoD, "Clinical Practice Guideline for Management of First-Episode Psychosis and Schizophrenia," [April 2023](#)

Administration: Oral

Administer with a meal (to decrease GI upset). Administer solution and suspension with supplied measuring devices. Administer reconstituted solution (packet [International product]) immediately after preparation.

ER tablets: Swallow whole; do not crush, cut, or chew. Administer once-daily doses with the evening meal.

Bariatric surgery: Some institutions may have specific protocols that conflict with these recommendations; refer to institutional protocols as appropriate. IR tablets and oral solution formulations are available. If safety and efficacy of metformin ER formulations can be effectively monitored, no change in formulation or administration is required after bariatric surgery; however, avoidance of Glumetza may be advisable as this formulation is designed to be retained in the stomach (Glumetza prescribing information 2014).

Administration: Pediatric

Oral: Administer with a meal (to decrease GI upset).

Immediate release: Administer in divided doses with meals. Administer oral solution and suspension with supplied measuring devices. Administer reconstituted solution (packet [International product]) immediately after preparation.

Extended release:

Oral suspension: Administer with evening meal. Shake well at least 10 seconds before use; use supplied dosing cup to measure dose. Rinse dosing cup with water only after each use to clean.

Tablet: Administer once-daily doses with the evening meal; swallow whole; do not cut, crush, or chew.

Dietary Considerations

Drug may cause GI upset; take with food (to decrease GI upset). Take at the same time(s) each day. Dietary modification based on ADA recommendations is a part of therapy.

Storage/Stability

Oral solution (immediate release): Store at 15°C to 30°C (59°F to 86°F).

Oral suspension (extended release): Store at 20°C to 25°C (68°F to 77°F); excursion permitted to 15°C to 30°C (59°F to 86°F).

Discard unused portion of reconstituted suspension after 100 days.

Tablets: Store at 20°C to 25°C (68°F to 77°F); excursion permitted to 15°C to 30°C (59°F to 86°F). Protect from light and moisture.

Preparation for Administration: Adult

Oral suspension (extended release): Reconstitute with accompanying diluent; shake continuously for ≥2 minutes.

Powder for oral solution (packet) [International product]: Mix contents of 1 packet (500 mg) in 150 mL water to obtain a clear to opalescent solution. Stir if necessary and administer immediately after preparation.

Preparation for Administration: Pediatric

Oral suspension (extended release): To reconstitute, pour the contents of the drug pellet bottle into the provided drug diluent bottle; shake continuously in an up and down motion for at least 2 minutes; resultant concentration is 500 mg/5 mL. **Note:** The provided diluent also contains metformin (9.46 g); do not use any other diluent for reconstitution.

Powder for oral solution (packet) [International product]: Mix contents of 1 packet (500 mg) in 150 mL water to obtain a clear to opalescent solution. Stir if necessary and administer immediately after preparation.

Medication Safety Issues

Sound-alike/look-alike issues:

MetFORMIN may be confused with metroNIDAZOLE

Glucophage may be confused with Glucotrol, Glutofac

International issues:

Dianben [Spain] may be confused with Diovan brand name for valsartan [US, Canada, and multiple international markets].

Glucon brand name for metformin [Malaysia, Singapore] is also the brand name for glucosamine [Singapore]

Contraindications

Hypersensitivity to metformin or any component of the formulation; severe renal dysfunction (eGFR <30 mL/minute/1.73 m²); acute or chronic metabolic acidosis with or without coma (including diabetic ketoacidosis).

Canadian labeling: Additional contraindications (not in US labeling): End-stage renal disease, patients on dialysis, or when renal function is unknown; unstable and/or insulin-dependent (type 1) diabetes mellitus; history of ketoacidosis with or without coma; history of lactic acidosis (regardless of precipitating factors); excessive alcohol intake (acute or chronic); severe hepatic dysfunction or clinical or laboratory evidence of hepatic disease; patients undergoing radiologic studies involving intravascular administration of iodinated contrast materials; cardiovascular collapse and disease states associated with hypoxemia, including cardiorespiratory insufficiency, which are often associated with hyperlactacidemia; stress conditions (eg, severe infection, trauma, surgery and postoperative recovery phase); severe dehydration; pregnancy; breastfeeding.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Disease-related concerns:

- **Bariatric surgery:** Altered absorption: Use of IR tablets or solution after surgery is generally preferred. ER tablets may have a reduced effect after gastric bypass or sleeve gastrectomy due to the direct bypass of the stomach and proximal small bowel with gastric bypass or a more rapid gastric emptying and proximal small bowel transit with sleeve gastrectomy (Mechanick 2020; Melissas 2013). After gastric bypass (Roux-en-Y gastric bypass [RYGB]), administration of IR tablets led to increased absorption (AUC_{0-∞} increased by 21%) and bioavailability (increased by 50%) (Padwal 2011). Lactate levels decrease after gastric bypass (RYGB)-induced weight loss irrespective of the use of metformin. Routinely lowering metformin dose after gastric bypass is not necessary as long as normal renal function is preserved (Deden 2018).
- **Heart failure:** Metformin may be used in patients with stable heart failure (ADA 2023). Use cautiously or avoid in hypoperfusion.
- **Hepatic impairment:** Use cautiously in patients at risk for lactic acidosis (Brackett 2010; Crowley 2017; Zhang 2014). An increased risk of mortality has been observed with higher metformin doses (eg, >1 g) and more severe liver dysfunction (eg, Child-Turcotte-Pugh C>B>>A) (Yen 2022). Cases of metformin-induced hepatotoxicity have also been reported

(LiverTox 2020; Zheng 2016).

- Renal impairment: Metformin is substantially excreted by the kidney; dosing adjustments may be required.
- Stress-related states: It may be necessary to discontinue metformin and administer insulin if the patient is exposed to stress (fever, trauma, infection, surgery).

Dosage form specific issues:

- ER tablet: A soft mass resembling the tablet may appear in the stool.
- Propylene glycol: Some dosage forms may contain propylene glycol; large amounts are potentially toxic and have been associated with hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Zar 2007). See manufacturer's labeling.

Other warnings/precautions:

- Appropriate use: Not indicated for use in patients with type 1 diabetes mellitus or with diabetic ketoacidosis.
- Iodinated contrast: Administration of iodinated contrast agents has been associated with postcontrast acute kidney injury (AKI); acute decreases in renal function have been associated with an increased risk of lactic acidosis due to reduced metformin excretion (ACR 2021; manufacturer's labeling). Current guidelines recommend temporarily withholding metformin prior to or at the time of iodinated contrast administration in patients with known AKI or severe chronic kidney disease (eGFR <30 mL/minute/1.73 m²), or who are undergoing arterial catheter studies (ACR 2021). Some experts also temporarily withhold metformin in patients with an eGFR <45 mL/minute/1.73 m² or risk factors for lactic acidosis (eg, hypotension, vascular instability, potential hypoperfusion) (Rudnick 2021; Wexler 2021b). If metformin is withheld, reassess kidney function ≥48 hours after contrast administration and resume therapy if kidney function is acceptable (ACR 2021).
- Surgical procedures: Metformin-containing products should be withheld the day of surgery; restart after renal function is stable (ADA 2023).

* See [Cautions in AHFS Essentials](#) for additional information.

Older Adult Considerations

Limited data suggest that metformin's total body clearance may be decreased and AUC and half-life increased in older adult patients; presumably due to decreased renal clearance. In one study of older adult subjects, its side effects could not be distinguished from tolbutamide, except for weight loss (Josephkutty 1990). Older adults with diabetes are at a higher risk of cognitive decline and institutionalization (ElSayed 2023).

Older adults are more likely to experience vitamin B₁₂ deficiency with long-term use of metformin (Kancherla 2017).

When extended-release metformin is being considered, swallowing ability should be assessed due to the inability to crush this formulation.

Reproductive Considerations

Metformin is approved for the treatment of type 2 diabetes mellitus; however, it is not preferred for use if pregnancy should occur. Regardless of treatment, patients with diabetes who could become pregnant should use effective contraception and those planning to become pregnant should continue contraception until glycemic control is achieved. When diet and exercise alone are not enough for the treatment of type 2 diabetes mellitus, metformin may be continued in select patients who are well controlled prior to pregnancy, such as those who are unwilling or unable to use insulin (ACOG 2018; ADA 2023).

Metformin has been studied in patients with polycystic ovarian syndrome (PCOS) who have reproductive symptoms such as irregular menstrual cycles and infertility.

- Adjunctive use of metformin may improve metabolic features of PCOS in patients when combination oral contraceptives and lifestyle modifications are not effective. Metformin may be most effective for patients with diabetes risk factors, impaired glucose tolerance, or a BMI ≥25 kg/m² (Teede 2018).

- Metformin (in combination with clomiphene) may be considered for ovulation induction in patients diagnosed with PCOS and anovulatory infertility (and no other infertility factors), who have a BMI ≥ 30 kg/m², and who wish to conceive. Pregnancy should be excluded prior to treatment (Teede 2018). Metformin monotherapy is no longer recommended for routine use for ovulation induction in patients with PCOS and anovulatory infertility but may be considered in patients who are unable or unwilling to take more effective agents (ASRM 2017; Teede 2018).
- Adjunctive use of metformin may reduce the risk of ovarian hyperstimulation syndrome (OHSS) and improve pregnancy rate in patients with PCOS and anovulatory infertility who are undergoing in vitro fertilization (IVF) or intracytoplasmic sperm injection (ICSI). Metformin should be discontinued once there is a positive pregnancy test unless it is needed for another indication (Teede 2018).

Pregnancy Considerations

Metformin crosses the placenta; concentrations may be comparable to or higher than those found in the maternal plasma (ADA 2023; Charles 2006; de Oliveira Baraldi 2011; Eyal 2010; Vanky 2005).

Outcome data following first trimester exposure from paternal (Wensink 2022) and maternal (Cao 2021; Cassina 2014; Gilbert 2006; Panchaud 2018; Scherneck 2018; Zeng 2016) use of metformin are available. Maternal and fetal/neonatal outcomes are influenced by underlying maternal disease (eg, gestational diabetes mellitus [GDM] or polycystic ovary syndrome [PCOS]) (Cao 2021; Panchaud 2018; Zeng 2016). An increased risk of birth defects or adverse fetal/neonatal outcomes has not been observed following maternal use of metformin for gestational diabetes mellitus or type 2 diabetes mellitus when glycemic control is maintained (Balani 2009; Coetzee 1984; Ekpebegh 2007; Niromanesh 2012; Rowan 2008; Rowan 2010; Terti 2008). Studies which evaluated children up to 12 years of age with previous in utero exposure to metformin have shown conflicting outcomes related to BMI and other growth parameters; however, results may be confounded by underlying maternal disease (eg, GDM, PCOS) and concomitant therapy (eg, insulin) (ADA 2023; Brand 2022; Paavilainen 2022; Skibinska 2021). Additional long-term data are needed (ADA 2023).

Due to pregnancy-induced physiologic changes, some pharmacokinetic properties of metformin may be altered. Changes in renal blood flow alter renal clearance; in one study this significantly decreased plasma concentrations of metformin for pregnant patients taking a dose of 1 g/day but not doses of 2 g/day. Changes in renal clearance correlated with changes in creatinine clearance (Eyal 2010; Liao 2020).

Poorly controlled diabetes during pregnancy is associated with an increased risk of adverse maternal and fetal outcomes, including diabetic ketoacidosis, preeclampsia, spontaneous abortion, preterm delivery, delivery complications, major malformations, stillbirth, and macrosomia (ACOG 2018). To prevent adverse outcomes, prior to conception and throughout pregnancy, maternal blood glucose and HbA_{1c} should be kept as close to target goals as possible but without causing significant hypoglycemia (ADA 2023).

Agents other than metformin are currently recommended to treat diabetes mellitus in during pregnancy (ADA 2023). Metformin may be considered for the treatment of GDM in patients who cannot safely or effectively use insulin (eg, due to cost, language barriers, comprehension, cultural issues). However, metformin monotherapy fails to provide adequate glycemic control in all patients. Metformin should not be used in patients with hypertension, preeclampsia, or those at risk for intrauterine growth restriction due to risk of fetal growth restriction or acidosis in the setting of placental restriction. Metformin is not the preferred treatment of type 2 diabetes mellitus during pregnancy (ADA 2023). Metformin, in addition to insulin, may be considered for some pregnant patients with type 2 diabetes mellitus due to potential maternal and neonatal benefits (eg, reduced insulin requirements, less maternal weight gain, fewer caesarean births, reduced incidence of macrosomia), although additional long-term data are needed (Benham 2021; Feig 2020).

PCOS is also associated with adverse pregnancy outcomes, including an increased risk of GDM, miscarriage, pregnancy-induced hypertension, preeclampsia, and preterm delivery. When used in the management of PCOS, metformin may reduce the risk of preterm delivery; however, the incidence of GDM, miscarriage, or preeclampsia are not affected (Cao 2021). Metformin monotherapy has not been found to improve the live birth rate in patients with PCOS when used for ovulation induction or in vitro fertilization (IVF) or intracytoplasmic sperm injection (ICSI) (Magzoub 2022; Tso 2020; Wu 2020). If metformin is used for ovulation

induction in patients with PCOS, treatment should be discontinued by the end of the first trimester (ADA 2023). In patients undergoing IVF/ICSI, metformin should be discontinued once there is a positive pregnancy test unless it is needed for another indication (Teede 2018).

Breastfeeding Considerations

Metformin is present in breast milk.

Data related to the presence of metformin in breast milk are available from multiple studies (Briggs 2005; Eyal 2010; Gardiner 2003; Hale 2002).

- One study included 5 women who initiated metformin 500 mg twice daily immediately postpartum. Maternal plasma and breast milk were sampled prior to and ~2 hours after the morning dose, 4 to 17 days after delivery. Average concentrations of metformin in breast milk were <0.09 to 0.47 mcg/mL. Using the actual maternal weight from the patient with the highest average breast milk concentration (0.47 mcg/mL), authors of the study calculated the relative infant dose (RID) of metformin to be 1.08%, compared to a weight-adjusted maternal dose of 6.55 mg/kg/day providing an estimated daily infant dose via breast milk of 0.07 mg/kg/day. Adverse events were not observed in the breastfed infants (Briggs 2005).
- Breast milk was sampled in a subset of postpartum women taking metformin 1,500 mg/day (n = 1) or 2,000 mg/day (n = 3) who were part of a larger pharmacokinetic study. Using breast milk collected over one dosing interval, the RID of metformin ranged from 0.14% to 0.43% of the weight-adjusted maternal dose (Eyal 2010).
- According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. Using the limited data reported in the literature, the RID of metformin is ~0.11% to 1% of the maternal weight-adjusted dose.
- In general, breastfeeding is considered acceptable when the RID is <10% (Anderson 2016; Ito 2000).

Metformin has been detected in the serum of some breastfeeding infants. Because breast milk concentrations of metformin stay relatively constant, avoiding breastfeeding around peak plasma concentrations in the mother would not be helpful in reducing metformin exposure to the infant (Briggs 2005; Eyal 2010; Gardiner 2003; Hale 2002).

Appropriate glycemic control is required for the establishment of lactation in patients with diabetes mellitus (Anderson 2018).

Breastfeeding provides metabolic benefits to mothers with diabetes mellitus as well as their infants; therefore, breastfeeding is encouraged (ACOG 201 2018; ADA 2023). According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother.

Briggs' Drugs in Pregnancy & Lactation

- [Metformin](#)

Adverse Reactions (Significant): Considerations

Gastrointestinal effects

Transient and/or reversible gastrointestinal (GI) adverse reactions, including **diarrhea**, **nausea**, **flatulence**, **dyspepsia**, **vomiting**, and **abdominal pain**, among others, are the most commonly reported adverse reactions with metformin use; intolerance to these reactions is often a reason for discontinuation of therapy (^{Ref}).

Mechanism: Dose-related; mechanism not fully understood. Multiple mechanisms have been proposed that may contribute, including alterations in transporters (eg, organic cation transporter 1 [OCT1]), direct serotonergic-like effects, increased gut motility, direct or indirect increases in glucagon-like peptide 1 (GLP-1), and disruptions in bile (^{Ref}).

Onset: Varied; typically occurs at initiation of therapy; however, late onset of diarrhea has been reported (^{Ref}). GI adverse reactions generally subside after several weeks of therapy; however, some patients require discontinuation.

Risk factors:

- Rapid dose escalation
- Use of IR formulations (^{Ref})
- Chronic asymptomatic gastritis (^{Ref})

- *Helicobacter pylori* infection ^(Ref)
- Concomitant use of OCT1-inhibiting agents (eg, verapamil, proton pump inhibitors) may be a potential risk factor ^(Ref)

Lactic acidosis

Postmarketing cases of metformin-associated **lactic acidosis** (MALA) have resulted in hypothermia, hypotension, resistant bradyarrhythmias, and death. The onset is often subtle, accompanied by nonspecific symptoms (eg, malaise, myalgia, respiratory distress, somnolence, abdominal pain); elevated blood lactate levels (>5 mmol/L); anion gap acidosis (without evidence of ketonuria or ketonemia); and/or increased lactate:pyruvate ratio. Lactic acidosis should be suspected in any patient with diabetes receiving metformin with evidence of acidosis but without evidence of ketoacidosis. Safety data for use in advanced heart failure (stage D) is lacking ^(Ref).

Mechanism: Dose-related; inhibition of mitochondrial electron transport, an effect that favors anaerobic metabolism and the accumulation of lactate ^(Ref), resulting in lactic acidosis without evidence of tissue hypoxia (type B lactic acidosis). In addition, if accumulation occurs (eg, in patients with kidney impairment) or if high doses are administered (eg, in overdose), glucose utilization decreases and hepatic production of lactate increases ^(Ref).

Onset: Varied; may occur at any time during treatment.

Risk factors: In general, risk is increased with increased metformin concentration, decreased lactate clearance, and/or increased lactate production.

- Kidney impairment; risk increases with the severity of kidney impairment
- Hepatic impairment
- Reduced tissue perfusion such as in unstable heart failure ^(Ref)
- Concomitant use of certain drugs that impair kidney function, cause significant hemodynamic changes, interfere with acid-base balance, or increase metformin accumulation
- Patients ≥65 years of age
- Radiologic study with contrast
- Surgery and other procedures due to potential for volume depletion, hypotension, and kidney impairment
- Hypoxic states (eg, acute heart failure, acute myocardial infarction, sepsis, shock)
- Excessive alcohol intake

Vitamin B12 deficiency

Long-term use of metformin has been associated with reversible **vitamin B₁₂ deficiency** and subsequent anemia and neuropathy ^(Ref).

Mechanism: Time-related; long-term use metformin interferes with the absorption of vitamin B₁₂ ^(Ref); potential mechanism includes interference of vitamin B₁₂-intrinsic factor absorption in the terminal ileum ^(Ref).

Onset: Delayed; occurs with chronic use of metformin.

Risk factors:

- Duration of therapy ^(Ref)
- Higher doses (eg, >1,000 mg/day) ^(Ref)
- Inadequate B₁₂ stores, poor underlying nutrition, and inadequate calcium intake or absorption ^(Ref)
- Older patients (eg, >65 years of age) ^(Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%: Gastrointestinal: Diarrhea (IR tablet: 53%; ER tablet: 10%) ([table 1](#)), flatulence (12%) ([table 2](#)), nausea and vomiting (IR tablet: 26%; ER tablet: 7%) ([table 3](#))

1% to 10%:

Cardiovascular: Chest discomfort, flushing, palpitations

Dermatologic: Diaphoresis, nail disease

Endocrine & metabolic: Hypoglycemia, vitamin B₁₂ deficiency (7%)

Gastrointestinal: Abdominal distention, abdominal distress (6%), abdominal pain, abnormal stools, dyspepsia (7%) ([table 4](#)), heartburn

Nervous system: Asthenia (9%), chills, dizziness, headache (6%)

Neuromuscular & skeletal: Myalgia

Respiratory: Dyspnea, flu-like symptoms, upper respiratory tract infection

Postmarketing:

Dermatologic: Fixed drug eruption (^{Ref}), lichen planus (^{Ref})

Endocrine & metabolic: Lactic acidosis (^{Ref})

Gastrointestinal: Pancreatitis (^{Ref})

Hematologic & oncologic: Hemolytic anemia (^{Ref})

Hepatic: Hepatic injury (cholestatic, hepatocellular, and mixed) (^{Ref})

Hypersensitivity: Drug reaction with eosinophilia and systemic symptoms (^{Ref}), hypersensitivity angiitis (^{Ref})

Nervous system: Encephalopathy (^{Ref})

* See [Cautions in AHFS Essentials](#) for additional information.

Allergy and Idiosyncratic Reactions

- [Biguanide Allergy](#)

Toxicology

- [Antidiabetic Agents, Biguanides](#)

Metabolism/Transport Effects

Substrate of OCT1, OCT2

Drug Interactions

[Open Interactions](#)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Open Interactions" button above.

Abemaciclib: May increase the serum concentration of MetFORMIN. *Risk C: Monitor therapy*

Alcohol (Ethyl): May enhance the adverse/toxic effect of MetFORMIN. Specifically, excessive alcohol ingestion (acute or chronic) may potentiate the risk of lactic acidosis. *Risk X: Avoid combination*

Alpha-Lipoic Acid: May enhance the hypoglycemic effect of Antidiabetic Agents. *Risk C: Monitor therapy*

Androgens: May enhance the hypoglycemic effect of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor therapy*

Arimoclomol: May increase the serum concentration of OCT2 Substrates (Clinically Relevant with Inhibitors). *Risk C: Monitor therapy*

Beta-Blockers (Beta1 Selective): May enhance the hypoglycemic effect of Antidiabetic Agents. *Risk C: Monitor therapy*

Beta-Blockers (Nonselective): May enhance the hypoglycemic effect of Antidiabetic Agents. Beta-Blockers (Nonselective) may diminish the therapeutic effect of Antidiabetic Agents. *Risk C: Monitor therapy*

- Bortezomib: May enhance the therapeutic effect of Antidiabetic Agents. Bortezomib may diminish the therapeutic effect of Antidiabetic Agents. *Risk C: Monitor therapy*
- Carbonic Anhydrase Inhibitors: May enhance the adverse/toxic effect of MetFORMIN. Specifically, the risk of developing lactic acidosis may be increased. *Risk C: Monitor therapy*
- Cephalexin: May increase the serum concentration of MetFORMIN. *Risk C: Monitor therapy*
- Cimetidine: May increase the serum concentration of MetFORMIN. Management: Consider alternatives to cimetidine in patients receiving metformin due to a potential for increased metformin concentrations and toxicity (including lactic acidosis). *Risk D: Consider therapy modification*
- Copanlisib: May diminish the therapeutic effect of MetFORMIN. Copanlisib may increase the serum concentration of MetFORMIN. *Risk C: Monitor therapy*
- Direct Acting Antiviral Agents (HCV): May enhance the hypoglycemic effect of Antidiabetic Agents. *Risk C: Monitor therapy*
- Dofetilide: MetFORMIN may increase the serum concentration of Dofetilide. *Risk C: Monitor therapy*
- Dolutegravir: May increase the serum concentration of MetFORMIN. Management: Consider alternatives to this combination or use of lower metformin doses. Carefully weigh the risk of metformin toxicities (including lactic acidosis) against the benefit of combining dolutegravir with metformin. *Risk D: Consider therapy modification*
- Etilefrine: May diminish the therapeutic effect of Antidiabetic Agents. *Risk C: Monitor therapy*
- Fedratinib: May increase the serum concentration of MATE1/2-K Substrates (Clinically Relevant with Inhibitors). *Risk C: Monitor therapy*
- Fedratinib: May increase the serum concentration of OCT2 Substrates (Clinically Relevant with Inhibitors). *Risk C: Monitor therapy*
- Fexinidazole: May increase the serum concentration of MATE1/2-K Substrates (Clinically Relevant with Inhibitors). Management: Avoid use of fexinidazole with MATE1/2-K substrates when possible. If combined, monitor for increased MATE1/2-K substrate toxicities. *Risk D: Consider therapy modification*
- Fexinidazole: May increase the serum concentration of OCT2 Substrates (Clinically Relevant with Inhibitors). Management: Avoid use of fexinidazole with OCT2 substrates when possible. If combined, monitor for increased OCT2 substrate toxicities. *Risk D: Consider therapy modification*
- Fludeoxyglucose F 18: MetFORMIN may diminish the diagnostic effect of Fludeoxyglucose F 18. Management: Consider holding metformin for 48 hours or longer prior to PET scans using fludeoxyglucose F18 (FDG-F18) when imaging of the colon or intestine is required. Consider increased monitoring of blood glucose when metformin is held. *Risk D: Consider therapy modification*
- Foslevodopa: May increase the serum concentration of MATE1/2-K Substrates (Clinically Relevant with Inhibitors). *Risk C: Monitor therapy*
- Gilteritinib: May increase the serum concentration of OCT1 Substrates (Clinically Relevant with Inhibitors). *Risk C: Monitor therapy*
- Givinostat: May increase the serum concentration of OCT2 Substrates (Clinically Relevant with Inhibitors). *Risk C: Monitor therapy*
- Glycopyrrolate (Systemic): May increase the serum concentration of MetFORMIN. *Risk C: Monitor therapy*
- Guanethidine: May enhance the hypoglycemic effect of Antidiabetic Agents. *Risk C: Monitor therapy*
- Guar Gum (Partially Hydrolyzed): May decrease the serum concentration of MetFORMIN. *Risk C: Monitor therapy*
- Hyperglycemia-Associated Agents: May diminish the therapeutic effect of Antidiabetic Agents. *Risk C: Monitor therapy*
- Hypoglycemia-Associated Agents: Antidiabetic Agents may enhance the hypoglycemic effect of Hypoglycemia-Associated Agents. *Risk C: Monitor therapy*
- Iodinated Contrast Agents: May enhance the adverse/toxic effect of MetFORMIN. Renal dysfunction that may be caused by iodinated contrast agents may lead to metformin-associated lactic acidosis. Management: Management advice varies. Refer to the full drug interaction monograph content for details. *Risk D: Consider therapy modification*
- Lamotrigine: May increase the serum concentration of MetFORMIN. Management: The lamotrigine Canadian product monograph states that coadministration of these drugs is not recommended. *Risk C: Monitor therapy*
- Maitake: May enhance the hypoglycemic effect of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor therapy*

MATE1/2-K Inhibitors: May increase the serum concentration of MetFORMIN. *Risk C: Monitor therapy*

Mavoxiafor: May decrease the serum concentration of MetFORMIN. *Risk C: Monitor therapy*

Methylol Cephalixin: May increase the serum concentration of MetFORMIN. *Risk C: Monitor therapy*

Monoamine Oxidase Inhibitors: May enhance the hypoglycemic effect of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor therapy*

Nonsteroidal Anti-Inflammatory Agents: May enhance the adverse/toxic effect of MetFORMIN. *Risk C: Monitor therapy*

Ombitasvir, Paritaprevir, and Ritonavir: May enhance the adverse/toxic effect of MetFORMIN. Specifically, the risk for lactic acidosis may be increased. *Risk C: Monitor therapy*

Ombitasvir, Paritaprevir, Ritonavir, and Dasabuvir: May enhance the adverse/toxic effect of MetFORMIN. Specifically, the risk for lactic acidosis may be increased. *Risk C: Monitor therapy*

Ondansetron: May increase the serum concentration of MetFORMIN. *Risk C: Monitor therapy*

Patiromer: May decrease the serum concentration of MetFORMIN. Management: Administer metformin at least 3 hours before or 3 hours after patiromer. *Risk D: Consider therapy modification*

Pegvisomant: May enhance the hypoglycemic effect of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor therapy*

Prothionamide: May enhance the hypoglycemic effect of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor therapy*

Quinolones: May enhance the hypoglycemic effect of Agents with Blood Glucose Lowering Effects. Quinolones may diminish the therapeutic effect of Agents with Blood Glucose Lowering Effects. Specifically, if an agent is being used to treat diabetes, loss of blood sugar control may occur with quinolone use. *Risk C: Monitor therapy*

Ranolazine: May increase the serum concentration of MetFORMIN. Management: Limit the metformin dose to a maximum of 1,700 mg per day when used together with ranolazine 1,000 mg twice daily. Monitor patients for metformin toxicities, including lactic acidosis and carefully weigh the risks and benefits of this combination. *Risk D: Consider therapy modification*

Reproterol: May diminish the therapeutic effect of Antidiabetic Agents. *Risk C: Monitor therapy*

Risdiplam: May increase the serum concentration of MATE1/2-K Substrates (Clinically Relevant with Inhibitors). Management: Avoid use of risdiplam with MATE substrates if possible. If the combination cannot be avoided, monitor closely for adverse effects. Consider a reduced dose of the MATE substrate according to that substrate's labeling if appropriate. *Risk D: Consider therapy modification*

Ritodrine: May diminish the therapeutic effect of Antidiabetic Agents. *Risk C: Monitor therapy*

Salicylates: May enhance the hypoglycemic effect of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor therapy*

Selective Serotonin Reuptake Inhibitors: May enhance the hypoglycemic effect of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor therapy*

Tafenoquine: May increase the serum concentration of MATE1/2-K Substrates (Clinically Relevant with Inhibitors). Management: Avoid use of MATE substrates with tafenoquine, and if the combination cannot be avoided, monitor closely for evidence of toxicity of the MATE substrate and consider a reduced dose of the MATE substrate according to that substrate's labeling. *Risk D: Consider therapy modification*

Tafenoquine: May increase the serum concentration of OCT2 Substrates (Clinically Relevant with Inhibitors). Management: Avoid use of OCT2 substrates with tafenoquine, and if the combination cannot be avoided, monitor closely for evidence of toxicity of the OCT2 substrate and consider a reduced dose of the OCT2 substrate according to that substrate's labeling. *Risk D: Consider therapy modification*

Thiazide and Thiazide-Like Diuretics: May diminish the therapeutic effect of Antidiabetic Agents. *Risk C: Monitor therapy*

Topiramate: May enhance the adverse/toxic effect of MetFORMIN. Specifically, the risk for lactic acidosis may be increased. MetFORMIN may increase the serum concentration of Topiramate. Topiramate may increase the serum concentration of MetFORMIN. *Risk C: Monitor therapy*

Verapamil: May diminish the therapeutic effect of MetFORMIN. *Risk C: Monitor therapy*

Vitamin K Antagonists (eg, warfarin): MetFORMIN may diminish the anticoagulant effect of Vitamin K Antagonists. Vitamin K Antagonists may enhance the hypoglycemic effect of MetFORMIN. *Risk C: Monitor therapy*

Food Interactions

Food decreases the extent and slightly delays the absorption. Management: Administer with a meal.

Monitoring Parameters

Urine for glucose and ketones, plasma glucose (individualize frequency based on treatment regimen, hypoglycemia risk, and other patient-specific factors) (ADA 2023). Monitor renal function (eGFR) prior to therapy initiation and at least annually or at least every 3 to 6 months if eGFR is <60 mL/minute/1.73 m² (KDIGO 2020) or at least every 1 to 3 months in patients with hepatic impairment (expert opinion). Initial and annual monitoring of hematologic parameters (eg, hemoglobin/hematocrit and red blood cell indices); monitor folate if megaloblastic anemia is suspected. Monitor vitamin B₁₂ serum concentrations every 1 to 2 years, particularly in patients who have been treated with metformin for ≥ 4 years, or in patients with peripheral neuropathy, anemia, or risk factors for vitamin B₁₂ deficiency (eg, malabsorption syndromes, reduced dietary intake) (ADA 2023; KDIGO 2020; manufacturer's labeling).

HbA_{1c}: Monitor at least twice yearly in patients who have stable glycemic control and are meeting treatment goals; monitor quarterly in patients in whom treatment goals have not been met, or with therapy change. **Note:** In patients prone to glycemic variability (eg, patients with insulin deficiency), or in patients whose HbA_{1c} is discordant with serum glucose levels or symptoms, consider evaluating HbA_{1c} in combination with blood glucose levels and/or a glucose management indicator (ADA 2023; KDIGO 2020).

Reference Range

Recommendations for glycemic control in patients with diabetes:

Nonpregnant adults with diabetes (AACE [Samson 2023]; ADA 2023):

HbA_{1c}: $<7\%$ (a more aggressive $<6.5\%$) or less aggressive $<8\%$ HbA_{1c} goal may be targeted based on patient-specific characteristics). **Note:** In patients using a continuous glucose monitoring system, a goal of time in range $>70\%$ with time below range $<4\%$ is recommended and is similar to a goal HbA_{1c} $<7\%$.

Preprandial capillary blood glucose: 80 to 130 mg/dL (SI: 4.4 to 7.2 mmol/L) (more or less stringent goals may be appropriate based on patient-specific characteristics).

Peak postprandial capillary blood glucose (~ 1 to 2 hours after a meal): <180 mg/dL (SI: <10 mmol/L) (more or less stringent goals may be appropriate based on patient-specific characteristics).

Older adults (≥ 65 years of age) (ADA 2023):

Note: Consider less strict targets in patients who are using insulin and/or insulin secretagogues (sulfonylureas, meglitinides) (ES [LeRoith 2019]).

HbA_{1c}: $<7\%$ to 7.5% (healthy); $<8\%$ (complex/intermediate health). **Note:** Individualization may be appropriate based on patient and caregiver preferences and/or presence of cognitive impairment. In patients with very complex or poor health (ie, limited remaining life expectancy), consider making therapy decisions based on avoidance of hypoglycemia and symptomatic hyperglycemia rather than HbA_{1c} level.

Preprandial capillary blood glucose: 80 to 130 mg/dL (SI: 4.4 to 7.2 mmol/L) (healthy); 90 to 150 mg/dL (SI: 5 to 8.3 mmol/L) (complex/intermediate health); 100 to 180 mg/dL (SI: 5.6 to 10 mmol/L) (very complex/poor health).

Bedtime capillary blood glucose: 80 to 180 mg/dL (SI: 4.4 to 10 mmol/L) (healthy); 100 to 180 mg/dL (SI: 5.6 to 10 mmol/L) (complex/intermediate health); 110 to 200 mg/dL (SI: 6.1 to 11.1 mmol/L) (very complex/poor health).

Pregnant patients:

HbA_{1c}: Pregestational diabetes (type 1 or type 2) (ADA 2023):

Preconception (patients planning for pregnancy): $<6.5\%$.

During pregnancy: $<6\%$ (if can be achieved without significant hypoglycemia) or $<7\%$ (if needed to prevent hypoglycemia).

Capillary blood glucose: **Note:** Less stringent targets may be appropriate if goals cannot be achieved without causing significant hypoglycemia (ADA 2023).

Gestational diabetes mellitus (ADA 2023):

Fasting: <95 mg/dL (SI: <5.3 mmol/L).

Postprandial: <140 mg/dL (SI: <7.8 mmol/L) (at 1 hour) **or** <120 mg/dL (SI: <6.7 mmol/L) (at 2 hours).

Pregestational diabetes mellitus (type 1 or type 2) (ADA 2023):

Fasting: 70 to 95 mg/dL (SI: 3.9 to 5.3 mmol/L).

Postprandial: 110 to 140 mg/dL (SI: 6.1 to 7.8 mmol/L) (at 1 hour) **or** 100 to 120 mg/dL (SI: 5.6 to 6.7 mmol/L) (at 2 hours).

Children and adolescents:

Preprandial glucose: 70 to 130 mg/dL (SI: 3.9 to 7.2 mmol/L) (ISPAD [Dimeglio 2018]).

Postprandial glucose: 90 to 180 mg/dL (SI: 5 to 10 mmol/L) (ISPAD [Dimeglio 2018]).

Bedtime/overnight glucose: 80 to 140 mg/dL (SI: 4.4 to 7.8 mmol/L) (ISPAD [Dimeglio 2018]).

HbA_{1c}: <7%; target should be individualized; a more stringent goal (<6.5%) may be reasonable if it can be achieved without significant hypoglycemia; less aggressive goals (<7.5% or <8%) may be appropriate in patients who cannot articulate symptoms of hypoglycemia, cannot check glucose frequently, have a history of severe hypoglycemia, or have extensive comorbid conditions (ADA 2023; ISPAD [Dimeglio 2018]).

Surgical patients (ISPAD [Jefferies 2018]):

Intraoperative: 90 to 180 mg/dL (SI: 5 to 10 mmol/L).

ICU, postsurgery: 140 to 180 mg/dL (SI: 7.8 to 10 mmol/L).

Classification of hypoglycemia (ADA 2023):

Level 1: 54 to 70 mg/dL (SI: 3 to 3.9 mmol/L); hypoglycemia alert value; initiate fast-acting carbohydrate (eg, glucose) treatment.

Level 2: <54 mg/dL (SI: <3 mmol/L); threshold for neuroglycopenic symptoms; requires immediate action.

Level 3: Hypoglycemia associated with a severe event characterized by altered mental and/or physical status requiring assistance.

Nursing Practice Points

Boxed warning information: This drug has a boxed warning.

Additional patient education: Teach patient:

- That long term use of metformin can cause vitamin B₁₂ deficiency. Report vitamin B₁₂ deficiency symptoms to their doctor (eg, weakness, lack of energy, pins and needles sensation, mouth irritation, vision changes, depression, or confusion).

Note: This information does not cover all factors that could impact the safe and effective use of the medication. Refer to the full drug monograph for more details.

Product Availability (US)

Riomet ER suspension has been discontinued in the United States for >1 year.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Oral, as hydrochloride:

Riomet: 500 mg/5 mL (118 mL, 473 mL) [contains propylene glycol; strawberry flavor]

Riomet: 500 mg/5 mL (118 mL, 473 mL) [contains saccharin calcium; cherry flavor]

Generic: 500 mg/5 mL (5 mL, 8.5 mL, 118 mL, 473 mL)

Suspension Reconstituted ER, Oral, as hydrochloride:

Riomet ER: 500 mg/5 mL (480 mL [DSC]) [contains methylparaben, propylene glycol, propylparaben]

Tablet, Oral, as hydrochloride:

Generic: 500 mg, 625 mg, 850 mg, 1000 mg

Tablet Extended Release 24 Hour, Oral, as hydrochloride:

Fortamet: 500 mg [DSC], 1000 mg [DSC]

Glumetza: 500 mg, 1000 mg

Generic: 500 mg, 750 mg, 1000 mg

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Tablet, Oral, as hydrochloride:

Glucophage: 500 mg, 850 mg

Glycon: 500 mg [DSC], 850 mg [DSC]

Generic: 500 mg, 850 mg, 1000 mg

Tablet Extended Release 24 Hour, Oral, as hydrochloride:

Glumetza: 500 mg

Glumetza: 1000 mg [contains fd&c blue #2 (indigotine, indigo carmine), fd&c red #40 (allura red ac dye), fd&c yellow #6 (sunset yellow)]

Generic: 500 mg, 1000 mg

Dosage Forms: International

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. **Note:** Availability of specific dosage forms may vary by country/region.

Solution, Oral, as hydrochloride: 850 mg/5 mL, 1,000 mg/5 mL

Solution Reconstituted, Oral, as hydrochloride [packet]: 500 mg

Generic Available (US)

May be product dependent

Pricing: US

Solution (metFORMIN HCl Oral)

500 mg/5 mL (per mL): \$2.46

Solution (Riomet Oral)

500 mg/5 mL (per mL): \$1.69

Tablet, 24-hour (Glumetza Oral)

500 mg (per each): \$61.78

1000 mg (per each): \$133.60

Tablet, 24-hour (metFORMIN HCl ER (MOD) Oral)

500 mg (per each): \$55.59 - \$55.99

1000 mg (per each): \$120.22 - \$120.23

Tablet, 24-hour (metFORMIN HCl ER (OSM) Oral)

500 mg (per each): \$4.52 - \$17.24

1000 mg (per each): \$6.38 - \$31.40

Tablet, 24-hour (metFORMIN HCl ER Oral)

500 mg (per each): \$0.05 - \$1.08

750 mg (per each): \$0.09 - \$1.61

Tablets (metFORMIN HCl Oral)

500 mg (per each): \$0.02 - \$1.80

625 mg (per each): \$37.06

850 mg (per each): \$0.03 - \$1.20

1000 mg (per each): \$0.03 - \$2.40

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Mechanism of Action

Decreases hepatic glucose production, decreases intestinal absorption of glucose and improves insulin sensitivity (increases peripheral glucose uptake and utilization)

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Within days; maximum effects up to 2 weeks.

Distribution: V_d : 654 ± 358 L; partitions into erythrocytes; concentrates in liver, kidney, and GI tract.

Protein binding: Negligible.

Metabolism: Not metabolized by the liver.

Bioavailability: Absolute: Fasting: 50% to 60%.

Half-life elimination: Plasma: 4 to 9 hours; Blood ~17.6 hours.

Time to peak, serum: Immediate release: 2 to 3 hours; ER tablet: 6 to 8 hours; ER suspension: 4.5 hours (range: 3.5 to 6.5 hours).

Excretion: Urine (90% as unchanged drug; active secretion).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Peak and systemic exposure is increased, and oral and renal clearance is decreased.

Pediatric: Obesity: In 22 pediatric patients who were obese (mean age: 14.5 ± 1.8 years; range: 11.1 to 17.5 years), pharmacokinetic analysis showed that clearance significantly increases and AUC decreases with increasing total body weight (van Rongen 2018).

Older adult: Total plasma clearance is decreased, half-life is prolonged, and C_{max} is increased.

Dental: Local Anesthetic/Vasoconstrictor Precautions

No information available to require special precautions

Dental: Effects on Dental Treatment

Key adverse event(s) related to dental treatment: Taste disorder.

Metformin-dependent patients with diabetes (noninsulin dependent, Type 2) should be appointed for dental treatment in morning in order to minimize chance of stress-induced hypoglycemia.

Dental: Effects on Bleeding

No information available to require special precautions

Related Information

- [Noninsulin Antidiabetic Agents Comparison Tables](#)
- [Oral Medications That Should Not Be Crushed or Altered](#)

- [Relative Infant Dose](#)

Index Terms

Glucophage; Glucophage XR; Metformin ER; Metformin HCl; Metformin HCl ER; Metformin Hydrochloride

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Brand Names: US

Fortamet [DSC]; Glumetza; Riomet; Riomet ER [DSC]

Brand Names: Canada

AG-Metformin; APO-MetFORMIN ER; APO-MetFORMIN [DSC]; Auro-Metformin; DOM-MetFORMIN [DSC]; Glucophage; Glumetza; Glycon [DSC]; JAMP-MetFORMIN; Mar-MetFORMIN; MAR-Metformin XR; MetFORMIN FC; MINT-MetFORMIN; MINT-Metformin XR; NRA-Metformin; PMS-MetFORMIN; PMSC-Metformin; PRO-MetFORMIN; PRZ-Metformin; RAN-MetFORMIN [DSC]; RATIO-MetFORMIN [DSC]; RIVA-MetFORMIN; SANDOZ MetFORMIN FC; Septa-MetFORMIN [DSC]; TEVA Metformin

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